



Master thesis

Macroinvertebrate communities of perennial and intermittent streams in Bioregion 14 (“Grazer Feld und Grabenland“, Austria: Styria) with a special focus on caddisflies

submitted by

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Graz, 23th January 2020

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Intermittierende Fließgewässer, also jene, die durch Phasen ohne Abfluss charakterisiert sind, werden durch die menschenbedingte Klimaerwärmung häufiger. Das Hauptziel der vorliegenden Arbeit ist es, Methoden zur Erkennung von hydrologischen Regimen in Fließgewässern anhand von Makrozoobenthos-Zönosen in einer biogeographischen Region Österreichs, Bioregion 14 – Grazer Feld und Grabenland – zu testen. Ein weiteres Ziel ist die Erweiterung autökologischen Wissens über Arten intermittierender Fließgewässer. Makrozoobenthos-Zönosen von acht naturnahen Probestellen mit unterschiedlichen hydrologischen Regimen (zwei perennierende, drei intermittierende mit beständigen Resttümpeln und drei intermittierende, die vollständig austrocknen) wurden im Zeitraum zwischen dem Frühjahr und Frühsommer 2019 mehrmals beprobt. Adultstadien von Ephemeropteren, Plecopteren und Trichopteren wurden gesammelt, um Flugzeiten zu beschreiben. Unterschiede in der Zusammensetzung der Makrozoobenthos-Zönosen zwischen intermittierenden und perennierenden Probestellen konnten gezeigt und Indikatortaxa für alle getesteten hydrologischen Fließgewässertypen ausgewiesen werden. Methoden, die auf biologischen Traits von Taxa des Makrozoobenthos basieren, stellten sich als geeignet zur Unterscheidung von hydrologischen Fließgewässertypen heraus. Der Test von internationalen Indices ergab, dass der *Biodrought index* am treffsichersten den hydrologischen Typ identifizieren konnte. Er scheint daher geeignet, in Österreich im Rahmen von angewandten Gewässeruntersuchungen eingesetzt zu werden. Terrestrische Habitate im Ufersubstrat konnten für Larven von *Ironoquia dubia* (Trichoptera: Limnephilidae) als Anpassung an das Trockenfallen beschrieben werden. Weiters werden Merkmale im Entwicklungszyklus und die Verbreitung von unzureichend untersuchten Bewohnern intermittierender Fließgewässer, z.B. von *Chaetopteryx rugulosa*, *Lithax obscurus* (Trichoptera), *Zwicknia* spp. (Plecoptera), *Siphonurus armatus* (Ephemeroptera) beschrieben. Biologische Traits geben einen Einblick in die Anpassungen von Makrozoobenthos-Zönosen an intermittierenden Abfluss. Die Anwendbarkeit der getesteten Methoden zur Erkennung hydrologischer Typen muss durch zukünftige Untersuchungen an zusätzlichen Fließgewässern innerhalb und außerhalb der Bioregion 14 verifiziert werden. Die Analyse des Makrozoobenthos, wirbellosen Bewohnern des Fließgewässerbettes, stellt prinzipiell eine effektive Methode dar, um Reaktionen von biotischen Zönosen auf intermittierenden Abfluss zu zeigen.

Abstract

Intermittent streams, characterised by flow cessation, are becoming more common due to anthropogenic global climate-warming. The main aim of the present study was to test methods for recognition of streams' hydrological regimes by means of macroinvertebrate communities in an Austrian biogeographic region, Bioregion 14 – *Grazer Basin and Grabenland*. Another aim was to extend autecological information on species inhabiting intermittent streams. Benthic macroinvertebrates of eight least-impacted stream sites with different hydrological regimes (two perennial, three intermittent which retain remnant pools, three intermittent which dry completely) were collected multiple times during spring and early summer 2019. Terrestrial adult stages of mayflies, stoneflies and caddisflies were captured to document flight periods. Macroinvertebrate community structure was shown to be significantly different between perennial and intermittent sites. Indicator taxa were defined for all hydrological regime types. Several macroinvertebrate-based methods which use biological taxa traits were shown to act as hydrological regime discriminators. The *Biodrought index*, among international indices tested, enabled most accurate identification of the

hydrological type. It seems to be suitable for use in the field of applied stream assessment. Larvae of *Isonychia dubia* (Trichoptera: Limnephilidae) were shown to be terrestrial and to aestivate buried in the bank substrate as adaption to streambed drying. Additionally, life-history traits and distribution of other poorly studied intermittent stream inhabitants, e.g. *Chaetopteryx rugulosa*, *Lithax obscurus* (Trichoptera), *Zwicknia* spp. (Plecoptera), *Siphonurus armatus* (Ephemeroptera) are described. Biological traits provide insight into the adaption of macroinvertebrate communities to intermittent flow. The applicability of the methods tested to recognise hydrological stream types should be verified in additional streams and beyond the study area in future investigations. The analysis of benthic macroinvertebrates generally constitutes an effective way to show responses of biotic communities to intermittent flow.

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Acknowledgements

I would first like to thank my supervisor Assoc. Prof. Dr. Wolfram Graf and my co-supervisor Dipl.-Ing. Dr.nat.techn. Patrick Leitner for the expert guidance of my master thesis. I would also like to thank Dipl.-Ing. Thomas Huber for his invaluable support in the laboratory and Dipl.-Ing. Florian Dossi for his support in statistics.

I would like to express my gratitude to the intermittent stream experts Marek Polášek and Petr Pařil (Masaryk University, Brno) for the help with the Biodrought index and related topics, and to Prof. Johann Waringer for his help concerning head capsule measurements of caddis flies.

All those who have successfully completed an academical thesis know how crucial fellow students are. Therefore, I want to thank Olivia Wilfling, Martin Tschikof and Claudia Dienstl who provided important data and their knowledge.

I would like to give special thanks to the water authorities of the Federal Province of Styria (Department *Gewässeraufsicht und Gewässerschutz*) in person of Mag. Barbara Friehs for giving me the possibility to gather valuable information for my thesis during three months of internship from 2017 to 2019. Among the numerous employees who supported me I would like to highlight Mag. Thomas Battisti whose help went far beyond the time of the internships. Furthermore Mag. Peter Rauch provided valuable information for the geological and hydrological description of the study area.

So many people were in some way supportive for my thesis: My father Günther enabled me to move from Graz to Vienna and to pursue my studies. My mother Herta provided me her car innumerable times for the field work. Helmut Schöggel's support was essential. I owe a very important debt to Michaela Hafner, who had to suffer most from my countless days and nights I spent for writing this thesis.

1 Rationale and objectives

The Water Framework Directive 2000/60/EG (European Commission 2000) is the legal basis of water resource policies in the European Union. The aim of the directive is to achieve or maintain at least “good ecological status” for all surface waters. Ecological status is assessed by analysing “Biological Quality Elements”, i.e. organism groups regarded indicative for environmental disturbances. Macroinvertebrates (invertebrate animals that are visible without magnification) are one BQE besides fish, diatoms, macrophytes and phytoplankton. In essence, these assessment methods are based on comparison of biotic communities with those that are not or minimally impacted by human activities (referred to as “reference conditions”) within one water body type. Subsequently, in Austria, like in each member state of the European Union, methods have been developed to assess ecological status. The Austrian method using macroinvertebrates (MI) in streams has been developed on this institute (BOKU; Ofenböck et al., 2005) and has been applied officially until today with some adaptations. However, not all types of streams can be assessed with this method. Intermittent streams (IS), streams that experience flow cessation, are among these types. Even though the occurrence of this hydrological stream type is known from most Austrian bioregions (Wimmer et al. 2012), only intermittent flysch streams of Vienna can be assessed currently (Ofenböck et al. 2018).

Leitner et al. (2016) and Leitner & Graf (2018) found evidence of the existence of ISs in Bioregion 14 when assessing several streams on behalf of the federal province of Styria. They showed that MI communities differed on a gradient of hydrological regimes. Additionally, known indicator taxa (mainly EPT taxa) for intermittent flow were found and could be partly confirmed by statistical means. Based on these findings, Sumper (2019) showed that the existing official assessment method for streams in Bioregion 14 is not suitable for application to ISs, and that the existing method has to be adapted for sampling in spring mainly because indicator taxa for ISs are most numerous in this season. However, the number of ISs included in these studies was low (four), and taxonomic resolution was not at species level (e.g. many Limnephilidae, Trichoptera) mainly due to the exclusive presence of early larval instars, if reliable determination in any larval stage is possible at all. Therefore, the ecological relevance of these findings has to be extended.

An important aspect when developing an assessment method is the **classification of sufficient stream types** that reflect natural variety (Stubbington et al. 2018). Defining only one “intermittent stream type” might not reflect the heterogeneity within this “type” considering the numerous hydrological parameters that characterise intermittence regimes (e.g. frequency, duration, spatial extent of drying). After validation of IS types, a standardised method to recognise those types based on MI samples taken during flowing phase has to be defined before developing a WFD-compliant assessment method for each stream type.

Methods to recognise ISs are not limited to the use of **indicator taxa**, but **structural** (i.e. taxonomic) **and functional (i.e. ecological and biological) traits** have been shown to act as effective hydrological regime discriminators (biological: e.g. Bonada et al. 2007; structural: e.g. Chadd et al. 2017; all three approaches combined: Straka et al. 2019). Functional community traits have the advantage to enable identification of environmental conditions (e.g. flow permanence) beyond taxonomic differences (Statzner et al. 1997) and thus are potentially applicable on a global scale.

The higher **temporal variability** of habitat conditions in ISs (e.g. discharge variation) compared to perennial streams results in higher temporal variability of biotic communities, which is also a potential source of recognising different hydrological regimes (Bonada et al. 2007a).

A prerequisite of using biological taxa traits is the high level of autecological information available, and consequently, missing information hampers effective characterisation. One such example in Bioregion 14 is *Ironoquia dubia* (Trichoptera: Limnephilidae). Detailed studies on this species are lacking and biological traits indicated in literature are inconsistent. One of the aims in this thesis is to contribute towards elucidating its life-history strategy for survival in ISs. Additionally, it has to be verified if other MI species inhabiting ISs in Bioregion 14 apply the same life-history strategies as in published literature from other regions.

Stating these circumstances, the objectives of this master thesis are:

- Increasing the number of sampling sites in ISs in Bioregion 14
- Increasing the taxonomic resolution of previous MI samples in Bioregion 14 by multiple samplings during spring and early summer and adult catches
- Identifying communities of intermittent and perennial streams including an intermittent type that retains chains of remnant pools and identify indicator taxa for all hydrological types
- Testing existing indices (DEHLI, Biodrought index) to track ISs for applicability in Bioregion 14
- Identifying hydrological stream types based on biological community traits that are known to promote survival in ISs
- Describing structural and functional changes in communities of the hydrological stream types during spring and early summer.
 - Increasing autecological information on selected species, especially *Ironoquia dubia*:
 - Describing its life-history strategies by field and laboratory observations: larval development, dormant stages, flight period, potential refuge use
 - Defining instars based on head capsule width
- Analysing faunal composition and life-history traits of EPT species between ISs of Bioregion 14 and other regions

2 Introduction

2.1 Importance of intermittent streams in a global and Austrian dimension

Intermittent streams (ISs) are the characteristic lotic ecosystems of arid and semiarid climate regions. In Australia 70% of river channels are considered intermittent (Sheldon et al. 2010). However, even in temperate regions intermittent flow makes up a considerable part of river networks: 25%-40% of the French stream network have intermittent flow (Snelder et al. 2013). On a global scale, a conservative estimate is that ISs constitute more than 30% of river length (Tooth 2000). As these estimates do not include low-order streams - which can make up more than 70% of river networks and are particularly prone to intermittence (Lowe & Likens 2005) – the proportion of ISs in the global stream network is likely to be greater than 50% (Datry et al. 2014b). Despite this, a concerted effort to study the ecology of ISs only started some 20 years ago (Datry et al. 2014b), even though pioneering works were published several decades ago especially from North America (e.g. Clifford 1966, Williams & Hynes 1977).

In addition to naturally intermittent streams, many **anthropogenic activities** such as impoundments and water abstraction have caused formerly perennial streams to become intermittent (Datry et al. 2014b). Similarly, anthropogenic **global climate warming** is likely to increase occurrence, duration and frequency of intermittence as many climate change-runoff models predict (Larned et al. 2010). Anthropogenic intermittence is not only likely to detrimentally effect aquatic communities, but to reduce ecosystem services provided by perennial streams (e.g. fisheries, Xenopoulos et al. 2005). Therefore, the need for its reduction and mitigation is urgent even from a purely human perspective. In **Austria**, important drivers for climate warming-induced reduction of stream discharge by natural dynamics are reduced precipitation, increased occurrence of severe droughts, reduction of groundwater levels (Dalla-Via 2008) and declining snow and glacier reserves in the Alps (Schädler 2008). Increased human use of water resources due to the effects of the climate warming is expected to exacerbate declines in discharge in Austria (Steininger et al. 2015). For example, crop failures increasingly urge Austrian farmers to irrigate agricultural land. For 2040 crop failures of up to 30% in years of extreme drought are predicted in certain low-altitude regions of Austria (Lower Austria: Weinviertel, Mostviertel; Northern and Southern Burgenland), which account for approximately 10% in the Mühlviertel, parts of Carinthia and **Southern Styria** (Strauss et al. 2013). As available ground- and surface water amounts are potentially sufficient to largely compensate for crop losses in these regions, future increases in agricultural irrigation can be expected to impact discharge in these regions (Steininger et al. 2015). In contrast, the decline of snow and glacier reserves in the Alps is likely to increase the probability that related melt-water fed streams dry in summer in the Alpine and foothill regions of Austria (Schädler 2008).

From the perspective of **nature conservation**, ISs harbour numerous regionally and globally endangered habitat specialist species as also shown for several Central European EPT species (e.g. Macadam 2016, Sommerhäuser & Timm 1994, Ehlert & Sommerhäuser 2002, Wolf 2007, Foltyn et al. 1996). Another reason to increase our understanding of ISs is their provision of numerous **ecosystem services** such as functioning as sink for floodwaters, sediments and organic matter (see Koundouri et al. 2017).

MI communities of intermittent streams in low altitude regions have not been studied systematically in Austria, even though numerous intermittent sections are known (Wimmer et al. 2012: none stated in Bioregion 14, though). Studies seem to exist from Bioregion 4 (Flysch) only (Philipp & Forster 2000, Battisti 2003, Dienstl 2019). Apart from this region, the spatially closest study areas to Bioregion 14, where extensive studies on ISs have been conducted are Germany and the Czech Republic (see chapter 5.3.14). Numerous studies from Germany not only describe community composition of ISs, but formerly unknown adaption strategies of many EPT species (e.g. Fiedler 1992, Sommerhäuser et al. 1997, Lillpopp et al. 1998, Fischer 2003).

Considering these circumstances, the necessity to incorporate ISs into ecological concepts, knowledge, methods and assessment is evident. To this day, intermittent rivers are clearly underrepresented in national management and assessment programs in European countries (Stubbington et al. 2018). Further, the mentioned occurrence of endangered habitat specialists in ISs requires stronger inclusion into protection programs.

2.2 Importance of intermittent streams for the understanding of stream ecology

The recent increase in interest for intermittent streams (ISs) and the substantial ecological differences when compared to perennial streams has given rise to **new concepts** in stream ecology that explain distribution of biotic communities in stream networks. IS networks are namely conceptualised as **shifting mosaics of lotic, lentic and terrestrial habitats** (Larned et al. 2010, Datry et al. 2014b; see chapter 2.3). Related hydrological discontinuities that result in longitudinal discontinuities of biotic communities are in conflict with the current concepts (and its successors) of stream ecology, for example the **River Continuum Concept** (Vannote et al. 1980). Instead, **meta-system concepts** (meta-communities, meta-populations, meta-ecosystems), which are new to stream ecology, have been tested for applicability in ISs and might explain how communities and populations are able to survive drying of their habitats (see chapter 2.4.1). Furthermore, exploring unknown **adaption strategies**, which enable ISs inhabitants to survive drying of their habitats is a major task in hydrobiology (see chapter 2.4.3).

2.3 Definition, classification and abiotic characterization of intermittent streams

2.3.1 Definitions and classification

Running waters emerge where earth's surface is inclined and where precipitation exceeds evaporation and infiltration, and therefore resulting in surface flow (Marcinek & Rosenkranz 1996). Running waters also form, where precipitation exceeds these processes only during certain periods of the year or irregularly. Thus, the following classification according to temporal consistency and timing of *surface* runoff can be established (after Marcinek & Rosenkranz 1996 and Sommerhäuser et al. 1998):

- (1) perennial
- (2) intermittent
 - a) periodic
 - i. summer-dry
 - ii. winter-dry
 - b) episodic
 - i. episodically dry
 - ii. episodically flowing (=ephemeral)

Perennial streams are characterised by permanent surface flow, whereas intermittent streams (ISs; *intermittere*: lat. interrupt, leave a gap; pons online dictionary 2019) exhibit temporary surface flow. **Periodic** streams have seasonal flowing and dry phases, which are irregular and unpredictable in **episodic** streams. **Summer-dry** streams have dry phases in the hydrological summer (May to October), **winter-dry** streams in the winter (November to April, e.g. alpine streams that freeze). Episodic streams can be **episodically dry** (only in some years) or **ephemeral** (frequent drying and long-lasting dry phases during the year: e.g. surfacewater-fed headwater streams, dessert streams). Even though most of these types are expected to occur in the study area of this thesis, only summer-dry streams are examined.

Unfortunately, the definition of some of these terms and many others in this context is inconsistent in literature, resulting in a “confusing plethora of terms” (Comín & Williams 1994). For example, non-perennial streams are subdivided into *intermittent rivers* and *ephemeral streams* (“**IRES**”), with *intermittent rivers* being the above-mentioned periodic streams and ephemeral streams being the episodic ones; as umbrella term for both, *temporary rivers and streams* is used in this case (e.g. Stubbington et al. 2018). Bonada et al. (2007a) relate “ephemeral” to periodic streams that completely dry and “intermittent” to periodic stream that retain remnant pools.

As the term “intermittent streams” to denote summer-dry streams is most common in recent English literature, this term is used in this thesis.

Numerous terms to describe the long-term temporal course (**regime**) of hydrological conditions exist in IS literature: aquatic, hydrological, flow, discharge, intermittence regime etc. Due to lacking definitions, this resulted in difficulties in perceiving differences between terms. The stream types investigated in this thesis are in accordance with types classified in Gallart et al. (2012). These types are based on the long-term schedule (regime) of “aquatic states” (e.g. flood, remnant pools, only

hyporheic flow, dry alluvium), which is consequently referred to as “aquatic regime”. Based on these regimes Gallart et al. (2012) classify four stream types: perennial, intermittent-pools, intermittent-dry and episodic-ephemeral. The first three of these types were investigated in this thesis and are characterised hereafter. The terminology of Gallart et al. (2012) was slightly changed: Stream types are referred to as **hydrological regime types** (shortened to **hydrological types**):

- **perennial** (per.): surface flow never ceases
- **intermittent+** (int+): periodic flow cessation typically occurring in summer, but remnant pools persist
- **intermittent-** (int-): periodic flow as in int+, but remnant pools vanish resulting in a completely dry stream bed

The terms “**discharge**” and “**flow**” are both used in this thesis and are defined hereafter. While the **flow** in a stream is related to running water irrespective of its quantity, **discharge** is used with reference to its amount (litres per second). For example, in intermittent streams discharge is reduced continuously before flow ceases.

For reasons of simplicity, the term “**stream**” is used for all natural running waters irrespective of their width, discharge, catchment area size etc. in this thesis.

2.3.2 Abiotic characterization

The above-mentioned classification of ISs is based on predictability of intermittent flow. However, many additional parameters that are ecologically relevant could be included for more detailed hydrological classification (Meyer et al. 2000). In Table 1 flow parameters are compiled including related states and relevance for biotic communities:

Table 1: Ecologically relevant flow parameters compiled from 1 Meyer et al. (2000), 2 Gallart et al. (2012), 3 Datry (2012), 4 Leigh & Datry (2016), 5 Bohle (2000), 6 Larned et al. (2010), 7 Stubbington et al. (2017), 8 Boulton (2003)

flow parameter	Parameter states	Effects on MI communities
predictability ¹	low: episodic streams	low taxa richness ⁴ , only opportunist species ⁵
	high: periodic streams	higher taxa richness ⁴ , life-history adaptations required ¹
hydrol. regime ²	duration ¹	e.g. days (mean drying event duration) ³
	flow permanence ³	% of time with surface flow ³
	frequency ¹	e.g. no. of dry events per year ³
	Long-term maximum intensity of drying ²	perennial (abundant riffles) intermittent+ (remnant pools) intermittent- (dry with or without hyporheic flow) episodic
		determine adaption strategies and taxon richness, e.g. duration: negative correlation with taxa richness ^{3 4}
		determine refuge use of species ⁸
spatial arrangement ¹	upstream aquatic refuges	recolonisation by drift possible
	downstream aquatic refuges	recolonization by upstream migration (swimming, crawling, flying) ⁷
	both	both
	distances of refuges ⁶	determine community composition acc. to different dispersal abilities
cause ¹	Natural: ⁶ seepage in porous beds evapotranspiration downward shifts in groundwater tables hillslope runoff recession freeze-up	no adverse effect to species adapted to drying conditions
	anthropogenic ¹	typically low predictability, causes low taxa richness (see above)

Besides an ecologically relevant classification of ISs, researchers attempted to develop hydrological models to understand the highly dynamic processes of biotic communities in ISs habitats.

Flow cessation causes hydrological decoupling between perennial reaches in a stream network and usually results in isolated lentic habitats (remnant pools). Advancing and retreating wetted fronts driven by groundwater level fluctuations and changing surface runoff cause continuous expansion and contraction of wetted reaches. This observation induced researchers to conceptualise IS networks as **shifting mosaic of lotic, lentic and terrestrial habitats** (Larned et al. 2010, Datry et al. 2014b).

2.4 Macroinvertebrate communities of intermittent streams

This chapter intends to summarise findings on the structure and functions of MI communities in intermittent streams (ISs). Starting off with how the structural variability of MI communities within and between IS networks is conceptualized in current literature (meta-community concept, chapter 2.4.1), it continues with the description of specific structural and functional features of IS communities (chapter 2.4.2). Finally, information on how MI species are able to survive flow intermittence of their habitats (adaption strategies, chapter 2.4.3) is provided.

2.4.1 Intermittent stream communities as meta-communities

A meta-community is “a set of local communities that are linked by dispersal of multiple potentially interacting species” (Leibold et al. 2004, p. 601). Consequently, the meta-community concept is likely to be applicable to ISs due to the disconnection and reconnection of aquatic habitats. Different distributions of species between intermittently isolated sections of ISs networks can be explained by meta-community concepts. Thereby, aspects such as dispersal/migration ability and mortality of species, different distances between local aquatic communities and distances to external colonist sources are considered (Larned et al. 2010). Dispersal and/or exceptional migration ability through multiple potential paths (water, air, hyporheic interstices and land; Stubbington et al. 2017; see chapter 2.4.3) of at least some species of local communities are a prerequisite for the exchange of species among local communities.

In contrast, the hydrological continuity of **perennial streams** is likely to favour other concepts, e.g. the **River Continuum Concept**, to explain distribution of biotic communities.

2.4.2 Structure and functional traits of macroinvertebrate communities in perennial and intermittent streams

Hydrological patterns like flow cessation are a primary driver of community composition around the world (Datry et al. 2014a). Generally, **taxa richness** decreases on intermittence gradients with intermittent stream (IS) communities being **nested subsets** of those of perennial streams (Datry 2012). These subsets mainly consist of **generalist and ubiquitous taxa** and establish due to the loss of taxa susceptible to drying (**perennial-flow specialists**) rather than the selection of **intermittent-flow specialists**. Nevertheless, as low numbers of these specialist species often occur exclusively in ISs they are good indicators of intermittent flow.

The mentioned decline in taxa richness can be explained by the **gradual and stepped hydrological decline** during an extreme drying event in a generally perennial stream (Fig. 1; Boulton 2003). It results from the gradual loss of in-stream habitats followed by the abrupt change of abiotic habitat conditions at critical thresholds. For example, when discharge decreases rheophilic taxa decrease gradually, but most disappear abruptly as soon as flow ceases due to low oxygen concentrations in the resulting pools and/or the need of flow for feeding (passive filter feeders). In contrast, lentic and limnophilic taxa which have higher resistance to low oxygen concentrations persist in remnant pools and even increase by colonisation from external sources, thus partly compensating for the continued loss of remnant rheophilic taxa.

Unlike in perennial streams, which might only dry at extreme events (Fig. 1), in ISs, taxa are not likely to be “lost” because they have adaptations to flow intermittence (e.g. desiccation-resistant stages; see chapter 2.4.3)

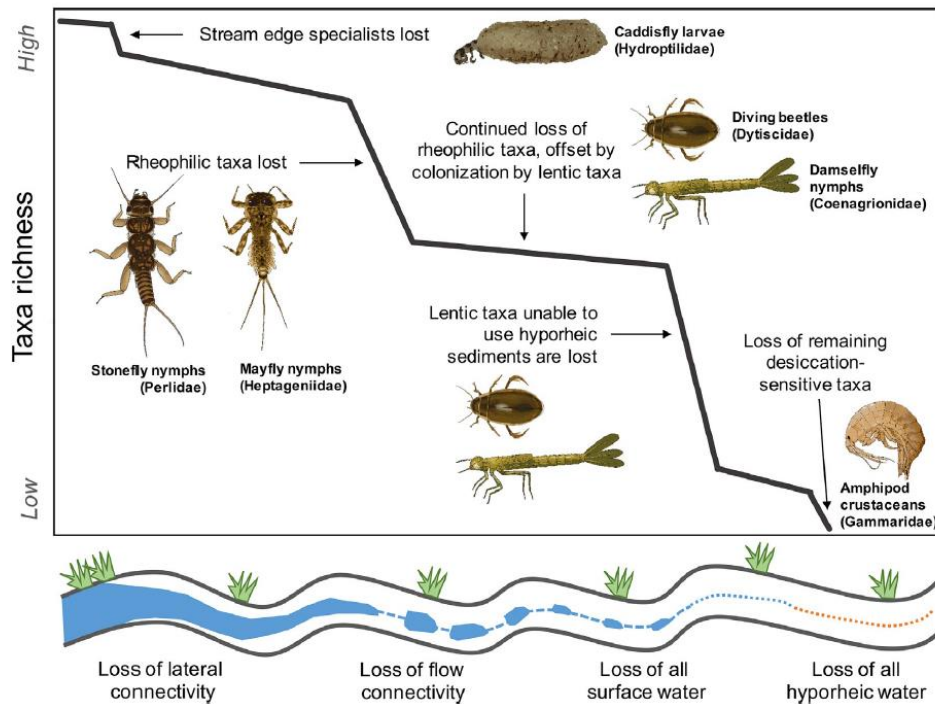


Fig. 1: Decline in taxa richness and changes of functional groups during a drying event (from Stubbington et al. 2017, after Boulton 2003)

This model is based on the Habitat Templet Theory (Southwood 1977). The theory indicates that spatio-temporal habitat variability shapes biological and ecological features (traits) of communities resulting in specific trait expressions (“trait states” or “trait modalities”; Stubbington et al. 2017 and Tachet et al. 2010, respectively) of related species. Accordingly, specific habitat conditions select organisms that cope with these conditions through their specific trait states. Transferred to IS ecology, different hydrological stream types are expected to have different communities.

Combining the Habitat Templet Theory and the model of Boulton (2003, Fig. 1) the following **implications for ISs** can be derived:

- (1) ISs which retain remnant pools (int+) select species with low oxygen-resistant aquatic stages for persistence in pools and high migration ability for following shifting aquatic refuges. ISs that dry completely (int-) select species with drying-resistant stages that are synchronised with seasonal dry periods.
- (2) Taxa richness declines on hydrological gradients such as perennial/int+/int-.
- (3) Community structure and traits (e.g. proportion of rheophilic and lentic taxa) change in a specific manner in the course of seasonal drying events.

When comparing **community traits** between ISs and perennial streams differences are largely related to higher temporal variability in ISs as derived from the model in Fig. 1. Accordingly, Bêche et al. (2006)

found that 39 of 73 MI traits changed significantly between flowing phase and pool phase of ISs, whereas none changed in comparable perennial streams. Higher variability in community functions in ISs can also be expected within the flowing phase of periodic streams as prospective strategies, e.g. emergence to terrestrial adult stages in good time before seasonal flow cessation, have evolved in many species.

Among **ecological traits** likely to be highly variable in ISs, **feeding groups** have to be mentioned: For example, predators typically increase as flow ceases because of the abundant availability of prey concentrated in remnant pools, whereas passive filter feeders decrease because they need flowing water for feeding (Bogan & Lytle 2007).

Biological traits, namely the focus of this thesis, are treated in the following chapter.

2.4.3 Invertebrate adoptions to intermittent flow

In order to survive intermittent flow numerous related traits have evolved in species. These traits encompass different **physiological, behavioural, morphological and/or life-history adaptations**. Usually species do not have one, but suits of co-evolved adaption traits that constitute the whole intermittent-flow survival strategy (or “response” to intermittent flow; Strachan et al. 2015).

Each adaption trait can be attributed to either **resilience** or **resistance** (Stubbington et al. 2017). In the context of ISs, resilience is the ability to recolonise a stream after flow resumption and resistance being the ability to tolerate flow cessation in the streambed. Some authors (e.g. Bohle et al. 2000, Meyer et al. 2000) further distinguish **opportunism**. Opportunists are species that benefit from habitats that develop after flow cessation (remnant pools, dry streambed) without being specifically adapted to ISs. They can be **aquatic predators** that feed on organisms from remnant pools (e.g. water beetles such as Dytiscidae, Gyrinidae, Hydrophilidae or water bugs such as Notonectidae, Nepidae) or **terrestrial predators** that feed on dormant IS inhabitants from the dry streambed (e.g. carabid beetles). Species that use remnant pools for **reproduction** by quick colonisation and short life-cycle duration (pioneer species) such as many Diptera species are also referred to as opportunists.

Resilience traits: morphological and behavioral adaptations

Resilience traits are characterised by exceptional **migration and/or dispersal ability**. Emigration and recolonisation can happen (1) **vertically** through the hyporheic zone, (2) **longitudinally** in the stream channel and (3) **laterally** (aerial) by flying (Meyer et al. 2000, Williams 1977, Delucchi 1989). While vertical and longitudinal migration typically occur within the water and are directed to aquatic upstream and downstream **refuges** (perennial reaches or remnant pools), lateral migration can also be directed to terrestrial “refuges” when emergence from aquatic larvae to long-lived terrestrial adults occurs such as in numerous limnephilid Trichoptera species of summer-dry streams. Vertical migration into the subsurface sediments of the hyporheic zone is enabled by appropriate **morphology** (e.g. elongate true fly larvae such as in Ceratopogonidae: Stanley et al. 1994; small body size, robust exoskeleton like in gammarids: Stubbington et al. 2017) and by the ability to actively migrate vertically. For example, vertical **burrowing** behaviour has been correlated with ISs (Bonada et al. 2007a). **Longitudinal** migration is favoured by high **drift, swimming** (e.g. amphipods) or **crawling** rates (Stubbington et al. 2017).

Resistance traits: physiological adaptations

Resistance (Fig. 2) requires exceptional physiological adaptations of organisms. The crucial adaptation to persistence in **pools** is the tolerance of low oxygen concentrations, which is favoured by, for example, **air breathing** by means of **spiracles** (e.g. adult and larval dytiscid water beetles, nepid and belostomatid water bugs) and/or **plastrons** (e.g. notonectid or naucorid water bugs), littoral preference (no need of flowing water due to resistance to low-oxygen concentrations) (Fig. 2: “pools”), **active gill movement** to produce a respiratory current (e.g. some mayfly species). **Desiccation-resistant stages** do not require pools, but usually in-stream microhabitat refuges with moist conditions (examples in Fig. 2: algal mats, crayfish burrows, woody debris, vegetation, large stones, leaf litter; Stubbington & Datry 2013).

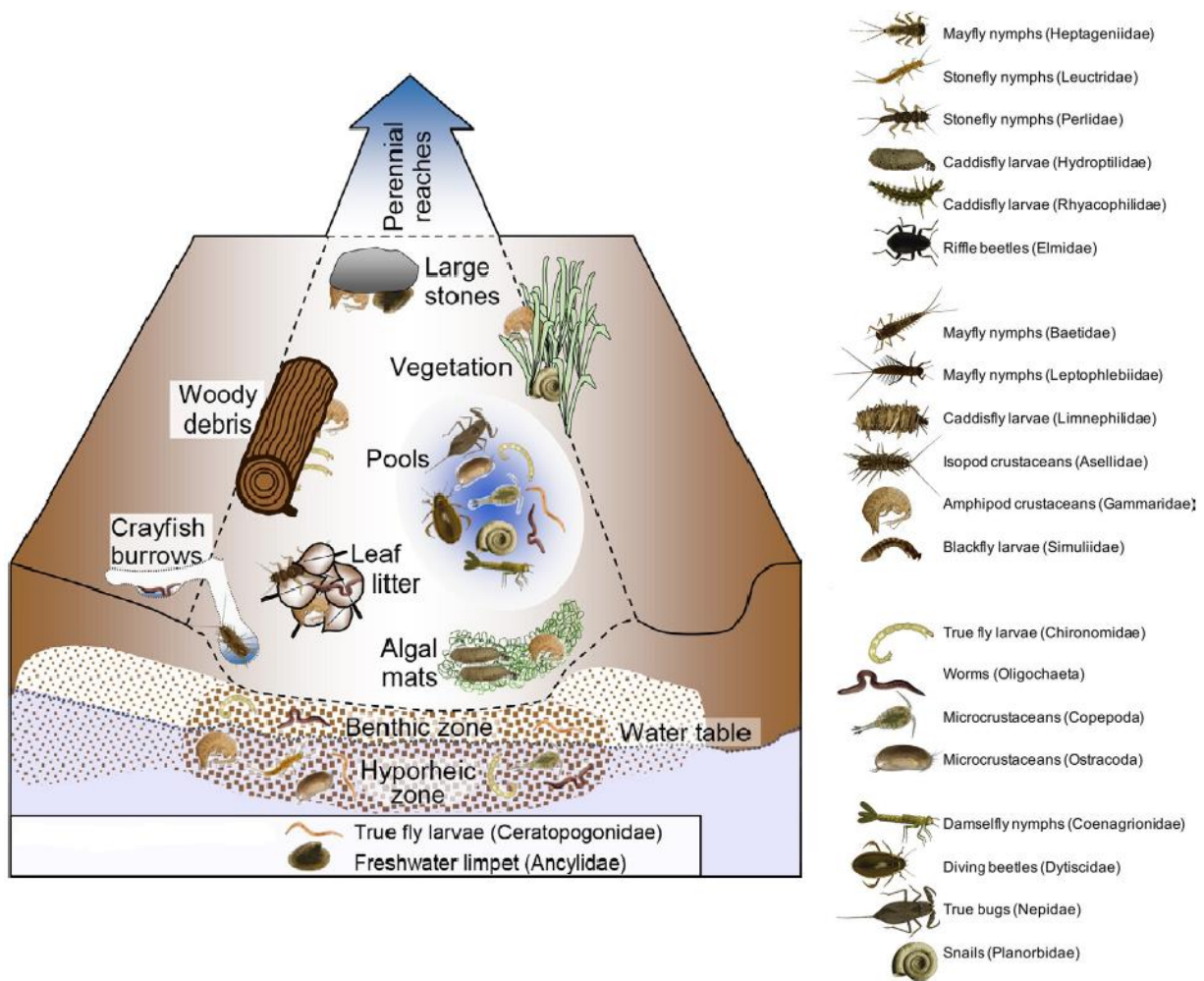


Fig. 2: In-stream refuges that promote resistance to flow cessation (after Stubbington et al. 2017)

Many MI species that are adapted to drying are not able to remain active and enter a stage of **dormancy** during dry phases. Dormancy regards any kind of negative deviation from the optimal species-specific developmental speed related to unfavourable environmental conditions (Müller 1970). Frequently, a classification of dormancy distinguishing only between quiescence and diapause is used. A more detailed classification according to Müller (1970) suggests four types of insect dormancy that constitute a series with increasing ecological adaption to seasonal changes of the environment such as the discharge of periodic streams:

- consecutive dormancy
 1. quiescence
 2. oligopause
 - prospective dormancy
 3. parapause
 4. eudiapause
- } diapause

Consecutive dormancy is initiated by environmental disturbance (e.g. stream bed drying), **prospective dormancy** has a genetically predetermined mechanism of initiation that happens before onset of environmental disturbance in a predetermined developmental stage. Therefore, the latter is restricted to inhabitants of ISs with predictable flow intermittence (periodic streams). **Quiescence** and **oligopause** are *consecutive dormancies* with **quiescence** being an immediate reaction to onset and end of disturbance, and **oligopause** being a gradual/delayed reaction and thus a “slight” **diapause**, which is defined as manifestation of dormancy.

Parapause is initiated obligatorily by reaching a certain life-cycle stage before the onset of unfavourable environmental conditions. It is maintained by a certain state of an environmental factor (e.g. long photoperiod, high temperature) and terminated by the change of the same factor when unfavourable environmental conditions still persist. In many Limnephilidae species (Trichoptera) inhabiting ISs, long-day induced diapause of ovaries in the pupal and adult stage in spring exists (Novák & Sehnal 1963): Adults emerge in spring with unmaturing gonads before flow ceases. Maturation of gonads starts as soon as day length falls below a critical value in summer and females oviposit in late summer and autumn when stream flow has resumed or is about to resume.

Eudiapause is nearly always initiated by the photoperiod, the change of which, however, has no impact on termination. Instead a longer period of lower temperatures terminates diapause, however, only when it has lasted for a certain time span. If under artificial conditions photoperiod does not change, development continues without diapause (eudiapause is facultatory). Eudiapause is known from species of the genus *Zwicknia* (Plecoptera: Capniidae; see discussion), which inhabit ISs.

Life-history adaptations

Based on life-history theory, **life-history adaptations** are defined as life cycle events (e.g. emergence, reproduction, growth) that are synchronised with environmental events (Lytle & Poff 2004) and usually consist of several species traits. Life-history adaptations of ISs inhabitants promote resilience by synchronisation of life-cycle events with flowing and dry phase of periodic streams and typically encompass numerous related traits. These traits encompass timing of reproduction, emergence of adult stage and diapause (Lytle & Poff 2004). For example, many univoltine MI inhabitants of summer-dry streams emerge as adult stage significantly before flow ceases in spring, aestivate in a dormant drying-resistant stage (eggs, adults or early-instar larvae) and continue development in autumn when flow resumes. **Shorter life-cycle duration** and/or **fast development** during flowing phase, reproduction traits like **terrestrial egg clutches** (Kampwerth 2010), **ovoviviparity**, **asexual reproduction** enhance rapid recolonization after flow resumption (Stubbington et al. 2017). The ability of populations to adapt life-stage durations to varying hydrological regimes (high **life-cycle plasticity**; Bohle 2000, Stubbington et al. 2017) resulting in, for example, **earlier flight periods** when compared to populations in perennial streams is also an observed life-history adaption to intermittent flow (Sommerhäuser et al. 1997). **Bet-hedging strategies** (risk spreading among the progeny), such as asynchronous egg hatching or different

larval development speed of individuals, that usually decrease fitness under stable environmental conditions but might increase the chance of at least a partial cohort to survive periods of no flow can enhance survival in ISs with unpredictable hydrological regime (Lytle & Poff 2004). Unsynchronised emergence to the adult stage (**long flight periods**) can also be part of bet-hedging strategies for survival of intermittent flow. The related disadvantage of lower availability of mating partners is compensated by minimising the chance of completely failed reproduction due to larval/pupal mortality caused by flow cessation (Graf et al. 2008).

2.5 Research questions and hypotheses

RQ1: Can hydrological stream types be separated statistically based on macroinvertebrate communities?

H1: Perennial streams, intermittent streams that retain remnant pools and streams that dry completely can be separated based on macroinvertebrate communities.

— Indicator species for all three hydrological stream types can be defined.

RQ2: Does the macroinvertebrate community structure change more markedly in intermittent than in perennial streams in the pre-drying period (spring/early summer) due to the unstable hydrological conditions?

H2: Temporal dynamics of the macroinvertebrate community structure are higher in intermittent than in perennial streams during spring and early summer.

RQ3: Can hydrological stream types be identified using biological macroinvertebrate community traits?

H3: The macroinvertebrate communities of intermittent streams have behavioural, physiological, morphological and/or life-history adaptations and these traits can be used to identify hydrological stream types:

- a. **Macroinvertebrate communities of intermittent streams can be identified by specific biological traits and can therefore be distinguished from those of perennial streams.**
- b. **Macroinvertebrate species have different (earlier and longer flight periods) in intermittent than in perennial stream communities.**

RQ4: Which strategies does *Ironoquia dubia* (Trichoptera: Limnephilidae) apply to survive intermittent flow? How does its larval and pupal development proceed?

H4: *Ironoquia dubia* shows dormant terrestrial larvae and pupae that bury in the bank substrate as reported by Fiedler (1992).

RQ5: What are important life-history traits of EPT species (e.g. *Chaetopteryx rugulosa*, *Lithax obscurus*, *Habrophlebia* spp., *Zwicknia* spp.) inhabiting intermittent streams in Bioregion 14?

RQ6: How are the following species distributed in Bioregion 14?

- *Zwicknia* species
- *Synagapetus moselyi*
- *Synagapetus krawanyi*
- *Oligostomis reticulata*
- *Lithax obscurus*
- *Ironoquia dubia*
- *Chaetopteryx rugulosa*

3 Materials and methods

3.1 Study area

3.1.1 Bioregion 14 – Grazer Basin and Grabenland

The study area is situated in Bioregion 14 (sensu Moog et al. 2001; Fig. 3) and belongs to the European Ecoregion *Dinaric Western Balkan*.

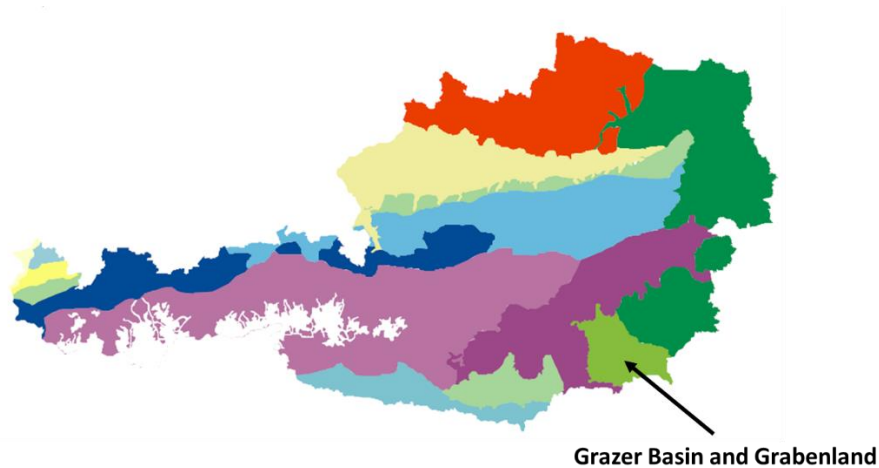


Fig. 3: Austrian Bioregions after Moog et al. (2001) (from Wimmer et al. 2012)

Bioregion 14 is a piedmont region of the Alps with altitudes from around 200 to 600 m (Fink 2000). The bioregion is dominated by *interfluvial landscape* (German: Riedelland), i.e. elongated narrow hill ridges between valleys with relatively constant altitude (Fink 2000). The most prominent valley belongs to the River Mur. In the West and North the bioregion is bordered by higher mountain ridges (Stryrian Randgebirge) with altitudes of up to 2200 m. In the East it is bordered by the Hungarian Plain (Ecoregion 11).

Bioregion 14 comprises of several landscapes (the *Eastern Styrian Grabenland*, the *Western Styrian Hügelland* and the *Grazer Basin* according to Fink 2000) that were combined to form a Bioregion based on similarity of MI communities (Schmidt-Kloiber 2002). Sampling sites of this thesis are restricted to the **Eastern Styrian Grabenland** and the **Grazer Basin** because the Western Styrian Hügelland was regarded to potentially have diverging MI communities due to diverging geology. This is one of the reasons why the inner differentiation of the Bioregion is currently being revised.

3.1.2 Geology

The Grazer Basin is a valley opening of the River Mur around the city of Graz. The Mur valley continues southwards and later eastwards towards the Austrian-Slovenian border and includes three additional valley openings (Leibnitzer, Murecker and Bad Radkersburger Basin; Fink 2000). It is dominated by Würmian lower terraces, which line the floodplain of the Mur (Fink 2000). Towards the East, higher quaternary terraces (e.g. Helfbrunner terrace, Schweinsbachwald terrace) follow to form the Eastern Styrian Grabenland. The Western Styrian Hügelland covers an area towards the West, which has a markedly diverging geology.

The **Eastern Styrian Grabenland** extends south- and eastwards from the Grazer Basin to the catchment of the River Raab in the East. Several nearly straight valleys of the larger streams (e.g. Schwarzaubach, Saßbach, Gnasbach) separated by low ridges ("interfluves", German: *Riedel*) that proceed in north-south direction are characteristic of this landscape. In the Southern part of the Grabenland, where seven sampling sites are situated, the ridges are parts of the quaternary higher terraces.

The rissian **Helfbrunner terrace** is characterised by a 2-5 m thick sequence of pebble and sand layers and an overlying 2 to 10 m thick loam cover (Flügel & Neubauer 1984). One of the intermittent streams sampled in this thesis is located in this terrace as a whole (Pfaffenbach). The other intermittent streams are situated in **colluvial sediments**, which are typically loose and stony substrates. These sediments favour infiltration of surface water to subterranean zones (Peter Rauch, personal communication). However, large parts of their catchments belong to other higher terraces and the mindelian **Schweinsbachwald terrace**. The latter is characterised by a layer of 3-5 m gravel with a loam cover that is usually thicker (up to 12 m) than the one of the Helfbrunner terrace (Flügel & Neubauer 1984). Streams situated in this terrace are less likely to penetrate the loam layer to reach more permeable substrates.

3.1.3 Climate

The whole Bioregion 14 is part of the **Illyric climate** province (Fink 2000) characterised by features of both submediterranean and Pannonian climates (Kilian et al. 1994). The influence of the Styrian Randgebirge is pivotal for this climate as it prevents westerly winds from bringing precipitation to the region. In contrast, low-pressure systems originating from the Adriatic Sea are often responsible for high amounts of precipitation.

Illyric climate is characterised by relatively high temperatures in summer and moderately cold winters (Kilian et al. 1994). Temperatures are similar to Pannonian climate regions of Austria, but precipitation is markedly higher (700 – 1000 mm per year). Most precipitation occurs in summer, but a second maximum is observed in autumn (submediterranean influence). In the East Styrian Grabenland minimum precipitation was reported from January and February (Wolking 1974).

3.2 Sampling sites

To support the description of sampling sites **drilling cores** taken in the surroundings of the sampling sites were interpreted with the assistance of Mag. Peter Rauch (Federal Province of Styria, Referat Gewässeraufsicht und Gewässerschutz). Related data were kindly provided by the Universalmuseum Joanneum, Geologie & Paläontologie. **Orthophotos** were generated online with the *Digitaler Atlas Steiermark: Gewässer & Wasserinformation*. The map in Fig. 4 was created in ArcGIS 10.4.1.

3.2.1 Selection of sampling sites and hydrological stream type

Eight sampling sites in five least-impacted streams in the Grazer Basin (n=2) and the East Styrian Hügelland (n=6) with different hydrological characteristics were selected in Bioregion 14 (Fig. 4, Table 2). **Selection criteria** encompassed natural morphology along the complete upstream course, no anthropogenic damming (data source for both: *Digitaler Atlas Steiermark: Gewässer & Wasserinformation*), predominantly forested catchment upstream of sites and not situated in the crenal or hypocrenal zone (catchment areas of sites range from 3.2 to 8.5 km²; Table 2). These criteria substantially reduced the number of potential sites as larger forested areas are rare due to intensive agricultural use in the Bioregion.

As gauging stations are not available in the selected stream sites and even if they were, they normally cannot distinguish between detailed site-specific hydrological states (e.g. presence of remnant pools; Costigan et al. 2017), alternative methods had to be applied to determine the hydrological regime type. **Temperature loggers** that were fixed to the streambed and measured at an interval of one hour revealed to be insufficient to identify hydrological states (dry, remnant pools, flowing). Instead, these states were noted and documented photographically during visits. Visits were purposefully carried out during longer dry periods in late spring and summer from 2017 to 2019 in order to estimate the long-term maximum drying state, which is crucial for community composition (Meyer et al. 2000). Several sections of each potential stream were surveyed in summer 2017 to find a representative site for sampling concerning both hydrology and morphology.

Table 2: Sampling sites and characteristics relevant for the hydrological regime; altit.: altitude (m a.s.l.), CAS: Catchment area size (km², from *Hydrographischer Dienst in Österreich* 2011, complemented by own measurements in *Digitaler Atlas Steiermark*), SO: stream order according to Strahler (from Wimmer & Moog 1994); *Cond. March*: conductivity of stream water on the sampling date in late March

Hydrological regime type	Sampling site	Site code	Coordinates (WGS84)	Altit.	CAS	SO	Cond. March (µS/cm)
Perennial	Gatterbach downstr.	<i>perGat</i>	46°47'2"N, 15°35'39"E	272	7.0	2	127
	Gepringbach	<i>perGep</i>	46°58'2"N, 15°23'8"E	336	7.2	2	246
intermittent+	Reichelgraben downstr.	<i>int+Rei</i>	46°46'15"N, 15°43'49"E	268	4.1	2	175
	Glauningbach upstr.	<i>int+Gla</i>	46°45'47"N, 15°47'12"E	256	6.2	2	248
	Gatterbach upstr.	<i>int+Gat</i>	46°48'3"N, 15°36'37"E	291	3.2	1	87
intermittent-	Reichelgraben upstr.	<i>int-Rei</i>	46°46'25"N, 15°43'49"E	272	3.6	2	178
	Pfaffenbach	<i>int-Pfa</i>	46°45'34"N, 15°37'36"E	271	8.5	2	83
	Glauningbach downstr.	<i>int-Gla</i>	46°45'24"N, 15°47'30"E	249	7.5	2	233

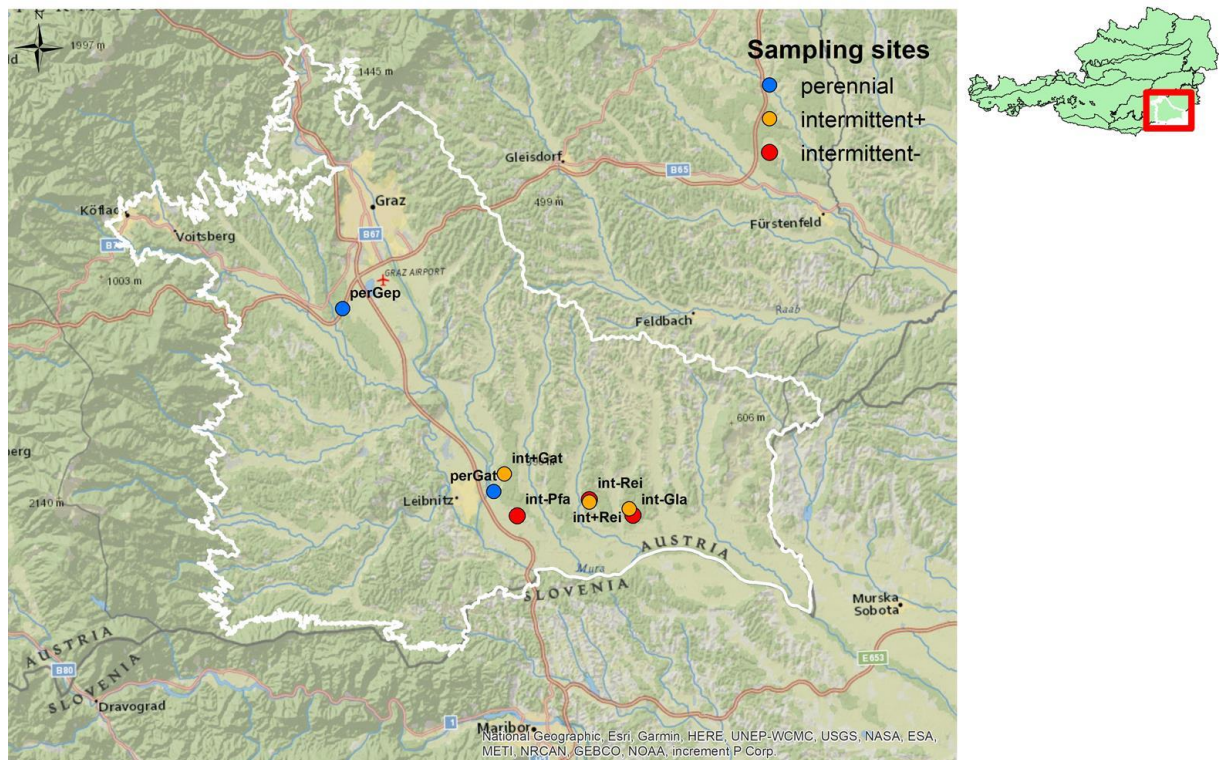


Fig. 4: Sampling sites characterised by hydrological regime type within Bioregion 14 (white border)

Selected sites were sampled four times from late **March to early July** (Table 3) in an approximate one-month interval to analyse structural and functional changes of MI communities. Late March was chosen as the initial date because this season is best for distinguishing between intermittent and perennial streams based on MIs in Bioregion 14 (Sumper 2019).

Table 3: Sampling dates; name of sites according to Table 2; *only EPT taxa were evaluated

Site	March	April*	June	July
<i>perGat</i>	26/3	26/4	3/6	12/7
<i>perGep</i>	23/3	26/4	30/5	8/7
<i>int+Rei</i>	26/3	26/4	2/6	8/7
<i>int+Gla</i>	26/3	26/4	2/6	8/7
<i>int+Gat</i>	23/3	26/4	2/6	8/7
<i>int-Rei</i>	26/3	26/4	2/6	8/7
<i>int-Pfa</i>	26/3	26/4	2/6	12/7
<i>int-Gla</i>	29/3	26/4	2/6	8/7

3.2.2 Perennial streams

Gepringbach

The sampling site in Gepringbach is located west of the village Unterpremstätten. It is situated on a higher **quaternary terrace** and therefore has eroded into a loam cover, which has a depth of 3 m according to a drilling core taken at a distance of about 150 m from the sampling site. A thin layer of **groundwater** (15 cm) at a depth of 45 cm from the surface suggests potential influence on the stream water. In the stream's catchment tertiary fine sediments, but also different forms of Palaeozoic limestone minerals can be found.

The **substrate** is dominated by pelal in deep sections (approximately 50% coverage in late March; Fig. 6, right) and consists of mesolithal, microlithal and akal in more shallow sections (Fig. 6, left). High availability of large woody debris is apparent. In the deep sections flow is substantially reduced and submerged macrophytes cover some of these sections (Fig. 6, right). The **bank vegetation**

is dominated by herbaceous vegetation, but it is also visibly rooted by trees. Flow cessation was never observed during 2018 and 2019.

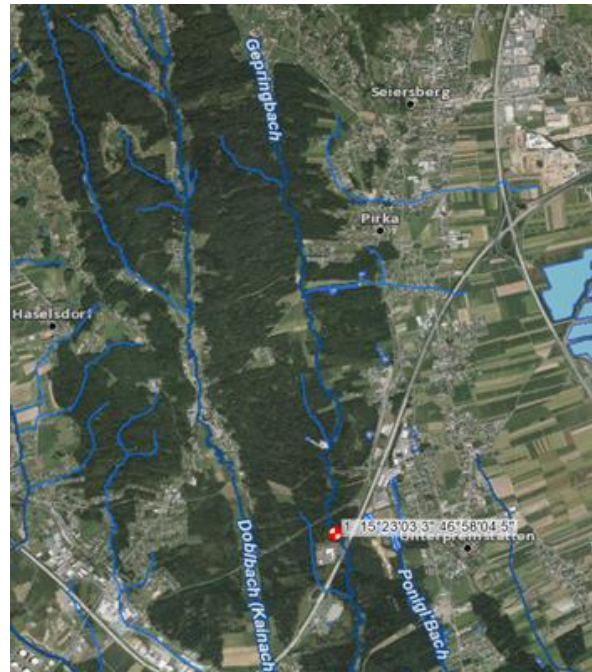


Fig. 5: Orthophoto of Gepringbach, the circle indicates the sampling site



Fig. 6: Sampling site in Gepringbach: left: 9th April 2018; right: 12th April 2018

Gatterbach downstream

The perennial downstream site of Gatterbach is located at a distance of 500 m from the Mur river in the village Gabersdorf. It is situated in colluvial sediments and close to the valley floor of the Mur. The section of the stream is surrounded by the Helfbrunner terrace and further upstream by the Schweinsbachwald terrace. A drilling core taken directly at the sampling site in 1978 revealed a thin layer of loam (40 cm) followed by a thicker layer of pebbles, sand and silt (2.1 m), which is likely to cover the surface of the streambed. As the streambed is deeply incised (Fig. 8) and the groundwater table resulted to be at 1.85 m in June of 1978 the surface water might be influenced by groundwater.



Fig. 7: Orthophoto of Gatterbach, the circle indicates the sampling site



Fig. 8: Sampling site at Gatterbach (downstream): 2 August 2018

3.2.3 Intermittent streams

Gatterbach upstream

The sampling site is located southwest of the village Neudorf an der Mur. The streambed is incised into colluvial sediments. The catchment upstream of the site corresponds to the Schweinsbachwald terrace. However, no nearby drilling cores are available for more detailed information.

The substrate was dominated by fine sediments (15% microlithal, 35% akal, 15% psammal and 5% argyllal/pelal in late March), but mesolithal was prominent in deep pools (30% coverage). The river course is pendulous to meandering and deep pools covered by a mixture of loam, mud and large stones are numerous. Large woody debris was abundant at the site. The banks are overgrown with herbaceous vegetation (e.g. ferns; Fig. 10, right) and trees, the roots of which shape the banks in certain sections.

Ceased flow was observed in the months of June to October 2019, but the stream section was never completely dry due to remnant pools with water depths of up to 1 m. Additionally, during some visits in summer, barely visible thin threads of water at the margins of some, but not all, riffle sections were observed (Fig. 11). The permanent presence of water in pools at ceased flow could not be attributed to a nearby permanent tributary and existed over long sections (several hundred metres) upstream and downstream of the sampling site.

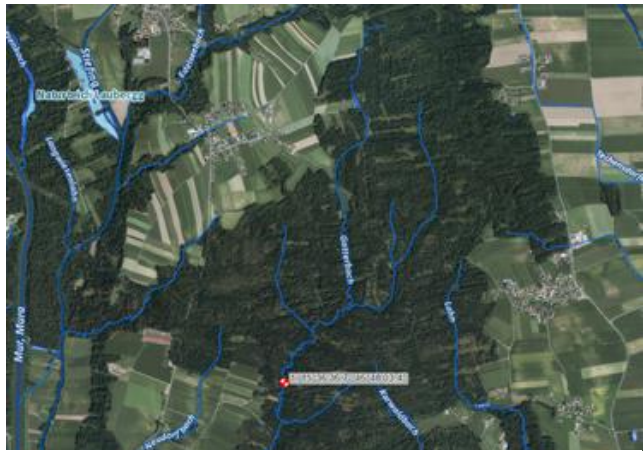


Fig. 9: Orthophoto of Gatterbach, the circle indicates the upstream sampling site



Fig. 10: Sampling site Gatterbach upstream: left: 9 September 2019; right: 14 June 2019



Fig. 11: Hardly visible rivulet at the margin of a riffle section at the sampling site Gatterbach upstream (8 July 2019)

Glauningbach upstream and downstream

Both sampling sites in Glauningbach are situated northwest of the village Ratschendorf in a relatively large forested area with some embedded fields for agricultural use and hay meadows. They are located in colluvial sediments with their catchment being part of higher terraces (Schweinsbachwald terrace and other higher terraces. Further upstream sections and all tributaries are situated directly in the terraces and were indeed observed to have a loamy substrate which might have made them less prone to complete drying. The stream course of the upstream site (int+) is pendulous, but meandering and deeply incised (Fig. 13) in the downstream site (int-). Heavily rooted banks dominate at the upstream site (Fig. 15) and naturally eroding banks also occur at the downstream site (Fig. 13, left). The substrate of the int+ site is dominated by coarse sediments (macro- and mesolithal with highest coverage), whereas the downstream substrate has exclusively fine sediments (akal and argyllal with highest shares: 45 and 25%, respectively; 5% psammal, 10% pelal and 15% microlithal in late March).

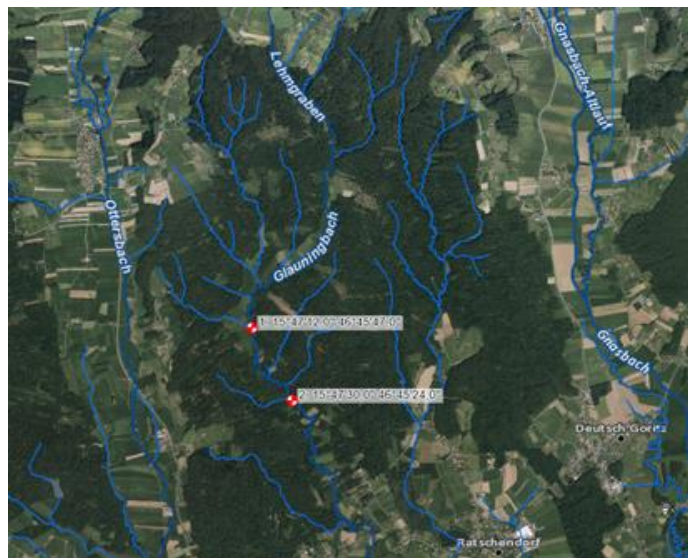


Fig. 12: Orthophoto of Glauningbach, the circles indicate sampling sites (upstream: int+, downstream: int-)



Fig. 13: Sampling site Glauningbach downstream (int-): left: 17th May 2019; right: 23th June 2019

The hydrological regime type of the int+ site was determined based on an observation on 30th August 2017: From numerous sections examined up- and downstream it was the only one to have remnant pools (Fig. 15). The assumed reason for this is a small seemingly perennial tributary which was flowing and might have caused Glauningbach to be wetted at deeper places along some 100 m downstream of the confluence. In 2019 ceased flow could only be observed in July, whereas at the downstream site in June, July and August.



Fig. 14: Sampling site Glauningbach upstream (int+): left: 26th April 2019; right: 30th July 2018



Fig. 15: Sampling site Glauningbach upstream on 30 August 2017

Reichelgraben upstream and downstream

Both sampling sites are situated north of the village Priebing. The distance between the sites is only approximately 300 m, but the hydrological characteristics were observed to be different in 2017: During a dry period at the end of August only one section of about 80 m was found to have remnant pools, whereas upstream and downstream sections were completely dry. This exception could be due to interflow from the hillside (transition from the valley floor to the higher terrace) which is directly at the margin of the streambed on the orographic right side (Fig. 17).

Both sites are situated in colluvial sediments. The catchment area is composed of the Schweinsbachwald terrace and other higher terraces. The substrate of both sites was dominated by mesolithal and microlithal: The latter dominated at the upstream site (60% coverage), mesolithal dominated in the downstream site (50% coverage). The fine fraction also had considerable shares, mainly as psammo-pelal (30% coverage downstream, 25% upstream).

The downstream section (int+) is deeply incised (approximately 2 m; Fig. 18). Bank substrate is dominated by rooted soil of the adjacent deciduous forest on the orographic right side and by loose soil on the left side due to lacking trees (Fig. 18). Riprap bank stabilisation is limited to one point bar (Fig. 17). At the int+ site ceased flow was observed once in late July 2019 and in late August 2017 (Fig. 18).

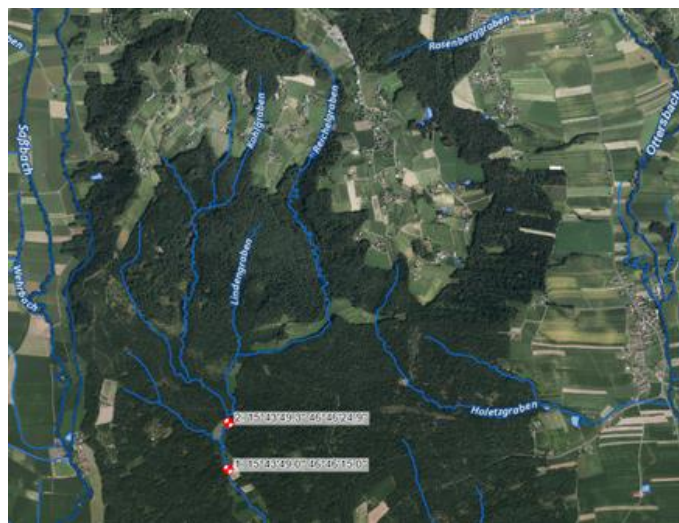


Fig. 16: Orthophoto of Reichelgraben, the circles indicate sampling sites (upstream: int-, downstream: int+)



Fig. 17: Sampling site Reichelgraben downstream (int+): left: 26th March 2019; right: 23th July 2019



Fig. 18: Sampling site Reichelgraben downstream (int+): 26th August 2017

In sharp contrast to the downstream site incision is low upstream (Fig. 19, Fig. 20). The channel width is also only about half of the other site. Bank substrate is dominated by herbaceous vegetation and occasional trees. Ceased flow was documented from all months from June through to October 2019 and from November in 2018.



Fig. 19: Sampling site Reichelgraben upstream (int-): left: 26th March 2019; right: 18th July 2019



Fig. 20: Sampling site Reichelgraben upstream (int-): 14th June 2019

Pfaffenbach

This sampling site is situated at a distance of approximately 4 km from the Mur near the village St. Veit am Vogau. The whole stream is located in the Helfbrunner terrace and accordingly, the **substrate** is dominated by loam (argyllal, approximately 30% coverage in late March) and other fine sediment fractions (psammal, akal). However, it seems that the loam layer is especially thin, namely 0.6 m according to a drilling core taken at a distance of about 100 m from the sampling site. Therefore, the stream might have partly eroded through the loam cover resulting in easier infiltration of surface water. Further, the stream is characterised by submerged macrophytes (*Fontinalis* sp.), natural eroding banks and visible roots from bank vegetation (Fig. 22). No-flow conditions were observed from June through to October 2018 and 2019.



Fig. 21: Orthophoto of Pfaffenbach, the circle indicates the sampling site

Water temperatures in winter 2018/19 were low (mean of daily means in December: 1.99, 0.86 in January, 2.13 °C in February) indicating no groundwater influence. This was confirmed by the analysis of the drilling core, which indicates that the groundwater table was at a depth of 10.5 m in 1981.



Fig. 22: Sampling site at Pfaffenbach: left: 12th October 2018; right: 8th December 2018

3.3 Field work and laboratory work

3.3.1 Multi-habitat sampling (MHS)

Benthic samples were taken following the procedure developed by the AQEM Consortium (2002): Major habitats (min. 5 % proportion) are sampled according to their proportional distribution on the streambed surface within a sampling section (MHS). Habitats were categorised according to mineral and organic choriotope types (Table 4, Table 5) and proportional coverages were noted.

Table 4: Mineral choriotope types in streams according to Moog et al. (1999)

<i>choriotope type (mineral)</i>	<i>grain size (cm)</i>	<i>verbal description</i>
hygropetric sites		water layers on solid substrates
megalthal	>40	upper sizes of large cobbles, boulders and blocks, bedrock
macrolithal	20–40	coarse blocks, cobbles, gravel and sand
mesolithal	6.3–20	fist to hand-sized cobbles with a variable percentage of gravel and sand
microlithal	2–6.3	coarse gravel (size of a pigeon egg to child's fist) with variable percentages of medium to fine gravel
akal	0.2–2	fine to medium-sized gravel
psammal	0.0063–0.2	sand
pelal	<0.0063	mud and sludge
argyllal	<0.00063	silt, loam, clay (inorganic)

Table 5: Organic choriotope types in streams according to Moog et al. (1999)

<i>choriotope type (organic)</i>	<i>verbal description</i>
xylal	tree trunks (dead wood), branches, roots
CPOM	deposits of coarse particulate organic matter (e.g. leaves)
FPOM	deposits of fine particulate organic matter
debris	organic and inorganic matter deposited within the splash zone area by wave motion and changing water levels, e.g. mussel shells, snail shells
submerged macrophytes	macrophytes, including moss and Characeae
emergent macrophytes	e.g. <i>Thypha</i> , <i>Carex</i> , <i>Phragmites</i>
living parts of terrestrial plants	fine roots, floating riparian vegetation

Samples were taken with a frame of 0.25 m width and 0.25 m height and a connected net with 500 µm mesh size. An area of 0.25 x 0.25 m directly upstream of the hand net was disturbed by kicking the substrate to a depth of 10-15 cm in order to dislodge the fauna and wash it into the net. Instead of 20 subsamples used for the standardised assessment of streams, 5 subsamples were taken. These

subsamples correspond to the five habitats with highest proportional coverages. When coverage was $\geq 40\%$ a second subsample of the habitat was taken. Subsamples were then transferred to a plastic container and preserved with formalin (4% final concentration).

3.3.2 MHS sample processing and determination

Two weeks after MHS sample collection they were processed in the laboratory. Sieving and sorting of the samples was executed according to the recommendations of the AQEM Consortium (2002). Specimen of different groups were determined at different taxonomic levels according to feasibility and availability of experts. Oligochaeta, Nematomorpha, Hydrachnidia and Turbellaria were not determined at a lower level, Diptera always at family level. The various determination literature consulted is listed in Table 6. Specimen were determined with a **stereomicroscope** (128x maximum magnification).

Table 6: Determination literature

Taxonomic group	Literature
Gastropoda	Glöer (2015)
Hirudinea	Eiseler (2010)
Ephemeroptera	Bauernfeind & Humpesch (2001), Eiseler (2005)
Odonata	Kohl (1998), Gerken & Sternberg (1999)
Plecoptera	Graf & Schmidt-Kloiber (2008), Zwick (2004)
Coleoptera	Eiseler & Hess (2013)
Megaloptera	Schmedtje & Kohmann (1988)
Trichoptera	Waringer & Graf (2011), Pitsch (1993), Hetrick et al. (1998; for first instar larvae of Goeridae)
Diptera	Sundermann & Lohse (2004), Tachet et al. (2010), Eiseler (2010)

Late instar larvae of *Chaetopteryx rugulosa* (Trichoptera) were determined by means of several unpublished characteristics that allow for separation from *C. fusca*, *C. villosa* and *C. major*.

3.3.3 Sampling and determination of adults (EPT species)

For validation of larval determinations, detection of additional species and description of flight periods **insect air nets** and **light traps** were used to catch adult specimen of EPT species in all seasons of 2017 to 2019.

Light traps consisted of a small soap water-filled plastic container (approximately 30 x 30 cm) and an 8 W or 18 W blacklight-blue fluorescent lamp fixed horizontally on it. Lamps were operated from dusk to dawn through a lead-acid battery including a twilight switch. Traps were placed onto dry bars of the streambed. Trapped specimens were preserved in 70% ethanol and determined with a stereomicroscope (128x maximum magnification).

For determination of adult **Trichoptera** Malicky (2004), Kumanski (1985), Kumanski (1987) and Nógrádi & Uherkovich (2002) were used. **Plecoptera** were determined with Graf & Schmidt-Kloiber (2008) and Murányi et al. (2014). **Ephemeroptera** were kindly determined by Ernst Bauernfeind (Naturhistorisches Museum Wien).

Based on adult records of selected EPT species distribution in Bioregion 14 was illustrated in maps that were generated in **ArcGIS 10.4.1**.

3.3.4 Life history of *Ironoquia dubia*

Starting from December 2018 specimens of *Ironoquia dubia* were collected in an approximate 1.5-week interval in Pfaffenbach and preserved in 70% ethanol. **Head capsule width** was measured to separate instars and to describe development rate based on mean widths from each sampling date. For instar identification specimen collected from other streams were included, but not formol-preserved specimen from MHS samples due to potential bias. Measurement of head capsule was done using a measuring eyepiece (Nikon SMZ-U UW10xA/24) on a stereomicroscope (Zeiss Stemi SV8) at its maximum magnification (64x, formally). The metric scale of the eyepiece was calibrated using a micrometer (Leica 10310345).

To find potential **terrestrial habitats** of larvae and pupae, substrate was excavated from the stream bank and adjacent areas with a shovel and sieved to find specimen. The vertical and horizontal distance from finding spots to the water surface was measured to compare it to most other representatives of limnephilid caddisflies that typically stay at places with low flow velocity at the stream edges. Air net and light trap catches were performed regularly from late spring to autumn to document the **flight period**.

For **observations** of larval and pupal stage, specimens were transferred to aquaria kept at room temperature. Aquaria were equipped to reflect the potential in-situ aquatic and terrestrial habitat conditions (Fig. 23). Water was aerated with an air pump. For the terrestrial department natural bank substrate from the sampled site (Pfaffenbach) was used and sprayed regularly to retain moisture.



Fig. 23: Aquarium used to rear larvae of *Ironoquia dubia*

3.4 Data analyses

3.4.1 Non-metric multidimensional scaling

Non-metric Multidimensional Scaling (NMS) is a multivariate ordination method that is applied to non-metric data that are independent from each other (Kruskal 1964) as are taxa lists of (semi-) quantitative MI samples. Based on similarities and dissimilarities of taxa lists the NMS algorithm ordines them in a two or three-dimensional point plot: The more similar taxa lists (=sampling sites), are the closer they are positioned as points. Stress values indicate how reliably distances between points correspond to (the extent of) their dissimilarities. Values <10 allow for interpretation of distances between points, values between 10 and 20 could lead to misinterpretation and plots with mean values >20 should not be used (see Leyer & Wesche 2008).

NMS analyses in this thesis were executed with the PC-ORD software (version 5.33, McCune & Mefford 2006). Abundances of taxa were log(x+1)- or presence/absence-transformed. To measure distances the Sørensen (Bray & Curtis) coefficient and 500 calculation runs were set.

The data basis for the NMS analysis were counts at genus or higher level. However, if two or more taxa within one genus were separated at each larval instar (no “sp. juv.” determinations, no “cf.”), these taxa were maintained; e.g. *Rhyacophila* s. str. and *R. hirticornis* were maintained, but *Hydropsyche saxonica*, *H. bulbifera* and *H. sp. juv.* were transferred to *Hydropsyche* spp.).

If determinations included different taxonomic levels, all being allocated in the highest systematic level, all were merged; e.g. Limnephilidae Gen. sp. juv., Limnephilinae Gen. sp. juv. and Limmnephilini Gen. sp. juv. *Nemoura/Nemurella* sp. juv. determined at one site (Gatterbach upstream) was excluded as both genera were determined in a more mature larval stage.

Table 7: Comparison of taxa from taxa lists (left column) and taxa used for NMS analyses (right column)

<i>Taxon</i>	<i>Taxon used for NMS</i>	<i>Taxon</i>	<i>Taxon used for NMS</i>
<i>Habrophlebia fusca</i>	<i>Habrophlebia</i> spp.	<i>Rhyacophila</i> s. str. sp. juv.	<i>Rhyacophila</i> s. str. spp.
<i>Habrophlebia lauta</i>		<i>Rhyacophila fasciata</i>	
<i>Habrophlebia</i> sp. juv.		<i>Rhyacophila hirticornis</i>	<i>Rhyacophila hirticornis</i>
<i>Orectochilus villosus</i>	Gyrinidae Gen. spp.	<i>Rhyacophila</i> s. str. sp. juv.	<i>Rhyacophila</i> s. str. spp.
Gyrinidae Gen. sp.		<i>Nemoura</i> sp.	<i>Nemoura</i> spp.
Corduliidae/Libellulidae Gen. sp. juv.	Corduliidae/Libellulidae	<i>Nemurella pictetii</i>	<i>Nemurella pictetii</i>
<i>Somatochlora</i> sp.	Gen. spp.	<i>Nemoura/Nemurella</i> sp. juv.	excluded
Limnephilidae Gen. sp. juv.	Limnephilidae Gen. spp.		
Limnephilinae Gen. sp. juv.			
Limmnephilini Gen. sp. juv.			

3.4.2 Indicator species analysis

To identify taxa statistically significantly associated with either perennial or intermittent streams (ISs) and all tested hydrological types separately, *Indicator species analysis* (ISA; Dufrêne & Legendre 1997) was performed with the PC-ORD software (version 5.33, McCune & Mefford 2006). ISA was applied to all samples and to samples of single dates. The indicator value (IV) calculated of a specific taxon ranges

from 0 to 100 depending on both its frequency and abundance within the tested groups (hydrological stream types here).

Taxa with statistically significant ($p < 0.05$) IV were designated as indicator taxa. In addition, when analysing all hydrological stream types separately, taxa with non-significant IV that occurred in all perennial streams ($n=2$) and only one non-perennial ($n=6$) are also considered indicative for perennial streams. For intermittent types, taxa that occurred in two of three streams within the related type and in no other one ($IV=66,7$) were included. When grouping both intermittent types non-significant taxa that occurred in three to six ISs ($IV = 50, 66.7, 83.3$ and 100 , respectively) were considered indicative for ISs and for perennial streams if they occurred in both sites and only one intermittent.

3.4.3 Cluster Analysis

To complement NMS analysis, cluster analyses were performed in the PC-ORD software using the same data (taxa and log- and presence/absence-transformed abundances of MI samples). Similar to NMS, Cluster analyses aim at finding groups of similar items within a dataset. The output of the analysis types used in this thesis shows a hierarchical structure of items (MI samples here) as a dendrogram, which reflects distinct groups/clusters: Very similar items are combined to small clusters, which are part of larger clusters with more dissimilar items. The used clustering method is agglomerative; i.e. starting with one item the other items are progressively added according to the degree of dissimilarity. Two-way cluster analyses were also performed with samples of only one date ($n=8$). This method of cluster analysis additionally shows relationships between samples and taxa by calculating and graphing clusters of taxa additionally (McCune & Mefford 2006).

The Sørensen (Bray & Curtis) coefficient was chosen to measure distances between samples. Flexible Beta ($\beta=-0.25$) was used as group linkage method.

3.4.4 Structural and functional community metrics/indices

Standard parametric and nonparametric **statistical tests** (one-way ANOVA, t-test; Kruskal-Wallis test, Mann-Whitney U test) were performed in SPSS (version 24) to detect statistically significant differences between hydrological stream types. Differences were tested between all three hydrological types (perennial, int+, int-) and between perennial and grouped intermittent types. These tests were applied to both metric/index values ($N=8$ for single sampling dates, $N=24$ for all three dates) and to classifications of taxa used for the metric/index calculation (N corresponds to the number of classified taxa). If eight values were compared, nonparametric Kruskal-Wallis (three hydrological stream types) or Mann-Whitney U test (two types) was used irrespective of their distribution. If 24 or more values were compared, normal distribution of values was tested to determine the use of the nonparametric or parametric tests.

The mean or median of values is shown in error bar charts depending on the group size of values. For groups of up to nine values the median was chosen to reduce the impact of outliers. If the median was used, the distribution of values was illustrated by showing their range as error bars. If the mean was used, the distribution was illustrated by the 95% confidence interval.

Determination of taxa richness

Pupae and larvae of the same taxon are counted as one, which also applies for adults and larvae. Apart from this, basically each taxon in the taxa list is counted, but only to a maximum that is defined by expert knowledge (Table 3), i.e. the maximum number of species potentially present in the related stream type of the region. For example, concerning the Trichoptera genus *Chaetopteryx* three species (*C. rugulosa*, *C. major* and *C. fusca*) are known to occur in the sampled stream types of the region. Therefore, when determined within the same sample, *C. sp. juv.*, *C. sp.*, *C. rugulosa*, *C. major* and *C. fusca* are counted to a maximum of three, irrespective of which of them were determined.

Table 8: Maximum counts for determination of taxa richness for relevant taxa

Taxon	Max. counts	present in taxa list	Taxon	Max. counts	present in taxa list
<i>Ecdyonurus</i> spp.	1	<i>E. sp.</i>	<i>Beraea</i> spp.	2	<i>B. sp. juv.</i> <i>B. pullata</i>
<i>Rhithrogena</i> spp.	1	<i>R. sp.</i>	<i>Hydropsyche</i> spp.	4	<i>H. sp. juv.</i> <i>H. saxonica</i> <i>H. saxonica</i> Pu. <i>H. angustipennis</i> <i>H. bulbifera</i>
<i>Baetis</i> spp.	2	<i>B. sp. juv.</i> <i>B. sp.</i> <i>B. rhodani</i>	<i>Rhyacophila</i> spp.	3	<i>R. hirticornis</i> <i>R. fasciata</i> <i>R. s. str. juv.</i>
<i>Habrophlebia</i> spp.	2	<i>H. sp. juv.</i> <i>H. lauta</i> <i>H. fusca</i>	Nemouridae Gen. spp.	3	<i>Nemoura</i> sp. <i>Nemoura/Nemurella</i> sp. juv. <i>Nemurella pictetii</i>
<i>Paraleptophlebia</i> spp.	1	<i>P. sp.</i>	Corduliidae/Libellulidae Gen. spp.	2	Corduliidae/Libellulidae Gen. sp. juv. <i>Somatochlora</i> sp.
<i>Chaetopteryx</i> spp.	3	<i>C. sp.</i> <i>C. sp. juv.</i> <i>C. fusca</i> <i>C. major</i> <i>C. rugulosa</i>			
<i>Athripsodes</i> spp.	2	<i>A. sp. Pu.</i> <i>A. bilineatus</i>			

Structural and functional metrics

Standard structural and functional community metrics calculated in the ECOPROF software (e.g. proportion of different higher taxonomic units, longitudinal zonation preferences; Moog et al. 2018, IHG 2019) were used to find significant correlations with hydrological stream types. Therefore, **Spearman correlation** analysis was performed in SPSS.

Flight period index

The flight period index (FPI) was developed to test if MI species have earlier flight periods (FPs) in intermittent than in perennial streams, which would enable sensitive life cycle stages to avoid harsh conditions associated with intermittent flow.

By analogy with the calculation of species specific sabrobic indices used for calculation of the community-level saprobic index according to Zelinka & Marvan (1961), a **species-specific flight-period index** (FPI_{species}) was calculated for each species based on ten-point assignments (a sum of ten points is allocated to the four seasons according to a species' flight period) available in the freshwater ecology database (Schmidt-Kloiber & Hering 2015; Table 9). The species-specific FPI is calculated by assigning an ordinal scale to the seasons (1 for winter, 2 for spring, 3 for summer and 4 for autumn):

$$FPI_{species} = \frac{1 \cdot p_{win} + 2 \cdot p_{spr} + 3 \cdot p_{sum} + 4 \cdot p_{aut}}{10}$$

$FPI_{species}$: species-specific flight period index
 $p_{win}, p_{spr}, p_{sum}, p_{aut}$: points assigned to winter, spring, summer and autumn (sum=10) available in the freshwater ecology database

The community-level FPI is an average of the species-specific FPIs weighted by abundances of species:

$$FPI_{community} = \frac{\sum_{i=1}^n FPI_{species,i} \cdot A_i}{\sum_{i=1}^n A_i}$$

$FPI_{community}$: community flight period index
 A_i : abundance of taxon i

Flight period types

Flight period (FP) types were analysed to test for higher proportions of species with early and/or extended FPs as hypothesised for intermittent compared to perennial streams.

Two classification systems of FPs were used. **FP types A-D** were developed by Sommerhäuser et al. (1997) based on Crichton (1971) and only involve Trichoptera species. Type A encompasses species with short FPs in spring (e.g. *Oligostomis reticulata*), type B species with long or short FPs but always mainly in summer (e.g. *Goera pilosa*), type C with long FP from spring to autumn involving an ovarial diapause (e.g. *Stenophylax permistus*) and type D with short FP in autumn (e.g. *Anabolia furcata*) (Table 9). **FP types 1-4** were developed in accordance with the types A-D, but applicable to all aquatic species based on indications in the freshwater ecology database (Schmidt-Kloiber & Hering 2015, parameters “emergence/flight period” and “duration emergence period”). Type 1 (*short in spring*) was assigned to species if highest scores are given to spring (ten-point assignment system in the parameter “emergence/flight period”) and the parameter “duration emergence period” has the state “short” (e.g. *Siphonurus armatus*). Type 2 (*short in spring and summer*) was assigned according to type 1 but at highest scores for summer (e.g. *Electrogena ujhelyii*). Type 3 (*long*) encompass species with long duration of FP (e.g. *Ephemerella ignita*, *Stenophylax permistus*). Type 4 (*short in autumn*) was assigned according to types 1 and 2 at highest scores for autumn (Table 9).

Taxa lists of single sampling dates and those with taxa of all dates were characterised with FP types based on presence/absence of species at sites. Statistical significance of differences between hydrological stream types was tested using both FP types of all species and proportions of the FP types.

Table 9: Flight period characteristics of EPT species used for calculations at community level; **FPI_{sp.}**: species-specific flight period index, calculation based on classifications in column “flight period”; **flight period**: timing/seasonality of flight periods classified with the ten-point-assignment system (win: winter, spr: spring, sum: summer, aut: autumn; freshwater ecology database: Ephemeroptera: Buffagni et al. 2019, Plecoptera: Graf et al. 2019b, Trichoptera: Graf et al. 2019a); **FP duration**: duration of flight period classified with single-category assignment (s: “short”, l: “long”; freshwater ecology database); **FP type (1-4)**: flight period type based on classifications in columns “flight period” and “FP duration”; **FP type (A-D)**: flight period type of caddisfly species according to Sommerhäuser et al. (1997); * classifications regarded to be erroneous in the database and corrected according to own observations, ** missing in the freshwater ecology database and classified according to own observations

species	FPI _{sp.}	flight period				FP duration	FP type (1-4)	FP type (A-D)
		win	spr	sum	aut			
<i>Amphinemura standfussi</i>	3.0		1	8	1	l	3	
<i>Anabolia furcata</i>	3.7			3	7	s	4	D
<i>Athripsodes bilineatus</i>	3.1		1	7	2	l	3	B
<i>Baetis rhodani</i>	2.5	1	4	4	1	l	3	
<i>Beraea pullata</i>	2.6		4	6		s	2	
<i>Beraeodes minutus</i>	2.8		8	2		s	1	A
<i>Brachyptera risi</i>	2.1	1	7	2		s	1	
<i>Centroptilum luteolum</i>	2.9		3	5	2	l	3	
<i>Chaetopteryx fusca</i> **	3.6	1		1	8	s	4	D
<i>Chaetopteryx major</i> **	3.9			1	9	s	4	D
<i>Chaetopteryx rugulosa</i>	3.4	2			8	s	4	D
<i>Cloeon dipterum</i>	2.7	1	3	4	2	l	3	
<i>Cynus trimaculatus</i>	2.9		3	5	2	l	3	B
<i>Electrogena ujhelyii</i>	2.6		4	6		s	2	
<i>Ephemerella ignita</i>	2.8		3	6	1	l	3	
<i>Glyptotaelius pellucidus</i>	2.6		4	6		l	3	C
<i>Goera pilosa</i>	2.8		3	6	1	s	2	B
<i>Habrophlebia fusca</i>	2.8		3	6	1	l	3	
<i>Habrophlebia lauta</i>	2.8		3	6	1	l	3	
<i>Halesus digitatus/tesselatus</i> **	3.8			2	8	s	4	D
<i>Hydatophylax infumatus</i>	-	-	-	-	-	-	-	-
<i>Hydropsyche angustipennis</i>	3.0		3	4	3	l	3	B
<i>Hydropsyche bulbifera</i>	3.0		2	6	2	l	3	B
<i>Hydropsyche saxonica</i>	2.2		8	2		s	2	B
<i>Isonychia dubia</i> *	3.7			3	7	s	4	D
<i>Isoperla tripartita</i>	2.3		7	3		s	1	
<i>Leptophlebia marginata</i>	2.2		8	2		s	1	
<i>Limnephilus auricula</i>	3.0		4	2	4	l	3	C
<i>Limnephilus lunatus</i>	3.0		4	2	4	l	3	C
<i>Limnephilus rhombicus</i>	3.0		4	2	4	l	3	C
<i>Lithax obscurus</i>	2.3		7	3		s	1	A
<i>Lype reducta</i>	2.8	1	3	3	3	l	3	B
<i>Mystacides nigra</i> *	2.8		3	6	1	l	3	B
<i>Nemurella pictetii</i>	2.8		3	6	1	l	3	
<i>Notidobia ciliaris</i>	2.1		9	1		s	1	A
<i>Oligostomis reticulata</i>	2.0		10			s	1	A
<i>Paraleptophlebia submarginata</i>	2.4		6	4		s	1	
<i>Plectrocnemia conspersa</i>	3.1		2	5	3	l	3	B
<i>Potamophylax rotundipennis</i>	2.8		3	6	1	s	2	B
<i>Rhyacophila fasciata</i>	2.7	1	3	4	2	l	3	B
<i>Rhyacophila hirticornis</i>	2.5		5	5		s	2	B
<i>Silo pallipes</i>	2.8		2	8		s	2	B
<i>Siphonurus armatus</i>	2.4		6	4		s	1	
<i>Stenophylax permistus</i>	3.0		3	4	3	l	3	C
<i>Synagapetus moselyi</i>	2.6		4	6		s	2	B
<i>Tinodes pallidulus</i>	3.0		3	4	3	l	3	B
<i>Zwicknia acuta</i>	1.8	2	8			s	1	

Biodrought index

In the past, **structural** (taxonomic) **and functional** (biological and ecological) **traits** of MI communities have been tested to discriminate between hydrological stream types including intermittent ones. Most of these attempts were based on samples from arid and semi-arid climate regions (e.g. Cid et al. 2016, Bonada et al. 2007, Bogan et al. 2015). Functional traits and their modalities used in European studies were often taken from Tachet et al. (2010 and preceding editions), e.g. resistance form, life-cycle features, feeding behaviour. Structural traits encompassed for example taxa richness, total abundance and different EPT metrics.

The occurrence of **indicator taxa** is another tool used to recognise hydrological stream types (e.g. Nadeau 2015). For a European temperate climate region (Czech Republic) Straka et al. (2019) developed a macroinvertebrate-community based multimetric index, the **Biodrought index** (BI) which integrates structural and functional metrics as well as the indicator taxa approach. In most cases, genus or species level is required for classifications of taxa. The index aims at detecting antecedent stream drying by classifying three “intermittence categories”: perennial, near-perennial (less than seven day drying event in the previous summer) and intermittent (more than seven days).

A BI exists for samples from spring (March to April) and from autumn (September to October). For each season three metrics are combined: For both the proportion of perennial **flow indicators** and proportion of individuals with high **body flexibility** are used, but $\ln(x+1)$ -transformed **total abundance** in the spring BI is substituted by *proportion of taxa with preference for organic substrates* in the autumn BI. The proportion of perennial flow indicators is based on presence/absence of both perennial and intermittent flow indicators including their **reliability** as flow indicators (2 for high, 1 for low reliability). Designation of indicator taxa and related reliability values was based on *Indicator species analysis* and on the autecological literature available. Finally, the index can be translated into **probabilities** of belonging to each *flow intermittence category* (sum=100%). For this purpose, probability density functions that reflect the distribution of index values within the intermittence categories of sites used for the index development are used.

BI calculation (including probabilities) of this thesis' samples was kindly executed by Marek Polášek, one of the authors of Straka et al. (2019).

DEHLI index

Another index that has been developed to detect MI community responses to supra-seasonal droughts of episodically dry streams (which only dry in some years) is tested in this thesis: the *Drought Effect of Habitat Loss on Invertebrates* index (**DEHLI**, Chadd et al. 2017). The index is based on the concept of continuous loss of benthic invertebrates due to the changes in habitat conditions and availability during streambed drying (Boulton 2003; see chapter 2.4.2). Taxa at family or genus level are assigned a Drought Intolerance Score (DIS, range: 1-10) which reflects the maximum stage of drying which related species are able to withstand. The drying stages described in Boulton (2003) are decreased flow/volume (DIS of 9 and 10), loss of lateral connectivity (7, 8), flow cessation (5, 6), early stages of pool shrinking (4-6), final stages of pool shrinking (1-3) and no surface water (1). The DEHLI index is calculated as the average DIS of scoring families/genera present in the sample (no abundances considered). The index is applicable to flowing-phase samples of each season and is likely to decrease with increasing drought duration in episodically dry streams. Its applicability to intermittent streams of Bioregion 14 is tested in this thesis.

3.4.5 Biological adaption traits

The basic source for defining biological adaption traits that promote persistence in intermittent streams (see chapter 2.4.3) was Stubbington et al. (2017: Table 4.3.1). Most of these traits were already translated into those defined by Tachet et al. (2010); for some traits, translation was done consulting primary research literature on intermittent streams that used Tachet's traits (Bonada et al. 2007a, b). Table 10 shows all these traits.

Table 10: Biological species traits that promote survival in intermittent streams translated into the biological traits (resilience and resistance traits) and associated modalities defined by Tachet et al. (2010)

Resilience trait	Trait modality	Resistance trait	Trait modality
dispersal	aerial active (flying) aquatic passive (drift)	resistance forms	eggs, gemmula, statoblasts diapause or quiescence ¹ housings against desiccation cocoons
locomotion and substrate relation	endobenthic (interstices) epibenthic (burrower) swimmer flier	respiration	plastron spiracle (aerial)
reproduction	ovoviviparity terrestrial clutches asexual reproduction		
life cycle duration	≤ 1 year		
maximum size	< 5 mm		

Trait modality classifications of taxa expressed by *affinity scores* (0 - 5) at family, genus or species level in Tachet et al. (2010) were used for taxa that occurred in the samples. Scores of resistance traits were summed for each taxon and the same was done with resilience traits. For example, the genus *Amphinemura* has affinity scores in the trait modalities resistant eggs (1) and diapause/dormancy (2) and therefore a summed score of 3 for resistance traits. It has affinity scores in the modalities aerial active dispersal (1), aquatic passive dispersal (2), life cycle-duration ≤ 1 year (3) and maximum size <5 mm (1) and therefore a summed score of 7 for resilience traits.

For each sample, these scores are summed to get the “community score” for resistance and resilience, respectively, based on presence/absence of taxa. Standardisation is achieved by dividing the community score by the number of taxa in the sample. The result is the **mean affinity score per taxon for traits providing resistance and resilience, respectively**. This value is viewed as a measure for the degree of resistance and resilience to intermittent flow in a community, respectively.

¹ The original term for this trait modality in Tachet et al. (2010) is “diapause or dormancy”, which is not compatible with the way these terms are defined in this thesis.

4 Results

4.1 Community structure

4.1.1 Taxa lists

A total of 117 taxa were identified. Only EPT taxa were evaluated from samples in late April; they can be found in Table 12 of this chapter. Taxa were determined at different taxonomic resolutions: Nematoda, Turbellaria, Oligochaeta and Hydrachnidia were never determined to a lower level, Diptera always at family level and EPT taxa at species level if possible. The remaining taxa were determined at different taxonomic levels in accordance with taxonomic expertise available.

Table 11: Taxa list with abundances (individuals per m²) of all samples; abbreviations: N: neozoa, iN: invasive neozoa (information taken from ECOPROF, Moog et al. 2018); P or Pu.: pupae, Ad.: adults

Taxa	perennial						intermittent+									intermittent-								
	Gepringbach			Gatterbach dwnstr.			Reichelgraben dwnstr.			Glauningbach upstr.			Gatterbach upstr.			Reichelgraben upstr.			Pfaffenbach			Glauningbach dwnstr.		
	23.3.	30.5.	08.7.	26.3.	3.6.	12.7.	26.3.	2.6.	8.7.	26.3.	2.6.	8.7.	23.3.	2.6.	8.7.	26.3.	2.6.	8.7.	26.3.	2.6.	12.7.	29.3.	2.6.	8.7.
TURBELLARIA Gen. sp.		19	138	93	13						42	3	118	51	26	694	70	67	6	58	48	16	35	83
NEMATODA Gen. sp.		13				3		51										6					3	3
GASTROPODA																								
HYDROBIIDAE																								
• <i>Bythinella</i> sp.										10	147	6				432	442	198				3		
• <i>Potamopyrgus antipodarum</i> (iN)	1850	4374	7562																					
Acroloxidae																								
• <i>Acroloxus lacustris</i>	38	6	70	74	42	22																		
Lymnaeidae																								
• <i>Radix labiata/balthica</i>		19	3	74	16	13		6		6	6	3	16	35	109	67	48	61		3		32	22	38
Physidae																								
• <i>Physella</i> sp. (N)	38	10	202				3		6							10		6			13			
Planorbidae																								
• <i>Ancylus fluviatilis</i>			3	3						70	182	330												
• <i>Ferrissia</i> sp. (N)													99	3										
• <i>Gyraulus</i> sp.	67	42	109	109						6	3										74			
Planorbidae Gen. sp.																			38	3				
BIVALVIA																								
Sphaeriidae																								

Hydrological regime type:	perennial						intermittent+									intermittent-								
Sampling site:	Gepringbach			Gatterbach dwnstr.			Reichelgraben dwnstr.			Glauningbach upstr.			Gatterbach upstr.			Reichelgraben upstr.			Pfaffenbach			Glauningbach dwnstr.		
Sampling date:	23.3.	30.5.	08.7.	26.3.	3.6.	12.7.	26.3.	2.6.	8.7.	26.3.	2.6.	8.7.	23.3.	2.6.	8.7.	26.3.	2.6.	8.7.	26.3.	2.6.	12.7.	29.3.	2.6.	8.7.
Taxa																								
• <i>Pisidium</i> sp.		35	64	19	83	13	179	157	394	54	838	38	1696	301	282	589	234	339	413	678	522	26	51	26
OLIGOCHAETA Gen. sp.	3290	1709	2368	832	1485	144	947	1677	1222	1197	1421	99	4026	3021	1862	2426	749	310	4112	2461	458	1242	1120	560
HIRUDINEA																								
Glossiphoniidae																								
• <i>Alboglossiphonia/Glossiphonia</i> sp.			6												3					29	38			
• <i>Helobdella stagnalis</i>																					3			
Haemopidae																								
• <i>Haemopsis sanguisuga</i>				3				3		3						16	10	3			10			
Erpobdellidae Gen. sp.				13												10				3				
AMPHIPODA																								
Gammaridae																								
• <i>Gammarus fossarum</i>	986	1494	11866	5325	3885	14851	2624	18474	13443	320	486	2291	9987	3181	3248	701	1549	1558	237	717	634	19	83	282
ISOPODA																								
Asellidae																								
• <i>Asellus aquaticus</i>		3	22						3							6	83	29	10	365	570			
DECAPODA																								
Astacidae																								
• <i>Pacifastacus leniusculus</i> (iN)									3															
HYDRACHNIDIA Gen. sp.		29	35	32		3	16	51	80			3	10	13	6		26			13	3	16		
EPHEMEROPTERA																								
Siphonuridae																								
• <i>Siphonurus armatus</i>							83	42		109	6		13			2442	13		2022	16		1578		
Baetidae																								
• <i>Baetis</i> sp.		29	10	3	739	70			6			16		13								16	19	
• <i>Baetis</i> sp. juv.			10	1040			6					45												
• <i>Baetis rhodani</i>	10		19	192			10												19			3		
• <i>Centroptilum luteolum</i>	3		29							83		3	13											
• <i>Cloeon dipterum</i>	35		102										3	51							637			
Heptageniidae																								
• <i>Ecdyonurus</i> sp.	35			80	6	6																		
• <i>Electrogena ujhelyii</i>		6	51	285	134	115	650	714	470	99		22	256	163	144		35	3				32	13	
• <i>Rhithrogena</i> sp.	3		13	134	10		6			10	3													
Leptophlebiidae																								
• <i>Habrophlebia fusca</i>	10	38	16			3		26	29		35	45	6	32	22		352	19		285	3		157	278
• <i>Habrophlebia lauta</i>				42	26		13	16	3		3		106				86							
• <i>Habrophlebia</i> sp. juv.	176		70	96	26		515	496	195	262	1622	976	624	205	102	3	947	42	6	646			1770	2045

Hydrological regime type: Sampling site: Sampling date:	perennial						intermittent+									intermittent-								
	Gepringbach			Gatterbach dwnstr.			Reichelgraben dwnstr.			Glauningbach upstr.			Gatterbach upstr.			Reichelgraben upstr.			Pfaffenbach			Glauningbach dwnstr.		
	23.3.	30.5.	08.7.	26.3.	3.6.	12.7.	26.3.	2.6.	8.7.	26.3.	2.6.	8.7.	23.3.	2.6.	8.7.	26.3.	2.6.	8.7.	26.3.	2.6.	12.7.	29.3.	2.6.	8.7.
Taxa																								
• <i>Leptophlebia marginata</i>													3											
• <i>Paraleptophlebia</i> sp.										3														
• <i>Paraleptophlebia submarginata</i>				3																				
Ephemeridae																								
• <i>Ephemera danica</i>	19	13	29	3						3														
Ephemerellidae																								
• <i>Ephemerella ignita</i>			109		16												16							
ODONATA																								
Calopterygidae																								
• <i>Calopteryx</i> sp.		3							3								3							
Lestidae																								
• <i>Chalcolestes viridis</i>																				26				
Platycnemididae																								
• <i>Platycnemis pennipes</i>		51	224																					
Coenagrionidae Gen. sp. juv.	74		13												54									
Corduliidae/Libellulidae Gen. sp. juv.	3	10								6						3			10			16	3	3
Aeshnidae																								
• <i>Aeshna</i> sp.	3																			42		3	6	
Gomphidae																								
• <i>Onychogomphus forcipatus</i>							3																	
Corduliidae																								
• <i>Somatochlora</i> sp.														13						6				
PLECOPTERA																								
Perlodidae																								
• <i>Isoperla tripartita</i>				3			32	22		179	6		35	3		730	13					272	67	
Taeniopterygidae																								
• <i>Brachyptera risi</i>																16					3			
Nemouridae																								
• <i>Amphinemura standfussi</i>							10			6						170	29				96	3		
• <i>Nemoura</i> sp.	10		38				3			29		16	70			749	3		691	3		250		
• <i>Nemoura/Nemurella</i> sp. juv.													64											
• <i>Nemurella pictetii</i>													13											
Capniidae																								
• <i>Zwicknia</i> cf. <i>acuta</i>							10									310					32			
Leuctridae																								
• <i>Leuctra</i> sp.			26										3											

Hydrological regime type:	perennial						intermittent+									intermittent-								
Sampling site:	Gepringbach			Gatterbach dwnstr.			Reichelgraben dwnstr.			Glauningbach upstr.			Gatterbach upstr.			Reichelgraben upstr.			Pfaffenbach			Glauningbach dwnstr.		
Sampling date:	23.3.	30.5.	08.7.	26.3.	3.6.	12.7.	26.3.	2.6.	8.7.	26.3.	2.6.	8.7.	23.3.	2.6.	8.7.	26.3.	2.6.	8.7.	26.3.	2.6.	12.7.	29.3.	2.6.	8.7.
Taxa																								
HETEROPTERA																								
Notonectidae																								
• <i>Notonecta</i> sp. Ad.																							19	
Corixidae Gen. sp.																							13	
Corixidae Gen. sp. Ad.																							6	
Hydrometridae																								
• <i>Hydrometra</i> sp. juv.																							10	
Veliidae																								
• <i>Velia</i> sp. Ad.									3															
Gerridae																								
• <i>Gerris</i> sp.			22												35								26	
• <i>Gerris</i> sp. Ad.																							42	
MEGALOPTERA																								
Sialidae																								
• <i>Sialis lutaria</i>			26												3								38	
COLEOPTERA																								
Dytiscidae																								
Dytiscidae Gen. sp.								3									3					90		6
Dytiscidae Gen. sp. Ad.						13										13						272		
• <i>Platambus maculatus</i>	3									3			13											
• <i>Platambus maculatus</i> Ad.			16		6																			
Elmidae																								
• <i>Elmis</i> sp.	3	333	102	128	358	67	22	42		6	13	6	86	29										
• <i>Elmis</i> sp. Ad.	102	346	461	6		22	3	10	10			6	13											
• <i>Limnius</i> sp.		29	22			3																		
• <i>Limnius</i> sp. Ad.	3	80	38			3							3											
• <i>Oulimnius tuberculatus</i>	32	102	26	6	13																			
• <i>Oulimnius tuberculatus</i> Ad.	19	29	160				3					6					6	3				3		3
Gyrinidae																								
Gyrinidae Gen. sp.									10													32		
• <i>Orectochilus villosus</i>				10																				
Haliplidae																								
• <i>Haliplus</i> sp.			3																					
• <i>Haliplus</i> sp. Ad.			3																					
Hydraenidae																								
• <i>Hydraena</i> sp. Ad.	10	154	186	19	26	112	118	166	163	35		131	64	64		26	29	64				19	10	3

Hydrological regime type: Sampling site: Sampling date:	perennial						intermittent+									intermittent-								
	Gepringbach			Gatterbach dwnstr.			Reichelgraben dwnstr.			Glauningbach upstr.			Gatterbach upstr.			Reichelgraben upstr.			Pfaffenbach			Glauningbach dwnstr.		
	23.3.	30.5.	08.7.	26.3.	3.6.	12.7.	26.3.	2.6.	8.7.	26.3.	2.6.	8.7.	23.3.	2.6.	8.7.	26.3.	2.6.	8.7.	26.3.	2.6.	12.7.	29.3.	2.6.	8.7.
Taxa																								
Hydrophilidae Gen. sp.																			3					
Dryopidae																								
• Pomatinus substriatus Ad.							3															13	45	
Scirtidae Gen. sp.				3				3				6	3				3	3						
TRICHOPTERA																								
Rhyacophilidae																								
• Rhyacophila fasciata		3P		6P																				
• Rhyacophila hirticornis													16			3						3		
Glossosomatidae																								
• Synagapetus moselyi							74	122, 54P		512	26, 54P		477	122	3P	109	3P					154	10, 86P	
Hydropsychidae																								
• Hydropsyche angustipennis				6				3																
• Hydropsyche bulbifera				10																				
• Hydropsyche saxonica	6	58	67	163	157, 6P	80	10	13				3												
• Hydropsyche sp. juv.	35	16	166	595	35	496	3					3												
Polycentropodidae																								
• Cyrnus trimaculatus	3		3																					
• Plectrocnemia conspersa			3P					10		6			6			3	6							
Psychomyiidae																								
• Lype reducta								19																
• Tinodes pallidulus				3	3																			
Phryganeidae																								
• Oligostomis reticulata											3			26			42	3	3P	112			22	13
Limnephilidae																								
• Anabolia furcata		106	38							3					6									
• Chaetopteryx cf. fusca		22	19								13			3	26									
• Chaetopteryx cf. major	13	10	26	6		3	6	35	10	35	26		93	189	19	10	38	10						
• Chaetopteryx rugulosa							3	10	6	13	90	35		29	3		19	10					3	
• Chaetopteryx sp.			16																					
• Chaetopteryx sp. juv.	42			6			19			70			291	141	26	13	3							
• Glyptotaelius pellucidus							3									131			3			19		
• Halesus digitatus/tesselatus	3		3	35	3																			
• Hydatophylax infumatus		3 P																						
• Ironoquia dubia										3			3			3			45	3		22		
• Limnephilidae Gen. sp. juv.	74			10			22			1235			202									10		
• Limnephilinae Gen. sp. juv.	61						42			99			310			3								

Hydrological regime type:	perennial						intermittent+									intermittent-								
Sampling site:	Gepringbach			Gatterbach dwnstr.			Reichelgraben dwnstr.			Glauningbach upstr.			Gatterbach upstr.			Reichelgraben upstr.			Pfaffenbach			Glauningbach dwnstr.		
Sampling date:	23.3.	30.5.	08.7.	26.3.	3.6.	12.7.	26.3.	2.6.	8.7.	26.3.	2.6.	8.7.	23.3.	2.6.	8.7.	26.3.	2.6.	8.7.	26.3.	2.6.	12.7.	29.3.	2.6.	8.7.
Taxa																								
• <i>Limnephilini</i> Gen. sp. juv.	3			3									26			3								
• <i>Limnephilus auricula</i>																3								
• <i>Limnephilus lunatus</i>	3							3						6										
• <i>Limnephilus rhombicus</i>	10												3					3						
• <i>Potamophylax rotundipennis</i>	19	3		54	6		6	13		67	26	13	90	3		10						13	3	
• <i>Stenophylax permistus</i>																45	3 P					38		
Goeridae																								
• <i>Goera pilosa</i>	3			3																				
• <i>Lithax obscurus</i>	182	93	6		64	42	13, 16P	3	90	16, 3P	746	58	186	13			6					38	42, 10P	45
• <i>Silo pallipes</i>	19																							
Leptoceridae																								
• <i>Athripsodes bilineatus</i>	48	96	35	10	3						3													
• <i>Athripsodes</i> sp.																						3P		
• <i>Mystacides nigra</i>	6	6																						
Sericostomatidae																								
• <i>Notidobia ciliaris</i>	3												6											
Beraeidae																								
• <i>Beraea pullata</i>													3											
• <i>Beraea</i> sp. juv.			26										3											
• <i>Beraeodes minutus</i>	240		29										394	19, 6P	886									
DIPTERA																								
Chironomidae Gen. sp.	723	1277	598	4387	787	355	2694	2803	992	2931	816	352	1613	621	493	1952	1043	208	326	458	618	1322	432	170
Chironomidae Gen. sp. Pu.	67	54	10	189		3		77		96		3	6	13		64		16	32	51	10	99	3	
Culicidae Gen. sp.																		3	6		19	16		
Simuliidae Gen. sp.				3, 32P		3, 3P	6					3				70		6				42		3
Ceratopogonidae Gen. sp.	13	182	3	54			26	102	29	83	67	19	109	26		128	115	64	13	118		32	278	10
Dixidae Gen. sp.			16			6			3			16						3						
Limoniidae/Pediciidae Gen. sp.	26	32	64	83	51	16	10	29	10	16	80	29	109	22		6			6	3		10	26	6
Psychodidae Gen. sp.		10	3	19		3						3				3	13	3				3		
Tabanidae Gen. sp.			13				3	10	10	6	6		26	13	3	6	3					6		
Tipulidae Gen. sp.																3						3		
Dolichopodidae Gen. sp.		3																						
sum	8426	10950	25418	14310	8000	16477	8218	25264	17194	7699	6771	4592	21312	8362	7434	11981	6029	3046	7997	6118	4205	5450	4320	3654
No. of taxa (total: 117)	45	39	53	45	26	24	37	30	25	37	26	29	43	27	24	39	31	26	21	21	24	31	28	21
No. of taxa (without "sp.")	23	14	8	24	2	27	22	14	13	19	15	11	18	18	9	15	13	8	8	8	8	18	17	11

Table 12: Taxa list with abundances (individuals per m²) of samples from late April; Pu: pupae

Taxa	Hydrological regime type:		perennial			intermittent+			intermittent-		
	Sampling site:		perGep	perGat.	int+Rei	int+Gla	int+Gat	int-Rei	int-Pfa	Int-Gla	
	Sampling date:		26.4.	26.4.	26.4.	26.4.	26.4.	26.4.	26.4.	26.4.	
EPHEMEROPTERA											
Siphonuridae											
• Siphonurus armatus				122	144			461	586	237	
Baetidae											
• Baetis sp.			1296								
• Baetis sp. juv.	26			154							
• Baetis rhodani	6										
• Cloeon dipterum	6					3					
Heptageniidae											
• Ecdyonurus sp.	6	67									
• Electrogena ujhelyii	1	317	339	13	237	3				3	
• Rhithrogena sp.	32	134	77	269	6						
Leptophlebiidae											
• Habrophlebia fusca	54		26	83	1	13					
• Habrophlebia lauta		115	51	6	1						
• Habrophlebia sp. juv.	38	16	282	179	61			123			
• Paraleptophlebia submarginata		3									
Ephemeridae											
• Ephemera danica	26	16									
Ephemerellidae											
• Ephemerella ignita			6								
PLECOPTERA											
Perlodidae											
• Isoperla tripartita			157	41	22	163				787	
Nemouridae											
• Amphinemura standfussi			118	13	3	163				192	
• Nemoura sp.				29	38	96	51			35	
TRICHOPTERA											
Rhyacophilidae											
• Rhyacophila fasciata	3										
• Rhyacophila hirticornis				3 Pu	6, 1 Pu						
• Rhyacophila s. str. sp. juv.	3										
Glossosomatidae											
• Synagapetus moselyi			163, 38 Pu	77, 6 Pu	73, 6 Pu	9				314	
Hydropsychidae											
• Hydropsyche saxonica	64, 1 Pu	12, 1 Pu	16								
• Hydropsyche sp. juv.	182	227	16								
Polycentropodidae											
• Plectrocnemia conspersa				13							
Psychomyiidae											
• Lype reducta		16									
• Tinodes pallidulus		6									
Phryganeidae											
• Oligostomis reticulata				26							
Limnephilidae											
• Anabolia furcata	6		6	3	19			1			
• Chaetopteryx fusca			3								
• Chaetopteryx major	29	16	99	64	173	32					
• Chaetopteryx rugulosa			1	317	77	19					
• Chaetopteryx sp. juv.	54	3	7	186	157	3					
• Glyphotaelius pellucidus					3 Pu					3	
• Halesus digitatus/tesselatus		16									
• Ironoquia dubia				3		35	19			48	
• Limnephilidae Gen. sp. juv.					32						
• Limnephilinae Gen. sp. juv.				25	16						
• Limnephilini Gen. sp. juv.					3						
• Limnephilus lunatus	13		3								
• Limnephilus rhombicus					3						
• Potamophylax rotundipennis	42	38	64	22	51	6				6	
• Stenophylax permistus						3, 16 Pu				13 Pu	

Hydrological regime type:	perennial		intermittent+			intermittent-		
Sampling site:	perGep	perGat.	int+Rei	int+Gla	int+Gat	int-Rei	int-Pfa	int-Gla
Sampling date:	26.4.	26.4.	26.4.	26.4.	26.4.	26.4.	26.4.	26.4.
Taxa								
Goeridae								
• <i>Goera pilosa</i>	3							
• <i>Lithax obscurus</i>	218	3 Pu	1 Pu		32, 1 Pu			13, 6 Pu
• <i>Silo pallipes</i>					6			
Leptoceridae								
• <i>Athripsodes bilineatus</i>	144	38						
• <i>Mystacides nigra</i>	3							
Sericostomatidae								
• <i>Notidobia ciliaris</i>	13, 3 Pu							
Beraeidae								
• <i>Beraeodes minutus</i>	6							
•								
sum	12	2586	183	225	1814	114	1869	1658
No. of taxa (total: 48)	24	18	21	2	23	13	5	11
No. of taxa (without "sp.")	17	12	16	15	16	11	3	1

4.1.1 Temporal dynamics of the community

Temporal dynamics in total abundance

80,400 specimens were collected in 24 samples (each five subsamples of 0.3125 m²). The average density was 10,100 ind./m² (SD 6,400, range 3,000–25,400). The five most abundant taxa were *Gammarus fossarum* (38%), Oligochaeta (15%), Chironomidae (11%), *Potamopyrgus antipodarum* (5%, only in Gepringbach) and *Habrophlebia* sp. juv. (5%).

Total abundance decreases along the hydrological gradient with int- streams having the lowest (Fig. 24, left; 12,600, 8,200 and 5,400 ind./m², respectively). Differences between groups are significant (Kruskal-Wallis test: Chi-square₂ = 10.428, p=0.005, N=24) but only due to differences between perennial and int-sites (pairwise comparison: p=0.007). Total abundances of grouped intermittent sites are also significantly different from perennial sites (Mann-Whitney U test: U=20.000, p=0.022, N=24). Nevertheless, variations within each hydrological type are considerable both considering all (Fig. 24, left) and single sampling dates (Fig. 24, right). However, in July differences were significant (Kruskal-Wallis test: Chi-square₂ = 5.556, p=0.025, N=8) related to perennial and int- sites (pairwise comparison: p=0.025). This is due to strong increases in total abundance in the perennial sites from June to July, which did not happen in int+ and int- sites.

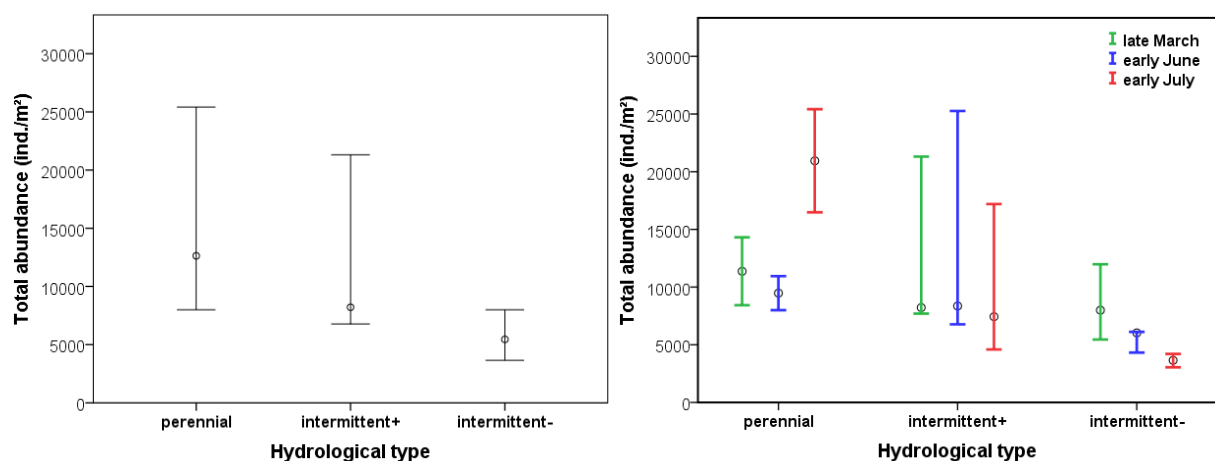


Fig. 24: left: Total abundance of all samples (perennial: n=6, int+: n=9, int-: n=9); right: Temporal change of total abundances (perennial: n=2 for each sampling period, int- and int+: n=3 for each sampling period); points indicate the median, error bars indicate the range

Temporal dynamics in abundance of dominant invertebrate groups

High increase in **Crustacea** (*Gammarus fossarum*, Fig. 25) was mainly responsible for the increase in total abundance from June to July in both perennial sites. In Gepringbach (perGep), molluscs (mainly the neozoon *Potamopyrgus antipodarum*) contributed to this increase additionally. Although the medians of abundances remained relatively constant in int+ sites (Fig. 24, right), abundances were highly different between sites and dates (Fig. 25). This was mainly due to temporal dynamics in *G. fossarum* (Fig. 26). Int- sites showed a clear trend towards decreasing total abundance from March to July, which was observed in all three sites. EPT taxa and oligochaets (Fig. 25, Fig. 27) were mainly responsible for this decrease.

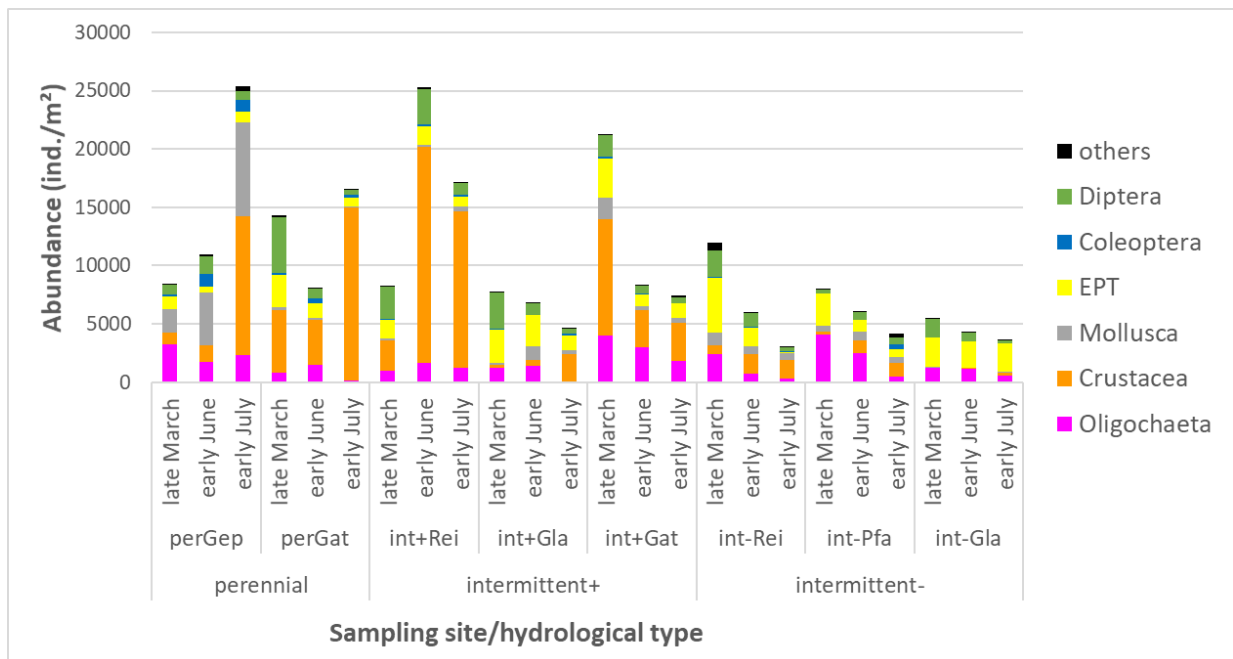


Fig. 25: Temporal dynamics of abundances of main taxonomic groups; others: Turbellaria, Nematomorpha, Hydrachnidia, Odonata, Heteroptera

Temporal dynamics in abundance of *Gammarus fossarum*

Generally, abundances of *G. fossarum* were lower in int- than in int+ and perennial sites. Temporal dynamics were heterogenous within the int+ type: In Reichelgraben a strong increase occurred from March to June, which, however, did not continue towards July. In contrast, in Gatterbach the highest abundance was stated in March. In Glauningbach temporal dynamics were similar to the perennial streams, but abundances were markedly lower. Both perennial sites are characterised by strong increases from June to July.

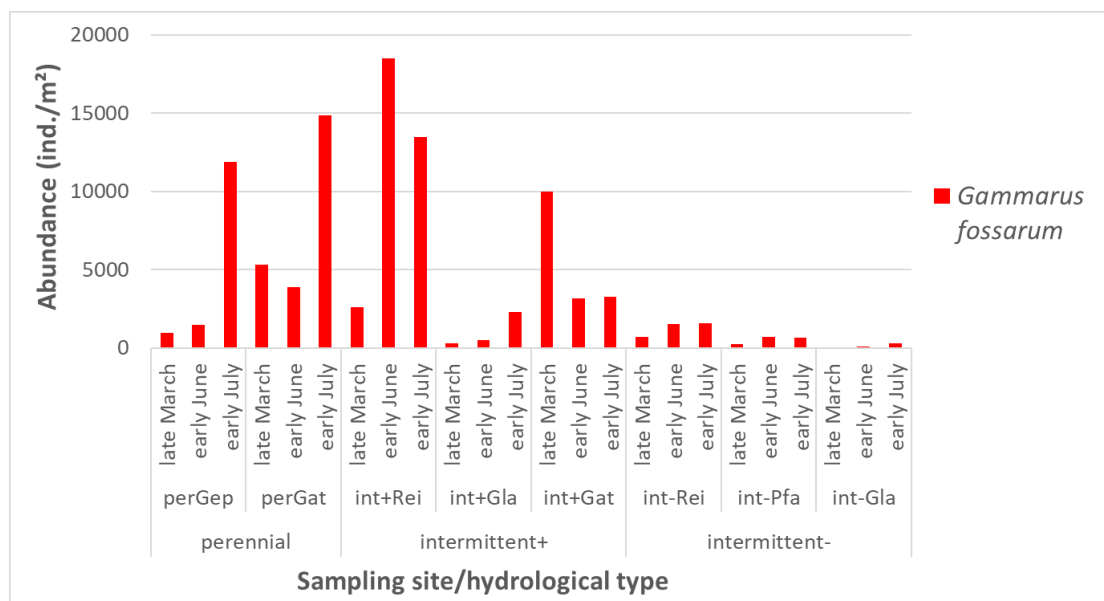


Fig. 26: Temporal dynamics of abundances of *Gammarus fossarum*

Temporal dynamics in abundance of EPT taxa

A general decrease in EPT abundance from March to July across all hydrological types was observed. This decrease was indistinct in int- Glauningbach (Fig. 27) because *Habrophlebia* (Ephemeroptera) appeared only as late as in the June samples, but in high abundances (June: 1,900, July 2,300 ind./m²; Fig. 28). In the two perennial streams temporal dynamics of EPT taxa was inconsistent (Fig. 27): In Gatterbach abundances were continuously reduced, whereas in Gepringbach an increase occurred from June to July.

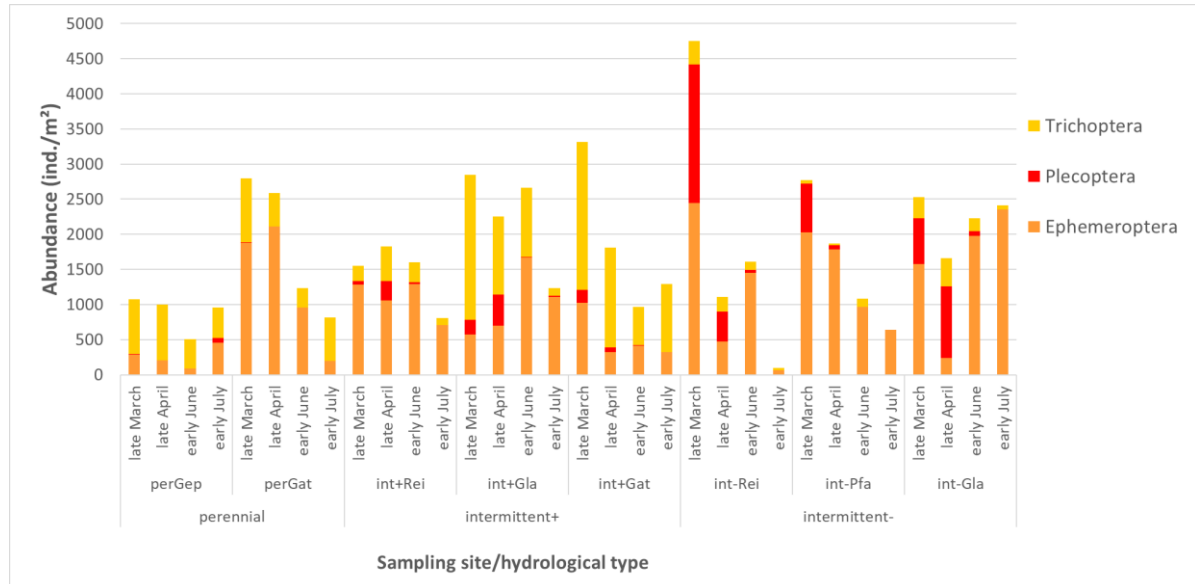


Fig. 27: Temporal dynamics of abundances of EPT taxa

Temporal dynamics in abundance of Ephemeroptera

High abundances of EPT taxa in March in int- sites were mainly due to *Siphonurus armatus* (**Siphonuridae**), which were substantially reduced in late April (Fig. 28, Fig. 29). This species has distinctly different abundances in the hydrological types: no specimens in perennial sites, relatively low abundances in int+, which even dominate total abundance in int- sites. The abovementioned late increase in *Habrophlebia* (**Leptophlebiidae**) in int- Glauningbach could also be seen in the other int- sites (in Pfaffenbach in late April already, Fig. 28) but not in the int+ sites, where it was already high in March. *Baetis* (**Baetidae**) and **Heptageniidae** are generally characteristic of the perennial sites and decreased from March to July. The increase in the Baetidae family and Ephemeroptera in general in Gepringbach towards July was due to *Cloeon dipterum* and *Ephemerella ignita* (**Ephemerellidae**). In the Heptageniidae family, reduction affected all occurring taxa, namely *Electrogena ujhelyii*, *Rhithrogena* sp. and *Ecdyonurus* sp. *E. ujhelyii* accounted for diverging dynamics of the family in two of the three int+ sites, where abundances were more constant across all sampling dates. The strong representation of the Baetidae family in Pfaffenbach (650 ind./m²) in July was due to *Cloeon dipterum* present in remnant pools. It is a typical pioneer species of stagnant waterbodies, thus having strongly diverging ecological preferences compared to other representatives of the family, which are rheophilic (e.g. *Baetis*) *Ephemerella danica* was the only representative of **Ephemeridae** and occurred in samples of both perennial streams and in int+ Glauningbach (one specimen in the March sample). Abundances were relatively low in all samples.

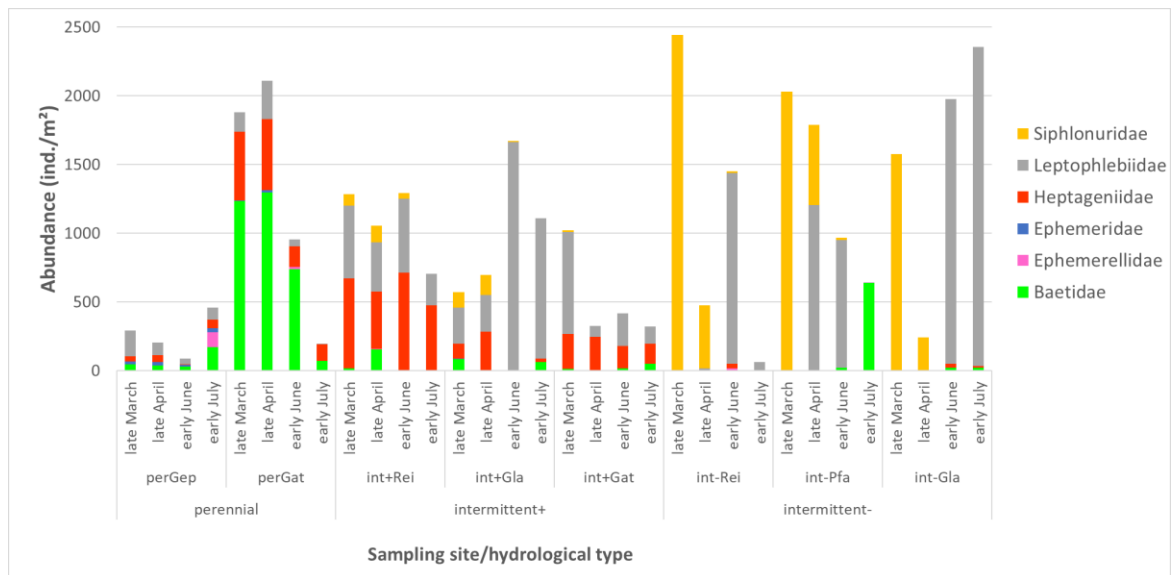


Fig. 28: Temporal dynamics of abundances of Ephemeroptera families (families with one occurring species only: Siphonuridae: *Siphonurus armatus*, Ephemeridae: *Ephemera danica*, Ephemerellidae: *Ephemerella ignita*)

Temporal dynamics in abundance of Plecoptera

Increasing abundances from perennial towards int- sites is striking in Plecoptera. Highest abundances of **Nemouridae** (mainly *Nemoura* sp.) were found in int- sites, which were reduced as soon as in late April. Abundances of **Isoperlidae** (only *Isoperla tripartita*) were also highest in the two stony int- sites. The **Capniidae** family was only represented by the genus *Zwicknia* in the March samples (Fig. 29). Adult catches suggest that only *Z. acuta* is present (see chapter 5.3.5). It occurred in int- Reichelgraben in relatively high abundance (310 ind./m²) and in minor abundances in int+ Reichelgraben and int-Glauningbach. However, abundances were reduced by emergence started before the sampling date. The same might also apply to *Brachyptera risi* (**Taeniopterygidae**) which was only present in low abundances in int- Reichelgraben and Glauningbach in late March.

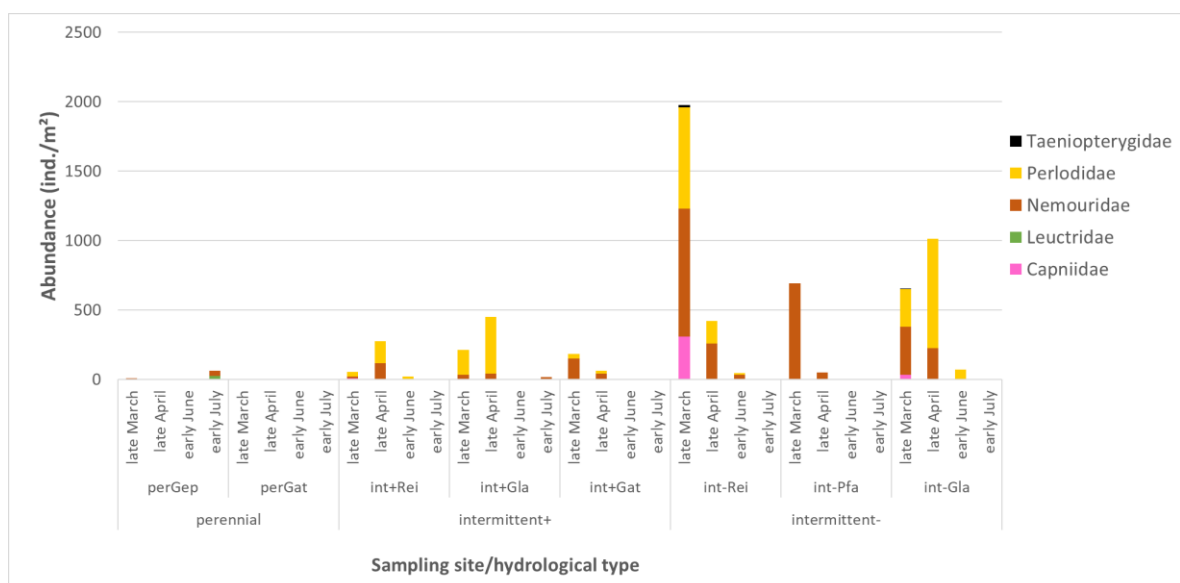


Fig. 29: Temporal dynamics of abundances of Plecoptera families (families with one occurring taxon only: Taeniopterygidae: *Brachyptera risi*, Perlodidae: *Isoperla tripartita*, Leuctridae: *Leuctra* sp., Capniidae: *Zwicknia* cf. *acuta*)

Temporal dynamics in abundance of Trichoptera

Hydropsyche (**Hydropsychidae**) had considerable abundances in both perennial sites and minor abundances in two of the int+ sites. Abundance in both perennial sites increased from June to July due to higher numbers of early instar larvae. Nevertheless, last instars of *H. saxonica*, the certainly most abundant species of the genus in the samples, were present on all dates, which is in line with the flight period from May to September observed during this thesis.

The **Limnephilidae** were present at all sites, but had clearly highest abundances in int+ sites. *Chaetopteryx* species generally constituted the major part of limnephilid specimens and were mainly responsible for the high family abundances in int+ sites. Abundances generally decreased continuously from March to July as more mature larval instars of *Chaetopteryx* became dominant.

Beraeodes minutus was the most abundant representative of the **Beraeidae** family. It occurred as last instar larvae in the March and April samples and as early instars in June and July of perennial Gepringbach and int+ Gatterbach (890 ind./m² in early July). Similarly, *Oligostomis reticulata* (**Phryganeidae**) was found as pupa in late March in Pfaffenbach and as early instar larvae from late April to early July in five of the six intermittent streams. The **Glossosomatidae** family was only represented by *Synagapetus moselyi*. It was present as larva in all intermittent streams but Pfaffenbach in March, April and June with highest abundances in March or April. Pupa were found in April and June. **Leptoceridae** were largely restricted to perennial sites with *Athripsodes bilineatus* being the most abundant species. It was sampled on all dates at the perennial sites with an abundance peak in late April.

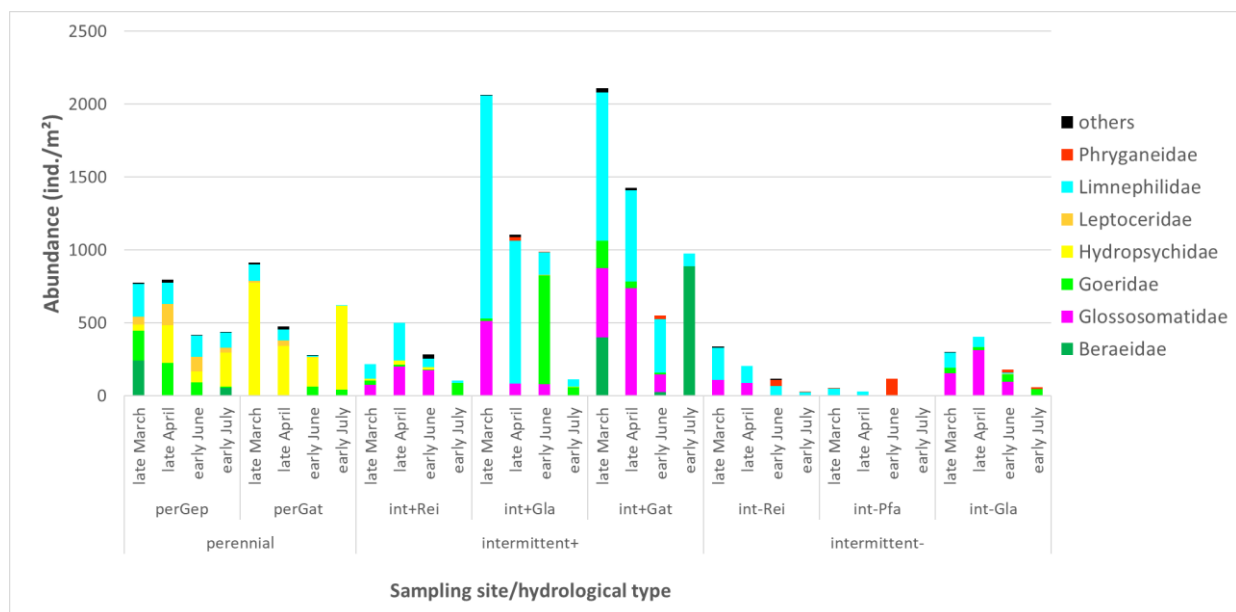


Fig. 30: Temporal dynamics of abundances of Trichoptera families (others: Psychomyiidae, Rhyacophilidae, Sericostomatidae, Polycentropodidae; families with one occurring species only: Phryganeidae: *Oligostomis reticulata*, Glossosomatidae: *Synagapetus moselyi*, Sericostomatidae: *Notidobia ciliaris*)

Temporal dynamics in taxa richness and taxonomic composition

A total of 116 taxa were collected, 81 from the two perennial, 83 from the three int+ and 73 from the three int- streams. The highest number of taxa in one sample was 53 (perennial Gepringbach in early July), the lowest was 21 (int- Pfaffenbach in early June and July and int- Glauningbach in July).

Taxa richness decreased along the hydrological gradient with int- streams having the lowest (Fig. 31, left; 42, 29 and 26 taxa, respectively). Differences between groups are significant (ANOVA: $F_{2,1} = 4.047$, $p = 0.033$) but only due to differences between perennial and int- sites (Bonferroni post-hoc test: $p = 0.030$). Taxa richness of grouped intermittent sites are also significantly different from perennial sites (t-test: $t_{22} = 2.624$, $p = 0.016$). Nevertheless, large overlap between hydrological types existed when taking together all sampling dates and no statistically significant differences between groups exist in single dates.

Trends towards specific changes in taxa richness in the course of spring are indistinct due to partly large overlap between dates within the perennial and the int- type (Fig. 31, right). However, consistent changes between sites towards a reduction of taxa richness from late March to early June was observed in the int+ sites, but without distinct further reduction towards early July.

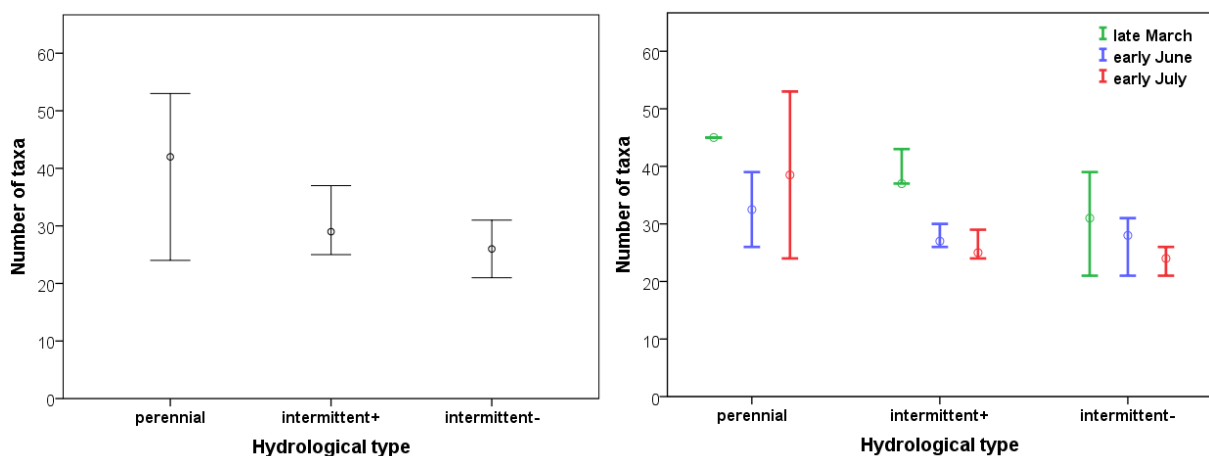


Fig. 31: left: Taxa richness of all samples (perennial: $n=6$, int+ and int-: $n=9$ each; right: Temporal change of taxa richness (perennial: $n=2$ for each sampling period, int- and int+: $n=3$ for each sampling period); points indicate the median, error bars indicate the range

EPT taxa richness was mainly responsible for changes in taxa richness of most sites (Fig. 32), which is, however, mainly because they were in general determined at species level unlike other groups. Taxonomic composition changes happened markedly from early June to early July in Pfaffenbach (int-Pfa, Fig. 32) due to the new presence of numerous pioneer species (dytiscid beetles, Heteroptera) and/or species with resistance to low oxygen concentrations (Hirudinea, Odonata). In July Pfaffenbach was the only stream which had disconnected pools, whereas the other intermittent streams exhibited low-flow conditions due to preceding rainfalls.

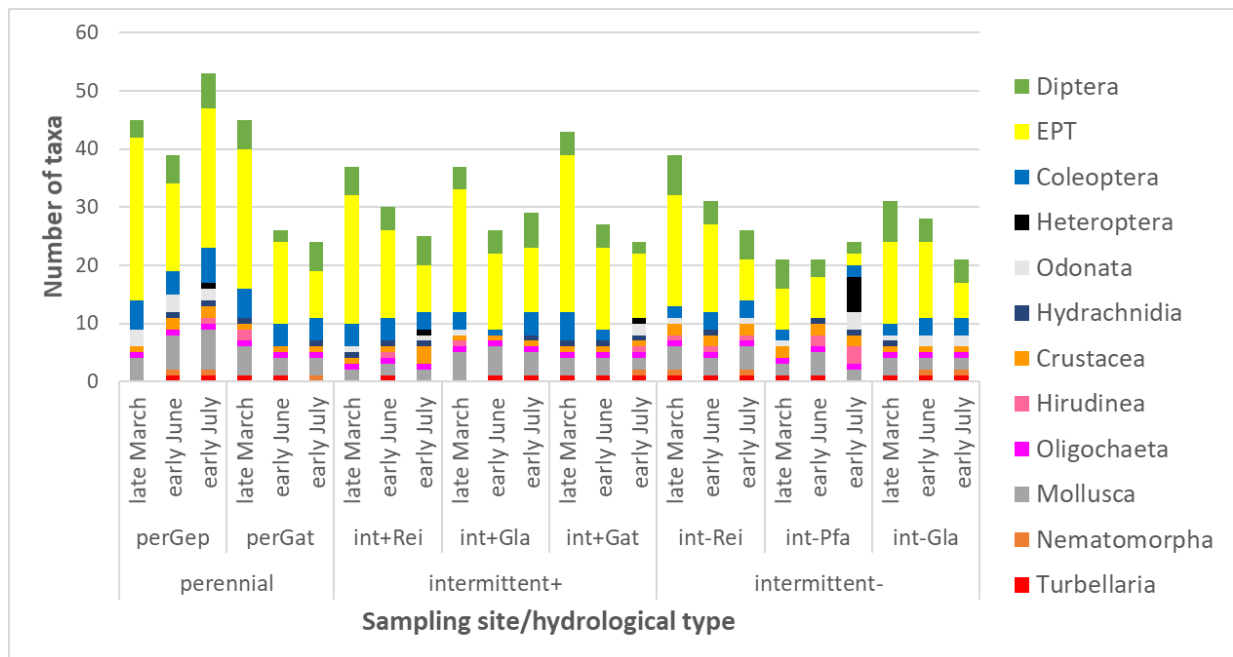


Fig. 32: Temporal change of taxa richness of macroinvertebrate groups

Temporal dynamics in richness of EPT taxa

EPT taxa were evaluated from an additional sampling date in late April.

Overall **EPT taxa richness** decreased along the hydrological gradient with int- streams having the lowest (21, 17.5 and 9.5 taxa, respectively; Fig. 33, left). In all of the sites EPT taxa richness was highest in late March (Fig. 33, right; Fig. 34). A continuous reduction in the course of spring and early summer is most distinct in int+ streams (Fig. 33, right). In the int- streams the reduction only occurred as late as in the July samples of all three sites (Fig. 34). The two perennial streams exhibited diverging patterns mainly from early June to early July: In Gatterbach the number decreased continuously, in Gepringbach the decrease until early June turned into a strong increase towards early July (from 15 to 24; Fig. 34) mainly due to mayflies.

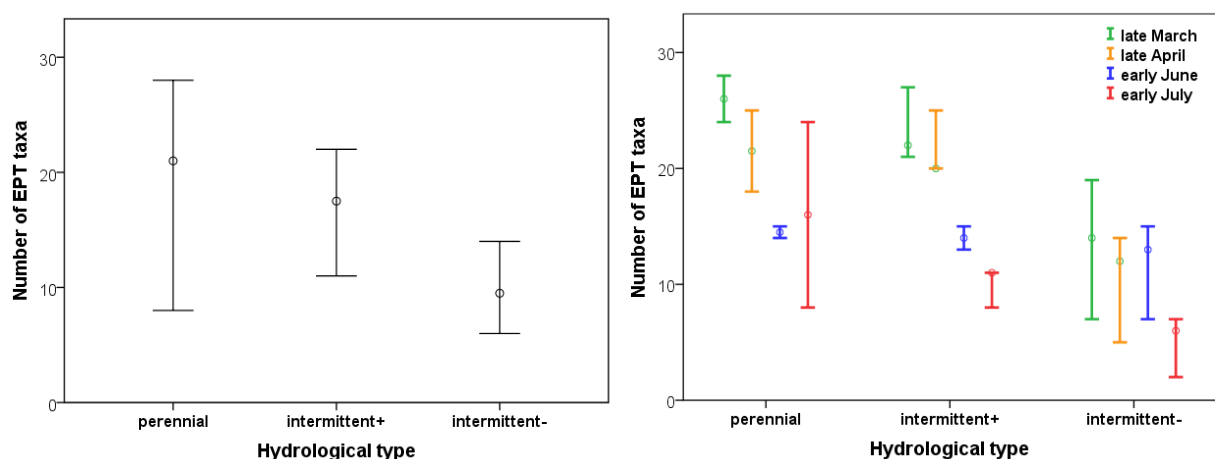


Fig. 33: left: EPT taxa richness of all samples (perennial: n=6, intermittent+ and intermittent-: n=9 each; right: Temporal change of EPT taxa richness (perennial: n=2 for each sampling period, int- and int+: n=3 for each sampling period); points indicate the median, error bars indicate the range

Concerning richness of the **Ephemeroptera** community, different patterns of changes in the course of spring and early summer were found especially between int+ and int- streams: In all three int- streams taxa increased continuously until early June before decreasing in July, whereas in int+ streams the pattern was generally inverse until June (decrease) and there was no significant further decrease from June to July (Fig. 34). Unlike *Siphonurus armatus* and *Rhithrogena* sp. which were only found until early June, *Electrogena ujhelyii*, *Habrophlebia* sp., *Cloeon dipterum* and *Baetis* spp. also occurred in July samples. The two perennial sites showed diverging patterns which, however, resembled more int+ than int- sites.

Plecoptera were represented by seven taxa (*Brachyptera risi*, *Nemoura* sp., *Nemurella pictetii*, *Amphinemura standfussi*, *Leuctra* sp., *Zwicknia* cf. *acuta* and *Isoperla tripartita*). Generally, Plecoptera occurred in higher abundances and richness in intermittent than in perennial sites. In the latter only *I. tripartita* (one specimen in March in Gatterbach), *Leuctra* sp. and *Nemoura* sp. were collected, which were also represented in IS samples. Plecoptera taxa richness generally decreased in both intermittent types from March to June. Among the July samples only one taxon was found in Glauningbach (int+) (*Nemoura* sp.).

In both intermittent types **Trichoptera** richness remained relatively constant until early June before dropping markedly in July. More distinct decreases from June to July were mitigated by reappearance of species as juvenile larvae (*Lithax obscurus* and *Oligostomis reticulata* in int- Glauningbach and Reichelgraben, *Beraeodes minutus* in int+ Gatterbach and perennial Gepringbach). As in Plecoptera and Ephemeroptera the two perennial sites diverged: In Gepringbach richness decreased from March to June (19, 16 and 11 taxa, respectively) and increased (13 taxa) in July, whereas in Gatterbach it decreased continuously (14, 10, 7 and 4, respectively).

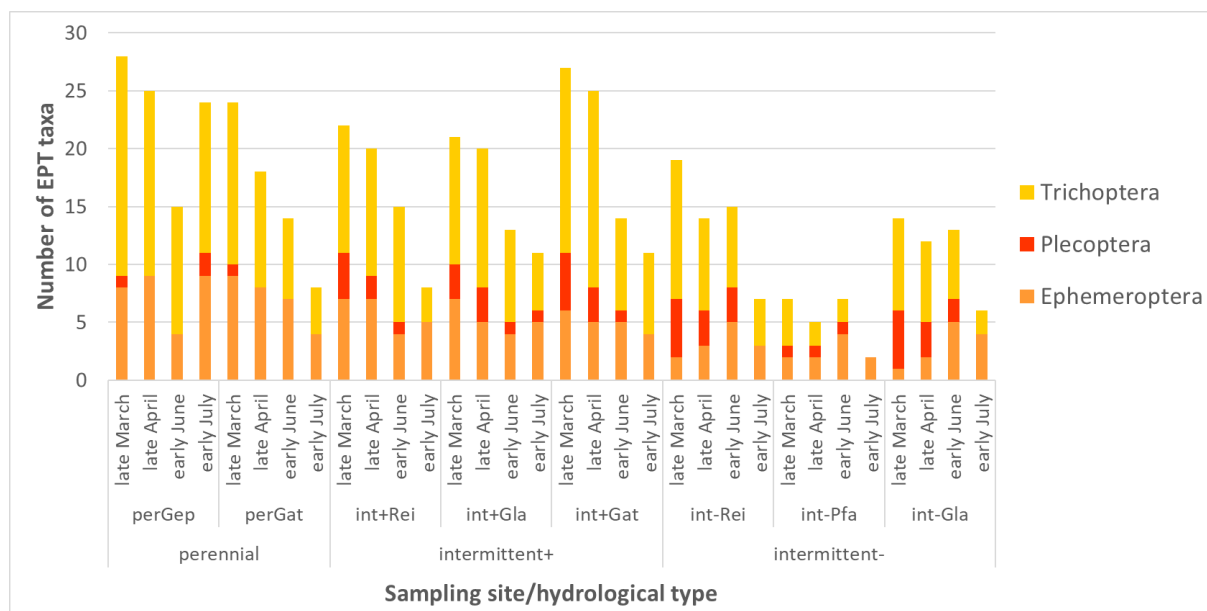


Fig. 34: Temporal change of EPT taxa richness at sampling sites

4.1.2 Similarity analyses

Non-metric multidimensional scaling (NMS) analyses that include all samples (all sites, all dates) show a clear separation of communities along the hydrological gradient (Fig. 35). Presence/absence-transformed abundances (Fig. 35, right) weakened the separation to some extent compared to log-transformation (Fig. 35, left), mainly between int+ and int- sites.

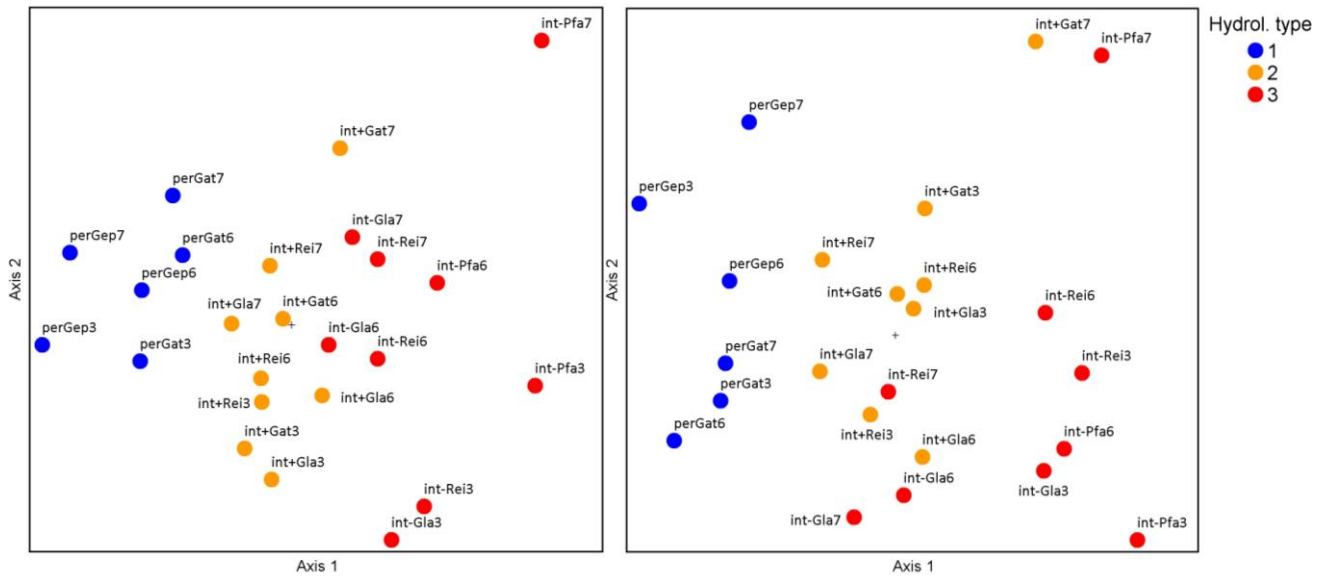


Fig. 35: NMS scatterplots; overlay: Hydrol. types = hydrological stream types (1 = perennial, 2 = int+, 3 = int-); left: abundances log(x+1)-transformed (stress= 10.850 for 3-dimensional solution); right: abundances presence/absence-transformed (stress=14.393 for 3-dimensional solution); numbers at the end of the code indicate sampling month

Cluster analysis confirmed the separation of communities along the hydrological gradient, but with three diverging samples from July (int+Gat7, int-Pfa7 and int+Gla7; Fig. 36 above). Perennial streams formed a cluster that is opposed to both intermittent types. The intermittent types formed several clusters, one consisting of all int+ sites (with two abovementioned exceptions). Presence/absence-transformed abundances separated perennial from intermittent sites in the same way, but int+ and int- sites were often distributed within the same clusters (Fig. 36 below).

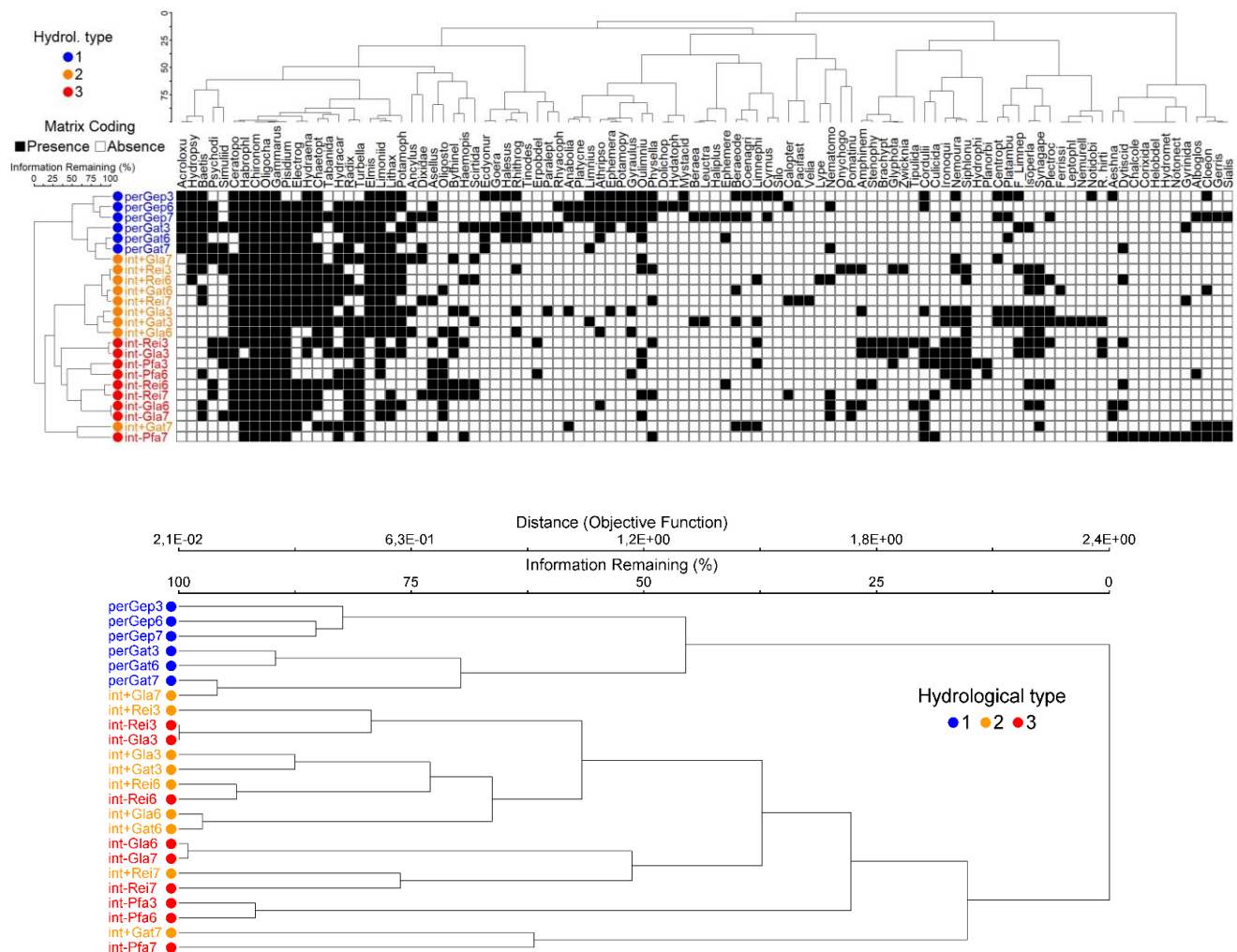


Fig. 36: Cluster analyses with all samples (n=24); overlay: hydrological stream types (1 = perennial, 2 = int+, 3 = int-); above: two-way cluster analysis with log(x+1)-transformed abundances; below: abundances presence/absence-transformed

In addition to clusters of sites, the **two-way cluster analysis** in Fig. 36 (above) groups taxa that tend to occur together and illustrates associations of taxa with sites. As hydrological stream types formed clusters, similar taxa compositions within the types can be recognised in the matrix. For example, *Acroloxus lacustris*, *Hydropsyche* spp., *Baetis* spp. and Psychodidae occurred in most perennial, but only in few intermittent stream samples (far left side of the matrix). Likewise, *Asellus aquaticus*, *Oligostomis reticulata*, *Bythinella* sp. and *Haemopsis sanguisuga* were characteristic of int- sites.

The **March** samples based on log-transformed abundances show separation of communities along the hydrological gradient with int+ sites being more closely related to the perennial sites (Fig. 37, above). Numerous taxa exclusively occurred in these two types but not in the int- type (e.g. *Elmis* sp., *Electrogena ujhelyii*, *Rhithrogena* sp.). Conversely, some taxa only occurred in the int- type (e.g. Culicidae, *Asellus aquaticus*). Nevertheless, the perennial is separated from the int+ type and is characterised by the exclusive occurrence of *Acroloxus lacustris*, *Ecdyonurus* sp. and *Goera pilosa*.

Sites with different hydrological types within one stream are not closely related: The int+ and the int- site of Reichelgraben are only 300 m away from each other, which is approximately 800 m between the two sites in Glauningbach.

Presence/absence-transformed taxa lists from March do not enable separation of hydrological types, but perennial sites form a cluster that is opposed to five of six intermittent sites (int+ and int- not

separated). Pfaffenbach (int-) is not associated to neither cluster and constitutes the “outgroup” that is opposed to all other sites (Fig. 37, below). It has by far the lowest taxa richness and in addition three exclusive taxa in the March samples (*Oligostomis reticulata*, an undetermined Planorbidae taxon and Hydrophilidae; Fig. 37, below). Lacking separation between int+ and int- based on presence/absence of taxa is likely due to many taxa in common between these hydrological types (Fig. 37, below).

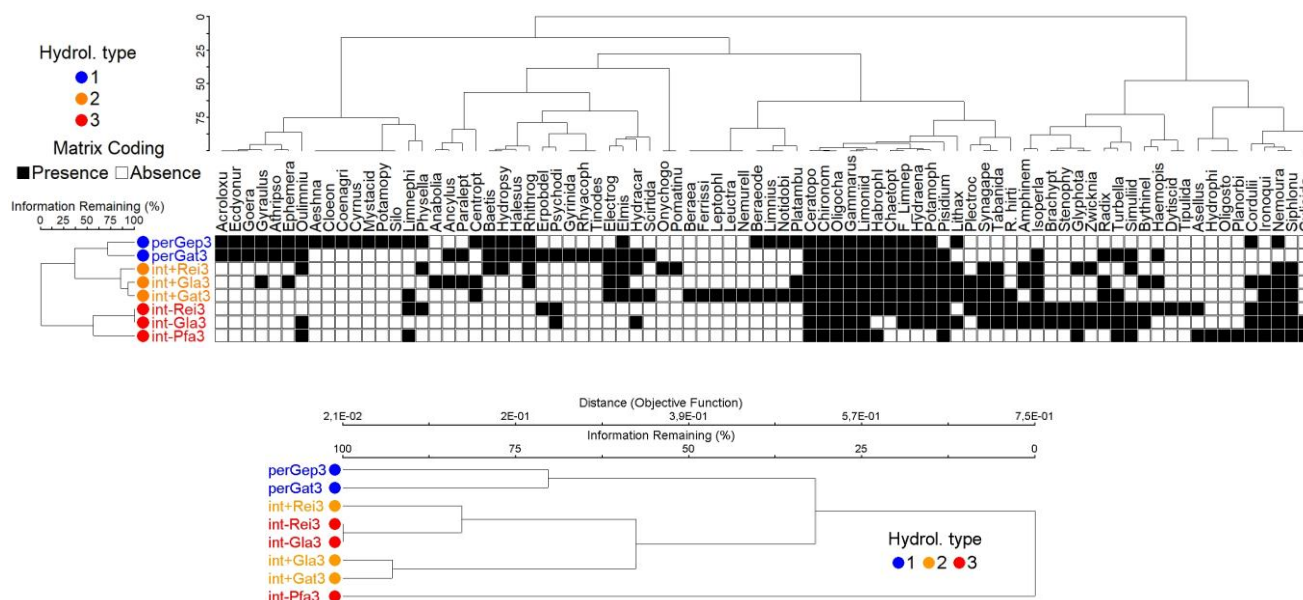


Fig. 37: Cluster analyses with samples from **late March** (n=8); overlay: hydrological regime types (1 = perennial, 2 = int+, 3 = int-); above: two-way cluster analysis with (x+1)-transformed abundances; below: abundances presence/absence-transformed

Samples from early **June** still enable separation of perennial and intermittent sites based on abundance data, but not between int+ and int- (Fig. 38, above). For example, *Lithax obscurus* and *Potamophylax rotundipennis* occurred in the int- site in Glauningbach and in all int+ sites, but not in the two other int- sites at this sampling date (but not in general). Only two taxa, *Acroloxus lacustris* and *Oulimnius tuberculatus*, exclusively occurred in both perennial sites. Subsequently, separation from all intermittent sites was not possible any longer based on presence/absence data (Fig. 38, below).

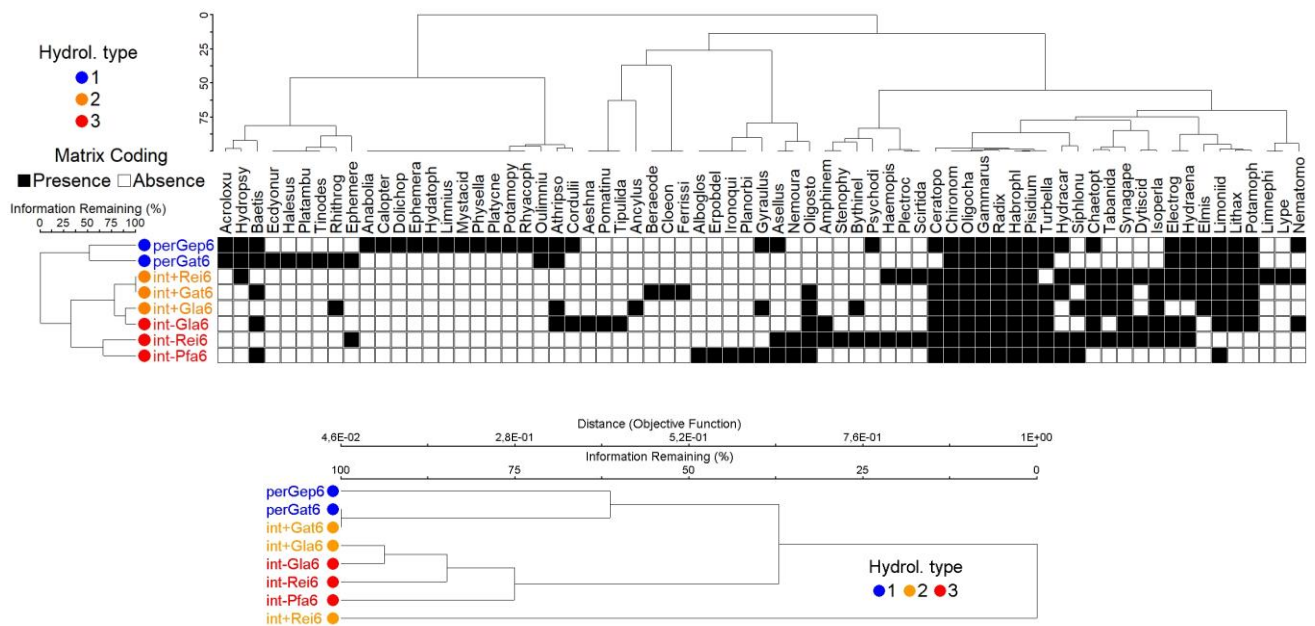


Fig. 38: Two-way cluster analysis with samples from early **June** (n=8); overlay: hydrological regime type (1=perennial, 2=int+, 3=int-); above: abundances log(x+1)-transformed, below: abundances presence/absence-transformed

July samples do not show any clustering of hydrological types (Fig. 39): The two perennial sites differ more markedly than in March and June: Gepringbach has more exclusive taxa compared to Gatterbach (30 in July and only 19 and 17 in June and March, respectively). Additionally, the perennial site in Gatterbach is closely related to the int+ site in Glauningbach in this sampling date. In both, for example, *Hydropsyche* sp., *Baetis* sp., Limoniidae/Pediciidae and Dixidae were sampled; taxa that are not shared by all other sites at this sampling date.

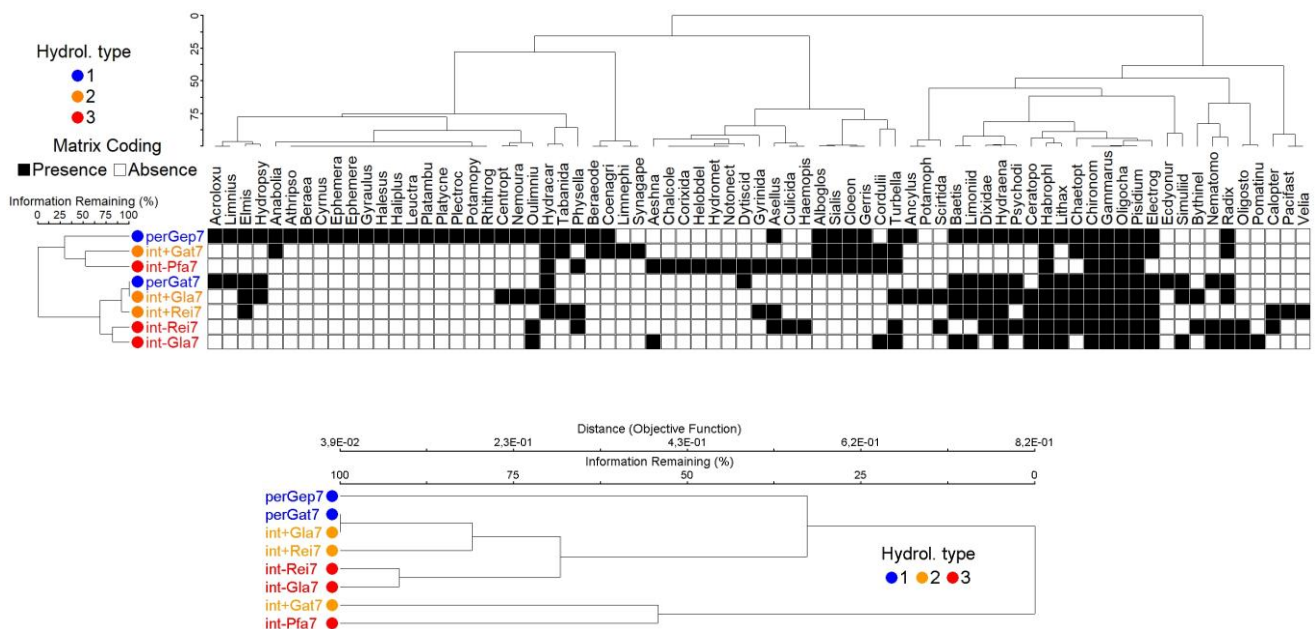


Fig. 39: Cluster analysis with samples from early **July** (n=8); overlay: hydrological regime type (1=perennial, 2=int+, 3=int-); above: abundances log(x+1)-transformed, below: abundances presence/absence-transformed

Indicator species analyses (ISA)

ISA was executed to find characteristic taxa for perennial or intermittent streams (int+ and int- grouped) and for each of the types separately. In addition, indicators for all and the single sampling dates were identified (Table 13).

For **all seasons**, 17 taxa were identified as significant **perennial-flow indicators** ($p \leq 0.05$: *Acroloxus lacustris*, *Hydropsyche* sp., *H. saxonica*, *Elmis* sp. and *Athripsodes bilineatus* had the highest indicator values) and four as significant **intermittent-flow indicators** (*Pisidium* sp., *Synagapetus moselyi*, *Habrophlebia* sp. and *Chaetopteryx rugulosa*). The number of intermittent-flow indicators was, however, higher for **specific sampling dates** and highest in late March for all hydrological types (13 perennial, 11 intermittent, 2 int+, 8 int-).

Table 13: List of hydrological regime indicators identified through *Indicator species analysis*: green indicates statistical significance ($p \leq 0.05$), yellow indicates high association with types based on presence/absence but without statistical significance; * only EPT taxa were evaluated

	taxon	all dates		late March		late April*		early June		early July	
		IV	p	IV	p	IV	p	IV	p	IV	p
Perennial flow indicators	<i>Acroloxus lacustris</i>	100	<0.001	100	0.0363			100	0.0373	100	0.0380
	<i>Hydropsyche</i> sp.	99.8	<0.001	99.8	0.0363	98.7	0.0391	100	0.0373	95.9	0.0380
	<i>Hydropsyche saxonica</i>	98.4	<0.001	98.1	0.0726	97.2	0.0391	98.1	0.0373	94.7	0.0380
	<i>Elmis</i> sp.	95.9	<0.001	84.6	0.0363			97.1	0.0373	86.7	0.0380
	<i>Athripsodes bilineatus</i>	82.9	<0.001	100	0.0363	100	0.0391	98.9	0.0719		
	<i>Baetis</i> sp.	81.6	<0.01	96.3	0.0726			98.8	0.0373		
	<i>Oulimnius tuberculatus</i>	81.5	<0.01	94.7	0.0363			100	0.0373		
	<i>Halesus digitatus/tessellatus</i>	66.7	<0.01	100	0.0363						
	<i>Ecdyonurus</i> sp.	66.7	<0.01	100	0.0363	100	0.0391				
	<i>Limnius</i> sp.	66.3	<0.01							100	0.0380
	<i>Ephemera danica</i>	65.6	<0.01	95.5	0.0712	100	0.0391				
	<i>Rhithrogena</i> sp.	64.1	<0.01								
	<i>Gyraulus</i> sp.	61.4	<0.01	98.8	0.0363						
	<i>Psychodidae</i> Gen. sp.	53.7	0.0395								
	<i>Potamopyrgus antipodarum</i>	50	0.0110								
	<i>Baetis rhodani</i>	47.7	0.0404	98.4	0.0726						
	<i>Platambus maculatus</i>	41.4	0.0384								
	<i>Goera pilosa</i>			100	0.0363						
Intermittent flow indicators (int+ and int-)	<i>Siphonurus armatus</i>			100	0.0363	83.3	0.1451	66.7	0.3225		
	<i>Pisidium</i> sp.	91.4	<0.01	98.1	0.0363						
	<i>Synagapetus moselyi</i>	61.1	0.0359	83.3	0.1433	83.3	0.1831				
	<i>Tabanidae</i> Gen. sp.			83.3	0.2135			66.7	0.3553		
	<i>Ironoquia dubia</i>			83.3	0.2529	66.7	0.3560				
	<i>Amphinemura standfussi</i>			66.7	0.3189	83.3	0.1850				
	<i>Glyptotaelius pellucidus</i>			66.7	0.3190						
	<i>Zwackia</i> cf. <i>acuta</i>			50	0.4267						
	<i>Bythinella</i> sp.			50	0.4270						
	<i>Plectrocnemia conspersa</i>			50	0.4603						
	<i>Rhyacophila hirticornis</i>			50	0.4636						
	<i>Habrophlebia</i> sp.	80.4	0.0157								
	<i>Chaetopteryx rugulosa</i>	61.1	0.0366			66.7	0.3277	83.3	0.2106	66.7	0.3898
	<i>Nemoura</i> sp.					83.3	0.1084				
	<i>Isoperla tripartita</i>					83.3	0.1456	83.3	0.2508		
	<i>Oligostomis reticulata</i>							83.3	0.2182		
Indicators for int+	<i>Chaetopteryx rugulosa</i>	76	<0.01	66.7	0.2457			85.1	0.0454		
	<i>Tabanidae</i> Gen. sp.	61	<0.01					90.0	0.0215		
	<i>Synagapetus moselyi</i>	62.2	0.0153					79.2	0.0341		
	<i>Electrogena ujhelyii</i>	64.1	0.0221								
	<i>Chaetopteryx major</i>	65.9	0.0231			77.2	0.0308				
	<i>Habrophlebia</i> sp.			77	0.0461						
	<i>Rhyacophila hirticornis</i>					66.7	0.2546				
Indicators for int-	<i>Glyptotaelius pellucidus</i>			98	0.0404						
	<i>Siphonurus armatus</i>			96.7	<0.01	82.9	0.0112				
	<i>Nemoura</i> sp.			93.5	<0.01	73.1	0.0200	66.7	0.2497		
	<i>Ironoquia dubia</i>			91.7	0.0329	97	0.0204				
	<i>Stenophylax permistus</i>			66.7	0.2448	66.7	0.2527				
	<i>Brachyptera risi</i>			66.7	0.2448						
	<i>Culicidae</i> Gen. sp.	44.4	0.0436	66.7	0.2556					66.7	0.2464
	<i>Asellus aquaticus</i>	64.2	0.0128	66.7	0.2543						
	<i>Oligostomis reticulata</i>	58.1	0.0165					85.9	0.0317	66.7	0.2447
	<i>Habrophlebia fusca</i>							84.1	0.0123		
	<i>Amphinemura standfussi</i>							66.7	0.2455		
	<i>Baetis rhodani</i>							66.7	0.2484		
	<i>Haemopsis sanguisuga</i>									66.7	0.2464
	<i>Aeshna</i> sp.									66.7	0.2518

4.2 Functional community traits

4.2.1 Biological adaption traits

Biological traits found to enable MI species to be resistant and/or resilient to intermittent flow were translated into trait modalities established by Tachet et al. (2010; Table 10). 79 out of 104 taxa could be classified at family, genus or species level. Average affinity scores of taxa for traits promoting resistance, resilience and the sum of both were compared between samples based on presence/absence of taxa. Affinity scores of taxa were tested for overall significant differences between hydrological stream types. Only affinity scores attributed to **resistance traits** from the **March and all samples** were **significantly different** between the three hydrological types (Kruskal-Wallis test: all dates: $\text{Chi-square}_2=10.909$, $p=0.004$, $N=482$; March: $\text{Chi-square}_2=6.742$, $p=0.034$, $N=194$). The mean scores per taxon increased along the hydrological gradient with perennial sites having the lowest (Fig. 40). **No statistically significant differences** in affinity scores between hydrological stream types were found in neither **resilience traits** nor the **sum of resilience and resistance traits**.

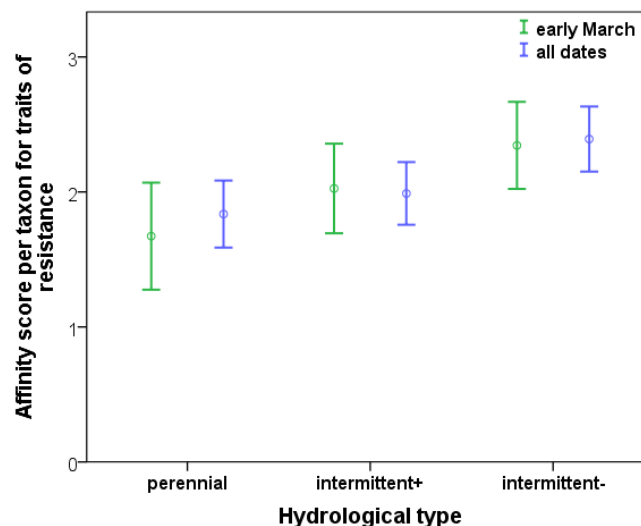


Fig. 40: Affinity scores (from Tachet et al. 2010) per taxon for biological traits providing **resistance** to intermittent flow (late March/all dates: perennial: $n=58/159$, int+: $n=78/175$, int-: $n=58/148$, $N=482/194$); points indicate means, error bars indicate 95% confidence interval

Based on the aforementioned overall differences in affinity scores for resistance traits, **mean affinity scores per taxon for traits of resistance** for the single sites were calculated (Fig. 41, all values in Appendix A). The mean affinity scores are only markedly different and do not overlap between perennial and int- sites in late March. No overlaps in values of perennial and int- sites can also be found based on combined taxa lists of all dates, but differences are distinctly slighter (Fig. 41). Nevertheless, Kruskal-Wallis test indicates significant differences based on combined taxa lists of all dates ($\text{Chi-square}_2=5.556$, $p=0.025$, pairwise comparison perennial vs. int-: $p=0.025$; $N=8$), but not based on the March samples ($\text{Chi-square}_2=3.778$, $p=0.200$, pairwise comparison: $p=0.053$; $N=8$).

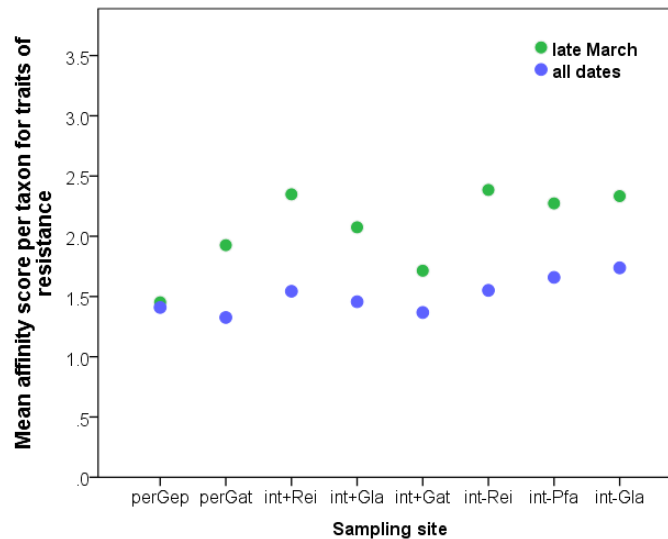


Fig. 41: Mean affinity scores (from Tachet et al. 2010) per taxon for biological traits providing **resistance** to intermittent flow at the sampling sites based on the March samples and combined taxa lists of all samples

4.2.2 Resistance vs. resilience

A trait analysis categorising MI taxa as *resilient* (RL), *resistant* (RS), *resilient and resistant* (RLRS), and *neither resilient nor resistant* (none) was performed according to Leigh et al. (2015) who used **family classifications** from Datry et al. (2014a). Proportions of these categories in each sample were calculated and compared between hydrological types. From a total of 104 sampled taxa at genus or higher level, 89 could be classified based on Datry et al. (2014a).

Only **proportions of RS** taxa were significantly different between the three hydrological types with taxa lists combining **all dates** (Kruskal-Wallis test: $\text{Chi-square}_2=5.442$, $p=0.029$, $N=8$) due to differences between perennial and int- sites (pairwise comparison: $p=0.032$). **Proportions of RL** taxa were different at $p \leq 0.1$ (Kruskal-Wallis test: $\text{Chi-square}_2=5.139$, $p=0.061$, $N=8$). Proportions of taxa with no strategies to intermittent flow were low in all hydrological types.

Considering single sampling dates, only the **March** samples exhibited significant differences in strategy categories RS and RL (Kruskal-Wallis test: $\text{Chi-square}_2=6.250$, $p=0.011$, $N=8$; test statistics are the same in RS and RL) due to differences between perennial and int- (pairwise comparison: $p=0.014$). Grouped intermittent types showed no statistically significant differences.

When comparing proportions of all dates ($N=24$; Fig. 42) significant differences were found between hydrological types in RL, RS and RSRL taxa, but not in taxa with no strategies (Kruskal-Wallis test: RL: $\text{Chi-square}_2=9.972$, $p=0.007$; RS: $\text{Chi-square}_2=8.189$, $p=0.017$; RLRS: $\text{Chi-square}_2=7.217$, $p=0.027$; none: $\text{Chi-square}_2=0.939$, $p=0.625$). Significance resulted from differences between perennial vs. int- sites in all three strategy categories (pairwise comparisons: RL: $p=0.002$, RS: $p=0.004$, RSRL: $p=0.012$) and from differences between perennial and int+ in RSRL taxa additionally ($p=0.024$).

In summary, int- streams had highest proportions of taxa with resistance and combined resistance/resilience strategies, but lowest with resilience strategies. In the perennial streams taxa with resilience strategies dominated. Low proportions of taxa with no strategies to survive intermittent flow occurred across all hydrological stream types.

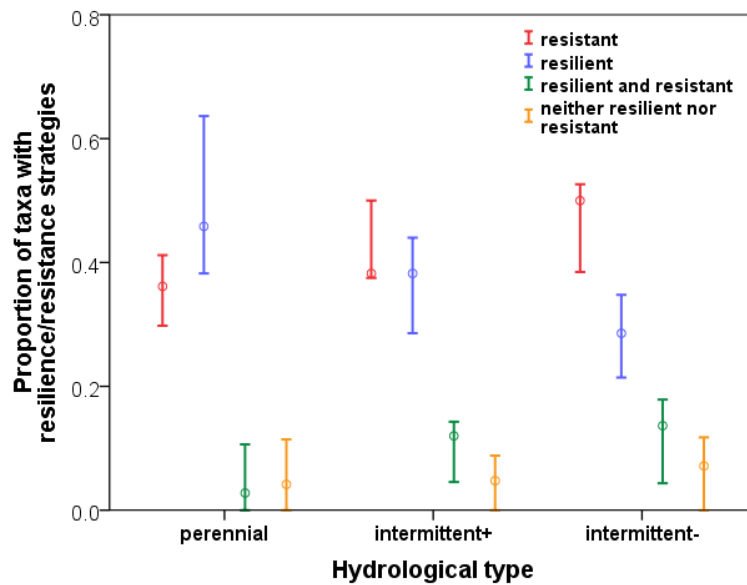


Fig. 42: Proportions of intermittent flow survival strategies after Leigh et al. (2015) based on family classifications from Datry et al. (2014a); samples from all dates (perennial: n=6, intermittent+: n=9, intermittent-: n=9); points indicate medians, error bars indicate the range

Proportions of strategy categories at the **single dates** are illustrated as mean proportions of sites in Fig. 43. This demonstrates that aforementioned differences in strategies between hydrological types were most distinct in late March. Despite statistically significant differences and clear changes along the hydrological gradient, proportions do not seem to be fundamentally different between hydrological types and neither do they substantially change in the course of spring and early summer. The increase of RL taxa in combination with a decrease in RLRS taxa from March to July in **int-** sites (RL: 23%-32%-36%, RLRS: 17% - 12% - 12%; March-June-July) is most conspicuous and did not occur in int+ and perennial sites.

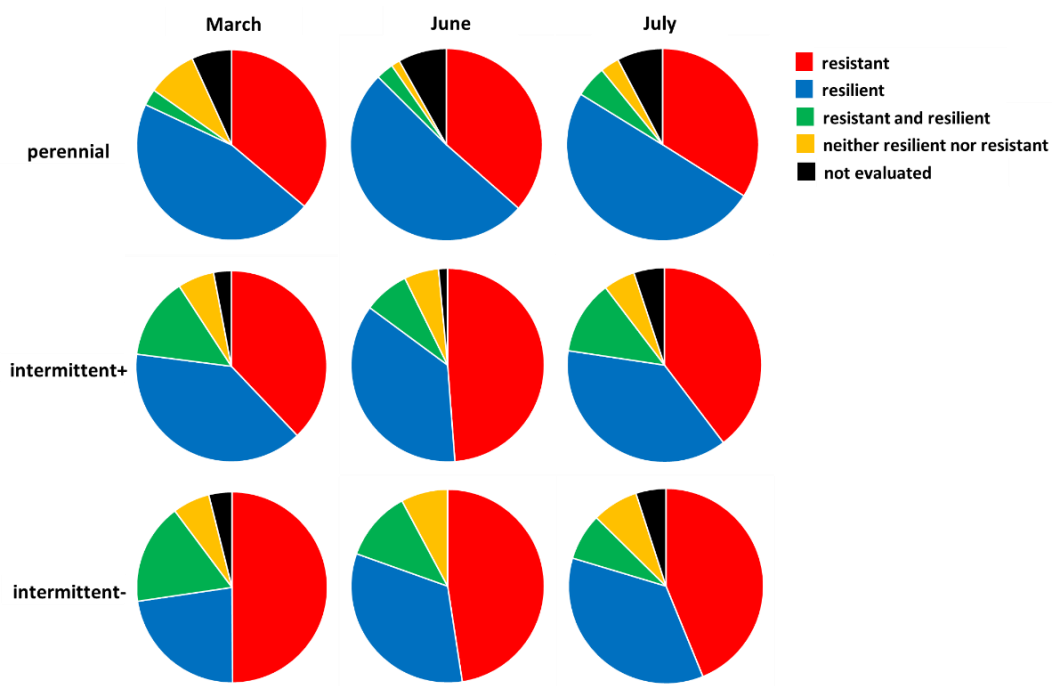


Fig. 43: Mean proportions survival strategies to intermittent flow separated by hydrological stream types and sampling dates (after Leigh et al. 2015)

4.2.3 Flight periods

Flight period indices (seasonality)

Species' flight periods (FPs) from the freshwater ecology database (Ephemeroptera: Buffagni et al. 2019, Plecoptera: Graf et al. 2019b, Trichoptera: Graf et al. 2019a) were applied to analyse differences in seasonality between hydrological stream types. FPs were only available for sampled EPT species and therefore the April samples could be included.

Species-specific flight period indices (FPI_{species} , range: 1-4) showed **no significant differences** between hydrological types based on presence/absence of species in late March (Kruskal-Wallis test: Chi-square₂=1.472, $p=0.479$ for all hydrological types; Mann-Whitney U test: $U=1503.500$, $p=0.305$ for per. vs. grouped intermittent types; $N=126$). Average FPI_{species} between sites of all hydrological types overlap at all sampling dates (Table 14). However, average FPI_{species} values of all three int- sites are lower than the other sites' values when using taxa of all dates, thus indicating earlier FPs. Differences are only slight and not significant (Kruskal-Wallis test: Chi-square₂=5.000, $p=0.075$).

Table 14: Mean **species-specific** flight period indices of all samples

<u>Hydrological type</u>	<u>site</u>	Avg. species-specific FPI				
		<u>late March</u>	<u>late April</u>	<u>early June</u>	<u>early July</u>	<u>all dates</u>
perennial	perGep	2.86	2.82	2.91	2.97	2.91
	perGat	2.87	2.89	2.82	2.76	2.84
intermittent+	int+Rei	2.73	2.92	2.80	2.94	2.85
	int+Gla	2.97	2.94	2.83	2.73	2.89
	int+Gat	2.79	2.90	2.87	3.13	2.85
intermittent-	int-Rei	2.83	3.01	2.87	2.83	2.78
	int-Pfa	2.75	3.15	2.70	2.75	2.82
	int-Gla	2.59	2.73	2.65	2.50	2.62

In contrast, the **community FPI** ($FPI_{\text{community}}$) which includes abundances of species, indicated significant differences in flight periods between the hydrological types in April (Kruskal-Wallis test: Chi-square₂=6.250, $p=0.011$; $N=8$) and at $p \leq 0.1$ in March (Kruskal-Wallis test: Chi-square₂=5.139, $p=0.061$; $N=8$). **Significantly earlier average flight periods** (lower FPI) resulted in int- compared to int+ sites (pairwise comparisons: $p=0.012$ for April, $p=0.091$ in March). These differences in $FPI_{\text{community}}$ between hydrological types (on average 0.31 between int+ and int- sites, 0.27 between perennial and int- in March, 0.23 between int+ and int- in April; Fig. 44) correspond to 21 to 28 days (one FPI point equals three months/one season).

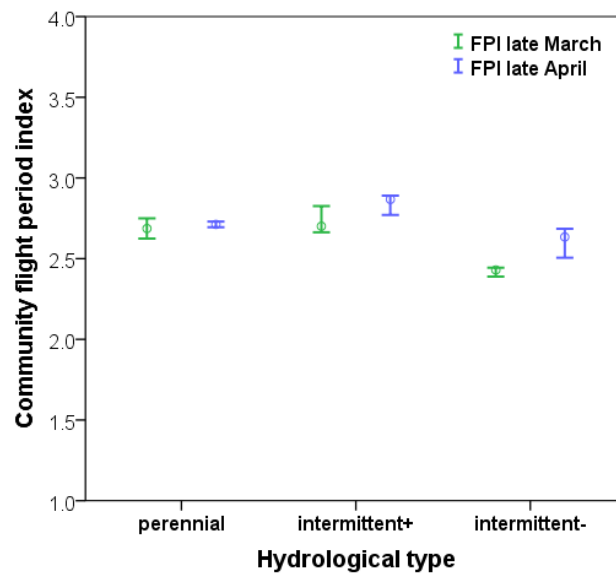


Fig. 44: Community flight period index of the March and April samples (perennial: n=2, int+ and int-: n=3); points indicate medians, error bars indicate the range

In contrast, the **June and July** samples showed no significant differences in $FPI_{community}$ (Kruskal-Wallis test: $\chi^2_{2}=0.222$, $p=0.946$; $N=8$; test statistics are the same in June and July) and overlaps of index values between the hydrological types (Table 15).

Table 15: Community flight period index (FPI) values of all samples

<u>Hydrological type</u>	<u>site</u>	community FPI			
		<u>late March</u>	<u>late April</u>	<u>early June</u>	<u>early July</u>
perennial	perGep	2.75	2.73	2.94	2.84
	perGat	2.62	2.69	2.42	2.44
intermittent+	int+Rei	2.66	2.77	2.70	2.65
	int+Gla	2.70	2.87	2.69	2.79
	int+Gat	2.83	2.89	3.07	2.83
intermittent-	int-Rei	2.39	2.63	2.81	2.91
	int-Pfa	2.43	2.69	2.71	2.70
	int-Gla	2.44	2.51	2.75	2.79

Grouped intermittent types showed no significant differences for none of the dates (Mann-Whitney U test: March: $U=4.000$, $p=0.643$; April: $U=6.000$, $p=1.000$; June and July: $U=7.000$, $p=1.000$) as values of perennial streams and int+ streams were more similar than those of the two intermittent types (Fig. 44).

Flight period types (duration and seasonality)

Flight period (FP) types were attributed to EPT species based on information in the freshwater ecology database (Ephemeroptera: Buffagni et al. 2019, Plecoptera: Graf et al. 2019b, Trichoptera: Graf et al. 2019a; parameters “emergence/FP” and “duration emergence period”). Species with short FP in spring (type 1), in spring and summer (type 2), in autumn (type 4) and with long FP (type 3) were distinguished and communities from the sampling sites were analysed for differences in these types based on presence/absence of species.

No significant differences were detected between FP types of hydrological stream types from single sampling dates and combined dates (Kruskal-Wallis tests, lowest p values from March samples: Chi-square₂ = 1.130, p=0.568 for all hydrological types; Mann-Whitney U test: U=1210.500, p=0.318 for per. vs. both int. types; N=113). Accordingly, differences in proportions of FP types between sites of the hydrological stream types were relatively low (Table 16, mean proportions of hydrological types in Fig. 45; values from April, July and all dates see Appendix B: Table B 1, Table B 2). However, **in late March**, species belonging to **type 1** (short FPs in spring) had **higher proportions in int- sites** than in perennial sites at p≤0.1 (Kruskal-Wallis test: Chi-square₂=4.929, p=0.057, pairwise comparison: p=0.080; N=8) and between perennial and grouped intermittent sites (Mann-Whitney U test: U=12.000, p=0.071). In **early June**, this type was also significantly higher in grouped intermittent sites than in perennial sites (Mann-Whitney U test: U=12.000, p=0.071) due to the reappearance of early instar larvae without diapausing eggs (e.g. *Oligostomis reticulata*, *Beraeodes minutus*, *Lithax obscurus*). **Type 3** (long FPs) had higher proportions in perennial than in int+ and int- sites in March (Kruskal-Wallis test: Chi-square₂=4.694, p=0.093, pairwise comparison perennial vs. int+: p=0.092, perennial vs. int-: p=0.080; N=8).

Table 16: Proportions of flight period types (%) from samples of late March and early June; type 1: short in spring, type 2: short in spring and summer, type 3: long, type 4: short in autumn; (*) differences at p≤0.1 between hydrological types tested with Kruskal-Wallis or Mann-Whitney U test (N=8)

<u>Hydrol. type</u>	<u>site</u>	<u>Proportion of FP type (March)</u>					<u>Proportion of FP type (June)</u>				
		<u>type 1 (*)</u>	<u>2</u>	<u>3 (*)</u>	<u>4</u>	<u>n</u>	<u>type 1 (*)</u>	<u>2</u>	<u>3</u>	<u>4</u>	<u>n</u>
perennial	perGep	21%	21%	47%	11%	19	17%	25%	33%	25%	12
	perGat	19%	25%	44%	13%	16	11%	33%	44%	11%	9
intermittent+	int+Rei	29%	29%	29%	14%	14	20%	27%	40%	13%	15
	int+Gla	29%	21%	21%	29%	14	33%	17%	25%	25%	12
	int+Gat	32%	26%	32%	11%	19	33%	25%	17%	25%	12
intermittent-	int-Rei	29%	21%	36%	14%	14	23%	15%	46%	15%	13
	int-Pfa	40%	0%	40%	20%	5	40%	0%	40%	20%	5
	int-Gla	42%	25%	25%	8%	12	30%	30%	30%	10%	10

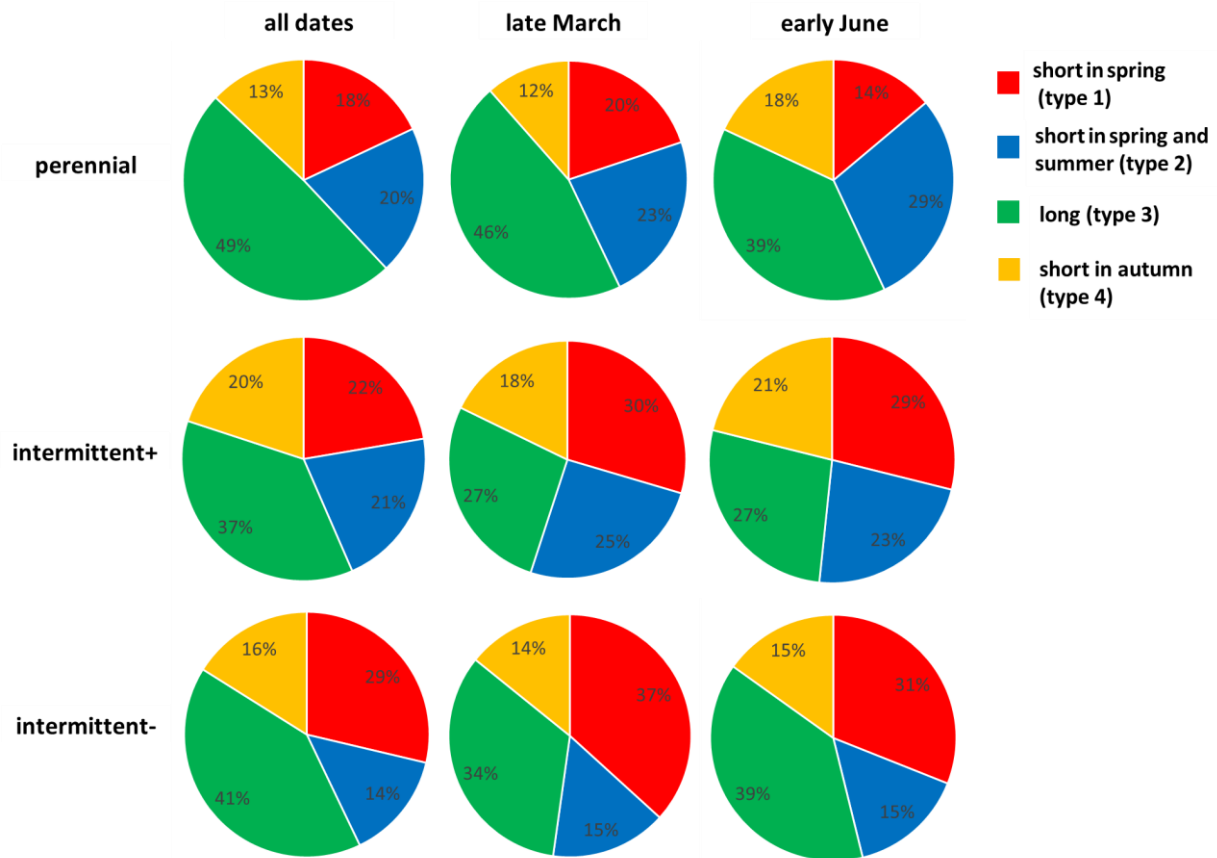


Fig. 45: Mean proportions of flight period types 1-4 of EPT species from Table 16; perennial sites: n=2, intermittent+: n=3, intermittent-: n=3

Flight period types of Trichoptera species

Flight period (FP) types according to Sommerhäuser et al. (1997) were attributed to caddisfly species: short in spring (type A), main FP in summer (type B), long from spring to autumn involving a diapause (type C) and short in autumn (type D). Caddisfly communities from the sampling sites were analysed for differences in these types based on presence/absence of species.

Significant differences were detected between flight period types of perennial and grouped intermittent types in **April** samples (Mann-Whitney U test: $U=549.500$, $p=0.038$; $N=62$). This is mainly due to species with short FPs in spring (type A) that had emerged from intermittent sites, but were still present in perennial sites (e.g. *Lithax obscurus*, *Beraeodes minutus*, *Notidobia ciliaris*).

Differences in proportions of FP types between sites of the hydrological stream types were mainly related to species with long FPs involving diapause (type C) and with main FP in summer (type B) (Table 17, mean proportions of hydrological types in Fig. 46; values from April, June and July see Appendix B: Table B 3 and Table B 4). Proportions of **type C** differed significantly between perennial and int- sites considering all dates (Kruskal-Wallis test: $\text{Chi-square}_2=5.377$, $p=0.036$; pairwise comparison: $p=0.033$; $N=8$) and at $p \leq 0.1$ in March (Kruskal-Wallis test: $\text{Chi-square}_2=5.122$, $p=0.082$; $N=8$), which was related to higher proportions in int- than in perennial and int+ sites (pairwise comparison int- vs. perennial: $p=0.070$, int- vs. int+: $p=0.043$). Proportions of **type B** were significantly higher in perennial compared to int- sites when considering all dates (Kruskal-Wallis test: $\text{Chi-square}_2=5.700$, $p=0.011$; pairwise comparison: $p=0.017$; $N=8$). Proportions of **type D** (short FP in autumn) were higher in grouped intermittent sites when considering all dates (Mann-Whitney U test: $U=11.500$, $p=0.071$; $N=8$). Unlike in

the analysis involving all EPT orders, differences in **type A** (FP in spring) were indistinct between hydrological types based on Trichoptera. However, only four species belong to this type in the samples. Comparisons of proportions **from late April, early June and early July** are not meaningful as numbers of caddisfly species were very low in some samples (e.g. n=2, n=2, n=0, respectively, in Pfaffenbach).

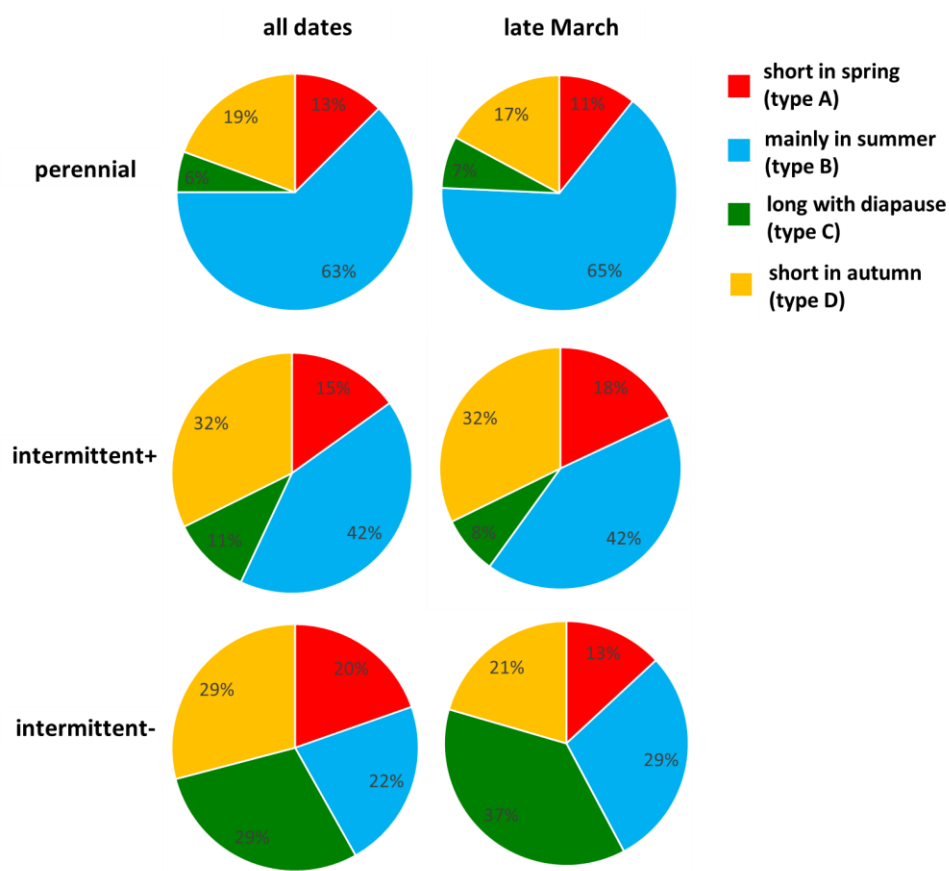


Fig. 46: Mean proportions of flight period types A-D of Trichoptera species; types according to Sommerhäuser et al. (1997); perennial sites: n=2, intermittent+: n=3, intermittent-: n=3

Table 17: Proportions of flight period types (%) of Trichoptera species from combined taxa lists of all dates and samples of late March; types according to Sommerhäuser et al. (1997): type A: short in spring, type B: mainly in summer, type C: long with diapause, type D: short in autumn; * statistically significant differences ($p < 0.05$) between hydrological types tested with Kruskal-Wallis- or Mann-Whitney U test ($N=8$), (*) differences at $p < 0.1$

Hydrol. type	site	Proportion of FP type (March)					Proportion of FP type (all dates)				
		type A	B	C (*)	D	n	type A	B *	C *	D (*)	n
perennial	perGep	21%	50%	14%	14%	14	17%	50%	11%	22%	18
	perGat	0%	80%	0%	20%	10	8%	75%	0%	17%	12
intermittent+	int+Rei	14%	43%	14%	29%	7	8%	46%	15%	31%	13
	int+Gla	13%	38%	0%	50%	8	15%	46%	0%	38%	13
	int+Gat	27%	45%	9%	18%	11	22%	33%	17%	28%	18
intermittent-	int-Rei	0%	44%	33%	22%	9	17%	33%	25%	25%	12
	int-Pfa	25%	0%	50%	25%	4	20%	0%	40%	40%	5
	int-Gla	14%	43%	29%	14%	7	22%	33%	22%	22%	9

4.2.4 DEHLI index

The DEHLI index (Chadd et al. 2017) indicates different extents of in-stream habitat loss due to supra-seasonal droughts in episodically dry streams based on *drought intolerance scores* (DISs) assigned at family or genus level. The index was calculated for all samples.

The numbers of taxa included in the index calculation ranged from 11 (Pfaffenbach, June) to 30 (Gepringbach, July) with an average of 18 taxa. DEHLI values ranged from 3.29 (Pfaffenbach, early July) to 5.58 (int+ Reichelgraben, late March) with an average of 5.00. The DIS taxa list for samples from late March can be found in Appendix C (Table C 2) as well as DEHLI index values of all samples (Table C 1).

Seven of eight sampling sites showed no significant differences neither between sites nor between sampling dates (Fig. 47): All index values indicate that drying proceeded to a maximum of an *early stage of pool shrinking* previous to sampling, even in the sites classified as perennial. Further, DEHLI values of around 5 indicate that mainly rheophilic (e.g. Hydropsychidae, Baetidae, Perlodidae) and *moderately tolerant lentic taxa* (e.g. *Habrophlebia*, Sialidae, Elmidae) have been lost. Pfaffenbach diverged from the other sites in all dates (Fig. 47): DEHLI values between 3 and 4 indicate that communities are characterised by many *highly tolerant lentic taxa* which indicate that drying might have proceeded to *final stages of pool shrinking*.

In summary, the DEHLI index does not differentiate between the hydrological stream types attributed to the investigated sites. Indicated maximum drying conditions prior to sampling do not correspond to observations either.

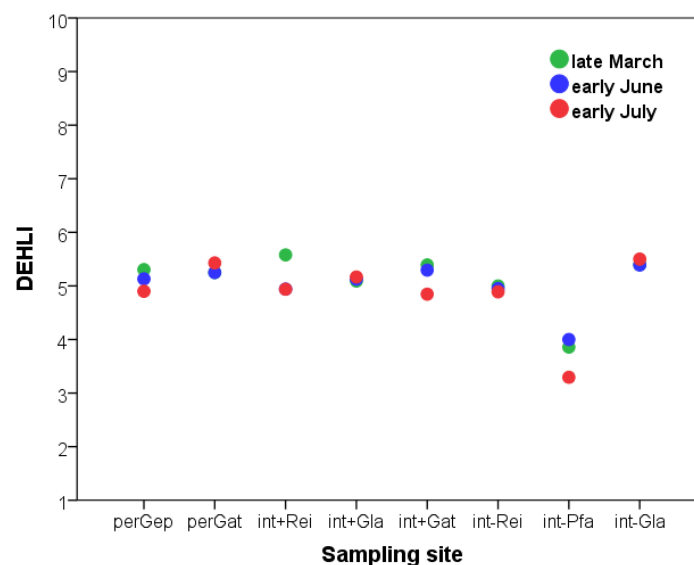


Fig. 47: DEHLI index values (range: 1-10) of all samples

4.2.5 Biodrought index

The Biodrought index (Straka et al. 2019) was developed to indicate antecedent drying events in streams integrating the occurrence of indicator taxa, structural and functional community metrics. It was only calculated for the samples of late March because validity of the index is restricted to March and April. Index values of the int- sites were markedly lower than all others (Fig. 48). One perennial site (perGat) had a clearly higher value than all others, the second was within the range of the int+ sites.

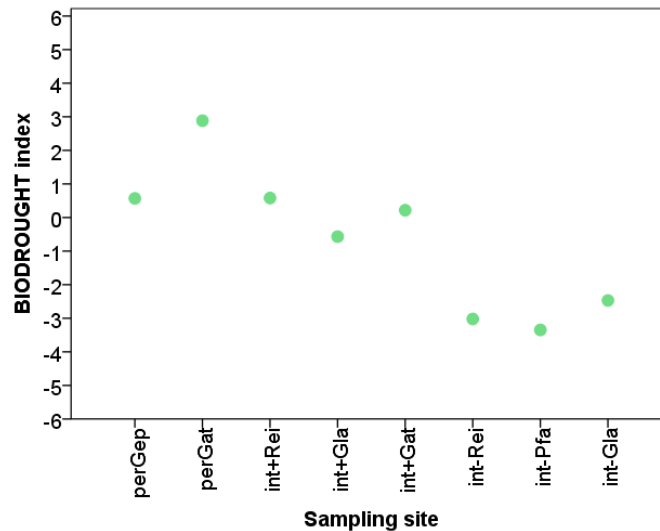


Fig. 48: Biodrought index of samples from late March; values kindly provided by Marek Polášek

Index values transferred to probabilities of belonging to *intermittence categories* (perennial, near-perennial, intermittent) showed high correspondence between all three int- sites and the *intermittent* type defined by Straka et al. (2019) (91, 100 and 98 % probability of being classified *intermittent*; Fig. 49). All three int+ sites resulted to have highest probabilities of belonging to the *near-perennial* type (55, 75 and 89%), but two of them had considerable probabilities for the perennial type (44% in Reichelgraben, 23% in Gatterbach). One of the perennial sites was classified “incorrectly” as it had the highest probability for the near-perennial type (56%). This was due to the low proportion of perennial-flow indicators (Appendix D). However, the other perennial site (Gatterbach) matched perfectly (100% probability).



Fig. 49: Probability of belonging to hydrological regime types according to the Biodrought index; values kindly provided by Marek Polášek

4.3 Life history of *Ironoquia dubia*

4.3.1 Instar definition

A total of 412 larvae were collected from seven different streams in Bioregion 14 from 2017 to 2019, the major part from Pfaffenbach (n=256). The head capsule width of each specimen was measured. Thereby, five larval instars, the usual in Trichoptera, were identified without overlaps in widths (Fig. 50). Head capsule widths ranged from 0.35 to 1.60 mm. Inclusion of unusual or inaccurately measured values was avoided by taking the inner 95% (the 2.5 to 97.5 percentile range) to define instars (Fig. 50, right).

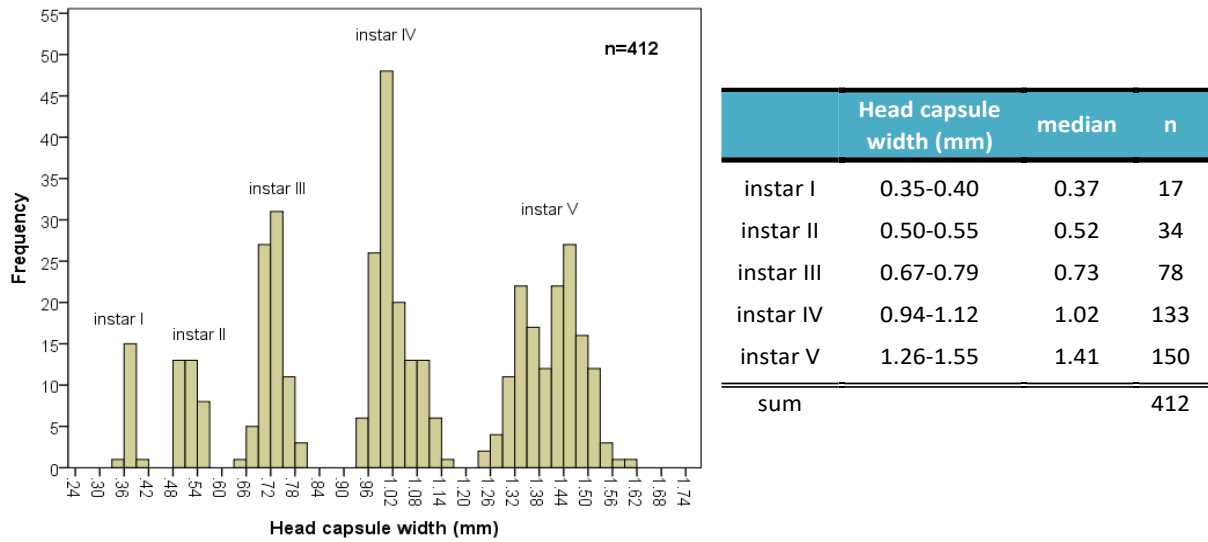


Fig. 50: left: Histogram of head capsule widths of *Ironoquia dubia* (class width: 0.03 mm); right: Instars of *Ironoquia dubia* defined by head capsule width (2.5 to 97.5 percentile range of measured values each instar)

4.3.2 Larval development

Specimens at Pfaffenbach were collected in intervals of approximately 1.5 weeks from early winter to late spring in 2019 to describe larval development. A total of 256 specimens were collected from nine dates and the head capsule widths were measured (Fig. 51).

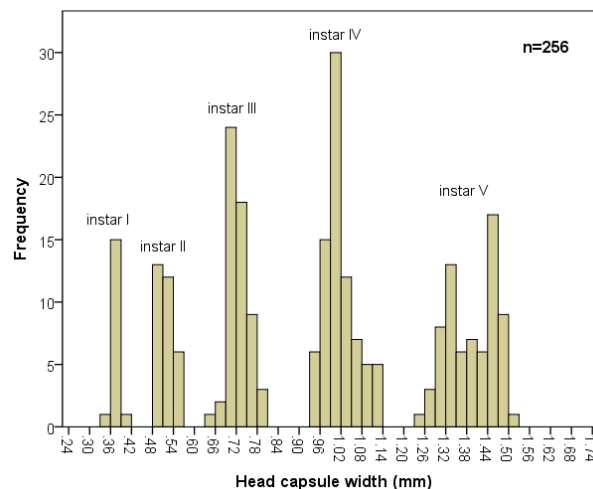


Fig. 51: Histogram of head capsule widths of *Ironoquia dubia* larvae (n=256) collected from Pfaffenbach in winter and spring 2019 (class width: 0.03 mm)

The first larvae (five first instars) were found on 13th February, the last (six fifth instars) on 16th June. No larvae were found later. The mean head capsule widths of collected larvae from each date is plotted in Fig. 52 (left): The developmental rate was relatively constant from the end of February to the end of April and decreased afterwards because most larvae (36 of 41) had reached the last instar on 24th April (Fig. 52, right). Therefore, an approximate larval development of 2 ½ months on average is indicated by these results.

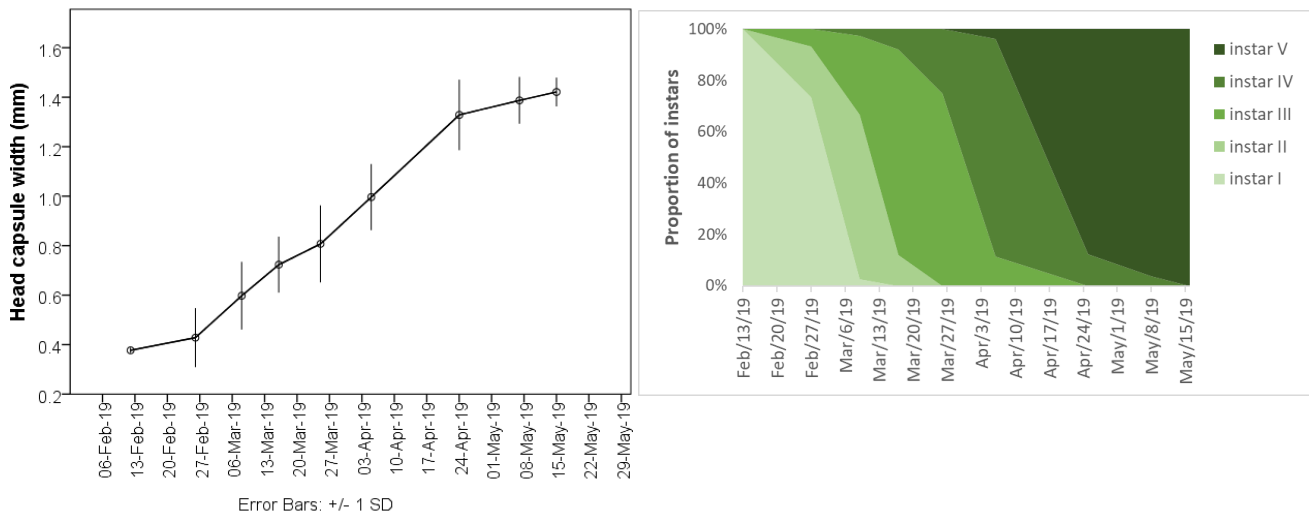


Fig. 52: left: Larval development of *Ironoquia dubia* in Pfaffenbach in spring of 2019 as change of mean head capsule width; N=156; sampling dates: 13th Feb (n=5), 27 Feb (n=15), 9th Mar (n=39), 17th Mar (n=25), 26th Mar (n=20), 6th Apr (n=78), 25th Apr (n=41), 8th May (n=27), 16th May (n=6); error bars indicate standard deviations; right: Proportion of instars

4.3.3 Field observations

Several attempts to collect **adults** with light traps and air net catches in late spring and summer of 2019 remained unsuccessful. Instead, in June, July and August numerous larvae were dug out of the bank substrate along the sampling site of this thesis. On 18th July eight larvae were found whose cases were not closed by silk. In contrast, on 25th August all seven cases found were closed. Two of them were revealed to be larvae by opening the cases, the rest was used for ex-situ observations. Two weeks later (8th September) four cases contained pupae. From the latter two dates, vertical and horizontal distance from the finding spots (n=10) to the water were measured: Vertical distances (height above the water surface) ranged from 15 to 30 cm (mean: 21) and horizontal distances to the wetted stream edge from 5 to 30 cm (mean: 19). The specimens were never buried deeper than an approximate 10 cm into the bank substrate. The bank substrate from which cases were extracted consisted in most cases of non-compacted, loose earth with sometimes small shares of loam situated in rooted areas (e.g. Fig. 53, lower left and right) or moss-covered protruding parts above undercut banks (Fig. 53, above). It was conspicuous that neither in areas with relatively dry nor with (near-)saturated bank substrate specimens were found. However, it can be assumed that some of the habitats potentially become inundated during higher flows in summer due to partly low distances from the water surface at the time of sampling (e.g. Fig. 53, above).



Fig. 53: Finding spots of buried *Ironoquia dubia* larvae and pupae at the sampling site in Pfaffenbach; red rings indicate spots; upper and lower left: 15 pupae on 8th September 2019, lower right: 7 larvae on 18th July 2019

In 2018 **adults** were found between 15th September and 11th October at Pfaffenbach, which was similar to the period of all catches during 2018 and 2019 in Bioregion 14 (11th September to 14th October). In Pfaffenbach, continuous flow started in late October in 2018, i.e. after the observed flight period. Despite an intense search on the dry streambed substrate and on marginal structures only one egg clutch attached to a branch at the stream edge at a height of 30 cm from the bed was found on 15th October 2019. However, this clutch could not be assigned to *I. dubia* with any certainty.

4.3.4 Ex-situ observations

Larvae were collected in early April 2019 from Pfaffenbach and reared in an aerated water-filled aquarium that contained substrate, plant material from the natural aquatic habitat and a terrestrial department (Fig. 23).

Larvae feeding on dead leaves were only rarely observed, whereas they usually started feeding on dead *Siphonurus* larvae as soon as they were offered to them.

To simulate habitat drying, water was not refilled and by the beginning of June all had been evaporated. On 28th May, when the water level had nearly dropped to zero some larvae were buried in the substrate

with the front end of the case pointing outwards. No individuals, however, were observed in the terrestrial structures inside the aquarium. In autumn none of them emerged as adult and all had died.



Fig. 54: Buried larvae of *Ironoquia dubia* in the aquarium, 28th May 2019

In addition, cases found on 18th July (n=7), 25th August (n=7) and 8th September (n=15) in the unsubmerged bank substrate were partly transferred to small containers filled with soil from finding spots and partly to Petri dishes filled with water. All specimens in Petri dishes died. All seven specimen from 18th July were observed to bury into the soil as soon as they were placed into the container. One of them emerged two months later on 26th September. Three adults dug out on 8th September emerged on 25th and 26th September (Fig. 55).



Fig. 55: left: Male of *Ironoquia dubia* emerged from the ex-situ habitat on 25th September 2019; right: Forewing pattern of the specimen

5 Discussion

5.1 Community structure as hydrological regime discriminator

Based on different hydrological states observed during longer periods of low precipitation in summer of 2017 and 2018 in streams of Bioregion 14, three **hydrological stream types** were postulated: perennial streams, intermittent streams that retain pools (int+) and those that dry completely (int-). These stream types have been clearly confirmed by NMS and cluster analysis of macroinvertebrate communities. In both analyses the intermittent type that retains pools (int+) had an intermediate position between perennial and intermittent sites that dry completely (int-). Additionally, numerous functional and structural metrics and indices changed along the hydrological gradient. These gradual changes in MI community composition along flow intermittence gradients have been abundantly reported in literature (e.g. Datry 2012, Bonada et al. 2007a). It could also be shown that communities of sampling sites within the same streams (e.g. 300 m distance between int+ and perennial Reichelgraben) were not closely related to each other (Fig. 36, Fig. 37). This suggests that despite the permanent faunal exchange between nearby water bodies and related overlaps in community composition hydrological characteristics outweigh these similarities.

Total abundance and **taxa richness** were shown to change along the hydrological gradient with int- sites being lowest in both metrics. Statistically significant differences between hydrological types were determined when involving all sampling dates, but not from samples of single dates (exception: total abundances in July). The negative correlation between flow intermittence and both metrics has been abundantly indicated in literature (e.g. Datry 2012, Řezníčková et al. 2007, Straka et al. 2019).

The overall negative correlation of **total abundance** with the hydrological stream types (perennial – int+ – int-) is confirmed by **Spearman correlation** (Table 18), but this correlation is not significant in the March samples. Straka et al. (2019) used total abundance together with body flexibility of taxa and hydrological regime indicators (*Biodrought index*) for separating hydrological stream types based on spring samples. The results of this thesis suggest that this metric might be problematic to separate all three tested hydrological stream types in Bioregion 14 based on samples from only one season. Site-specific temporal dynamics of *Gammarus fossarum* during the sampling period from spring to early summer had a high impact on total abundance in int+ and perennial sites so that trends between these stream types only seem to become evident if samples at sites are taken at different seasons.

Table 18: Metrics that show strongest and statistically significant correlations (Spearman) with the hydrological classification (1: perennial, 2: int+, 3: int-); used metrics correspond to standard metrics in ECOPROF (Moog et al. 2018); number of sites in late March: 8, all dates: 24; r_s : Spearman correlation coefficient; * correlations significant at $p \leq 0.05$, ** significant at $p \leq 0.01$; ¹ results are not considered relevant due to low taxonomic resolution

Metric type	Metric name	Metric description	r_s late March	r_s all dates
abundance	P-IND	abundance of Plecoptera	0,945**	
	COL-IND	abundance of Coleoptera	-0,824*	-0,640**
	GES-IND	total abundance		-0,666**
	AMP-IND	abundance of Amphipoda		-0,634**
	CRU-IND	abundance of Crustacea		-0,603**
	T-IND	abundance of Trichoptera		-0,596**
relative abundance	% P-IND	% Plecoptera individuals	0,945**	
	% E-IND	% Ephemeroptera individuals	0,819*	0,659**
	% EP-IND	% EP individuals	0,819*	0,684**
	% EPT-IND	% EPT individuals	0,743*	0,510*
taxa richness	E-TAXA	no. of Ephemeroptera taxa	-0,956**	-0,647**
	COL-TAXA	no. of Coleoptera taxa	-0,927**	-0,716**
	EPT-TAXA	no. of EPT taxa	-0,882**	-0,527**
	GES-TAXA	total taxa richness	-0,803*	-0,452*
	T-TAXA	no. of Trichoptera taxa	-0,735*	-0,527**
taxonomic composition	% OLIDIP-TAXA ¹	% Oligochaeta + Diptera taxa	0,882**	0,463*
	% D-TAXA ¹	% Diptera taxa	0,819*	0,42*
	% T-TAXA	% Trichoptera taxa	-0,735*	-0,455*
	% P-TAXA	% Plecoptera taxa	0,722*	
	% EPT-TAXA	% EPT taxa		-0,447*

Contrary to prediction, **temporal dynamics** of total abundances in the course of spring and early summer were not more pronounced in intermittent sites. On the contrary, the two perennial sites were characterised by a strong increase from June to July. This was due to *Gammarus fossarum*, which is known to have peaks in reproductive activity in spring and early summer (Pöckel et al. 2003). In this period the increase in *G. fossarum* did not happen in intermittent sites, except for Glauningbach (int+), which had generally relatively low abundances of this species. Int- site had a gradual reduction in total abundance in common, which, however, was also the case in two int+ site.

In summary, a potential source of separating perennial from intermittent streams based on temporal dynamics in total abundance is an increase in late spring/early summer, which only occurs in perennial sites.

Abundance of *Gammarus fossarum* was significantly and negatively correlated with the hydrological regime, across all sampling dates (Table 18: metric *AMP-IND*). Its gradual reduction along the hydrological gradient can be explained by relatively high persistence in remnant pools, which is lower in moist subsurface interstices (Pařil et al. 2019; see chapter 5.3.1).

The strongest correlation ($r_s=0.945$) in March among standard metrics from ECOPROF (Moog et al. 2018; Table 18) was determined for the **abundance and relative abundance of Plecoptera**, which, however, was not significant across all sampling dates due to emergence during spring (e.g. *Nemoura cinerea*, *Isoperla tripartita*). Leitner & Graf (2018) also reported the relative abundance of Plecoptera as strongest correlation among abundance metrics in samples from Bioregion 14. **Abundance of Coleoptera** shows

the second strongest correlation in March, which was also significant across all sampling dates. Additionally, relative **abundance of Ephemeroptera** and **combined with Plecoptera** are also significantly and positively correlated with the hydrological types in March and across all samples.

The overall significant and negative correlation of **taxa richness** along the hydrological regime types was confirmed by Spearman correlation (Table 18, metric *GES-TAXA*). The correlation is distinctly stronger in the March samples, which suggest that taxa richness might be a useful discriminator of hydrological stream types for spring samples in Bioregion 14. However, differences were relatively slight in the March samples between all hydrological stream types. Higher taxonomic resolution has the potential to enlarge differences. Based on more than 100 samples from the Czech Republic Straka et al. (2019) found minor differences between perennial and the near-perennial sites (equivalent to int+), but taxa richness was distinctly lower in int- sites. Therefore, it was not used for the separation of the three hydrological stream types.

Among the tested standard metrics from ECOPROF, the **number of Ephemeroptera, Coleoptera and EPT taxa** show the strongest correlations in the March samples related to richness metrics with int- sites having the lowest numbers of these taxonomic groups (Table 18). These metrics also statistically significantly correlate across all sampling dates.

Proportions of taxa (metric type *taxonomic composition* in Table 18) most strongly correlated in the **sum of Diptera and Oligochaeta** and **Diptera** with int- streams having highest values. However, taxonomic resolution was low in these taxa (Oligochaeta: order level, Diptera: family level). Surprisingly, Leitner & Graf (2018) also found a strong correlation with the proportion of Diptera taxa, but with lowest values in intermittent streams. This is most probably due to different taxonomic resolution in Chironomidae and Simuliidae (family vs. species level). **Proportions of Trichoptera, Plecoptera and EPT taxa** also showed significant correlations in the March samples and the importance of these metrics are thought to be higher as taxonomic resolution was mostly species level.

5.2 Functional community traits as hydrological stream discriminators

5.2.1 Ecological traits

In addition to structural, functional metrics available in ECOPROF (Moog et al. 2018) were also subject to Spearman correlation analysis. The strongest correlation resulted from **passive filter feeders** (Table 19). However, proportions of these feeding type were based on single taxa (mainly *Hydropsyche* spp.) and were low (maximum 4.1%, mean 0.4%). Therefore, this result must be interpreted with caution even if it could be justified by the need of flow for passive filter feeders, which is permanently available in perennial in contrast to intermittent streams. **Grazers** and **shredders** are also significantly and negatively correlated with the hydrological gradient in both late March and across all samples. This correlation in the case of grazers was also reported from Bioregion 14 by Leitner & Graf (2018). They also found detritus feeders to be significantly associated with intermittent streams, which was in line with findings in Bonada et al. (2007a). In this thesis, this was also the case involving all samples, but the correlation was relatively weak ($r_s=0.502$). Two other metrics related to feeding types were significantly and positively correlated with the hydrological gradient: the RETI and PETI. They indicate the proportion feeding types that are typical of rhithral (shredders and grazers, RETI) and potamal (filter and detritus feeders, PETI) sections (IHG 2019). The RETI was negatively correlated with intermittence indicating higher proportions of potamal-like feeding types in intermittent compared to perennial sites. The PETI confirmed this correlation by giving the converse result, therefore both are in line with the correlations found in the single feeding types.

One metric related to saprobic preferences of species was correlated with the hydrological types: Xenosaprobic preferences were highest in int- streams, which is surprising because remnant pools of intermittent streams typically have high saprobity and accordingly low oxygen concentration (e.g. Boulton 2003). Several species that were designated as indicator species of intermittent streams have indeed considerable xenosaprobic preferences: e.g. *Bythinella* sp., *Amphinemura standfussi*, *Chaetopteryx* spp., *Lithax obscurus*. However, this correlation must be interpreted with caution as *Siphonurus armatus*, the most abundant EPT species in int- sites (e.g. 1600 – 2400 ind./m² in March), is not classified, but probably has a mainly beta-mesosaprobic preference (Bauernfeind et al. 2017).

Table 19: Metrics that show the strongest and statistically significant correlations (Spearman) with the hydrological classification (1: perennial, 2: int+, 3: int-); used metrics correspond to standard metrics in ECOPROF (Moog et al. 2018); number of sites in late March: 8, all dates: 24; r_s : Spearman correlation coefficient; * correlations significant at $p \leq 0.05$, ** significant at $p \leq 0.01$; ¹ result is not considered relevant due to low proportions of this feeding type in all samples

Metric type	Metric name	Metric description	r_s late March	r_s all dates
feeding types	PFIL ¹	passive filter feeders	-0,918**	-0,781**
	WEI	grazers	-0,756*	-0,592**
	ZKL	shredders	-0,756*	-0,549**
	PETI	Potamon feeding type index (prop. of filterers and detritus feeders)	0,756*	0,662**
	RETI	Rhithron feeding type index (prop. of grazers and shredders)	-0,756*	-0,662**
	DET	prop. of detritus feeders		0.502*
saprobity	XENO	Prop. of xenosapr. preference	0,718*	0,418*

5.2.2 Biological traits

Adaption traits

Based on taxa **classifications in Tachet et al. (2010)** traits promoting resistance to intermittent flow (respiration through plastron and spiracles, resistance forms) differed statistically significantly between hydrological stream types based on samples from late March and taxa lists that contain all sampled taxa at sites. In contrast, traits enabling resilience from intermittent flow were not significantly different. This result is consistent with the finding that in temperate climates, community resistance to flow intermittence compared to resilience is more important than in desert and Mediterranean streams (Stubbington & Datry 2013). Subsequently differences in community resistance are expected to be higher between different hydrological regimes in temperate climates.

Affinity to resistance traits of taxa increased along the hydrological gradient (perennial – int+ – int-), which is confirmed by Spearman correlation (March: $r_s=0.189$, $p=0.008$, $N=195$; all dates: $r_s=0.147$, $p=0.001$, $N=482$). Accordingly, **mean affinity scores per taxon in the community** also increased along the hydrological gradient (Spearman correlation: March: $r_s=0.731$, $p\leq 0.05$; all dates: $r_s=0.882$, $p\leq 0.01$; $N=8$). This metric was tested to function as hydrological regime discriminator. However, values overlapped between perennial and int+ sites and between int+ and int- sites (Fig. 41), but not between perennial and int- sites. This result indicates that resistance to ceased flow in remnant pools or the dry stream bed is a more important feature for survival in int- streams than in int+ streams.

In summary, high mean affinity scores per taxon for traits providing resistance to intermittent flow based on classifications in Tachet et al. (2010) might be a good metric to recognise intermittent flow (int+ or int-), but low values also seem to result from int+ sites. Due to the low number of perennial streams in this thesis ($n=2$) the range of the metric in this hydrological type could not be shown appropriately and should be determined in future studies.

The importance of resistance traits for separating hydrological types was also found in an **analysis according to Leigh et al. (2015)**, which uses taxa classifications at family and sometimes genus level. In this analysis, statistically significant differences in proportions of strategies additionally resulted from resilience and combined strategies, which were most distinct in the March samples. All three strategy categories changed along the hydrological gradient (resilience lowest, resistance highest and combined highest in int- sites) and Spearman correlation indicates significance in the March samples (resistance: $r_s=0.945$, $p\leq 0.001$, resilience: $r_s=-0.945$, $p\leq 0.001$, combined: $r_s=0.718$, $p=0.045$; $N=8$). Despite statistical significance these differences and gradients are not regarded suitable to function as hydrological regime discriminators because they are relatively slight (Fig. 43). Additionally, even perennial streams resulted to have high proportions of taxa with adaption to intermittent flow (85% on average) which seems doubtful and might be due to the coarse taxonomic level. Williams (2006, p. 144) highlights the importance of species level when comparing intermittent and perennial water bodies within a biogeographic region.

Therefore, the trait analysis according to Leigh et al. (2015) is not recommended to be used as hydrological regime discriminator in Bioregion 14 based on own findings.

Flight periods

Four analyses were conducted to test the prediction of **earlier flight periods (FPs)** in intermittent than in perennial streams. All of them showed trends towards earlier FPs of EPT species along the hydrological gradient, but differences were partly low. Three of the analyses were based on presence and absence

of EPT species in the benthic samples: the average species-specific flight period index (FPI), FP types of EPT species and FP types of Trichoptera. FP types of EPT species showed most distinct differences: On average, 20% of species in perennial, 30% in int+ and 37% in int- sites belonged to the spring FP type in samples from late March. The analysis based on species' abundances in the benthic samples and seasonal occurrence of FPs indicated in literature (the community FPI) suggests that individuals inhabiting int- sites have on average around 25 days earlier FPS than in perennial streams. No earlier FPs resulted from int+ streams, which suggests that remnant pools are a significant refuge for aquatic stages of MIs during summer (see Chester & Robson 2011). The analysis of caddisfly species' FP types only showed minor differences along the hydrological gradient in species with early FPs in spring (on average 20% in int-, 15% int+ and 13% in perennial sites).

Contrary to the hypothesis, species with **long flight periods** had higher proportions in perennial than in intermittent streams. A possible explanation for this might be the existence of different types of long flight periods that were not distinguished. In the consulted database (freshwaterecology database, category "duration emergence period") long flight periods are indicated for both long-lived diapausing adults of caddisfly species and for shorter-lived species without diapause but with asynchronous emergence. In the first group, emergence (and aquatic development) in spring happens synchronously and therefore has a short duration according to this conception. This type was indeed shown to have higher proportions in int- sites (on average 29% of caddisfly species in int-, 11% in int+ and 6% in perennial sites). The reason for the high proportions of EPT species with long flight periods in permanent streams was the presence of many caddisfly and mayfly species with presumably asynchronous emergence, e.g. *Ephemerella ignita*, *Baetis rhodani*, *Athripsodes bilineatus*, *Hydropsyche* spp.

In summary, no evidence was found for long flight periods in general as significant bet-hedging strategy (risk spread among the progeny) for survival of intermittent flow in Bioregion 14. Autecological information on synchronicity of emergence rather than on duration of flight period are thought to have a higher potential to separate hydrological stream types. However, risk spreading might only be significant strategy in intermittent streams with unpredictable hydrological regime as indicated in Lytle & Poff (2004).

Short flight periods in autumn were also included in the analyses. Unlike indicated in Sommerhäuser et al. (1997), this FP type was not restricted to perennial streams, on the contrary, there were slightly higher proportions in intermittent sites (on average 29% of caddisflies in int-, 32% in int+ and 19% in perennial sites). For example, *Isonychia dubia* and *Chaetopteryx rugulosa* only occurred in intermittent sites. Slightly higher proportions of autumn fliers in intermittent sites also resulted from the analysis of all EPT species (on average 13% perennial, 20% in int+, 16% in int-). As in Sommerhäuser et al. (1997) caddisfly species with main **flight period in summer** without adult diapause dominated in perennial sites (on average 63% in perennial, 42% in int+ and 22% in int- sites) and seem to be good indicators for separating hydrological types.

The identified differences in FP types between hydrological stream types are also found in the indicator taxa designated by the *Indicator species analysis*: Intermittent stream indicators have predominantly long FPs and FPs in spring. This pattern is reflected in FPs of both Trichoptera and all EPT species (Fig. 56). Similar numbers of species with short FPs in autumn and main FP in summer are among perennial and intermittent flow indicators.

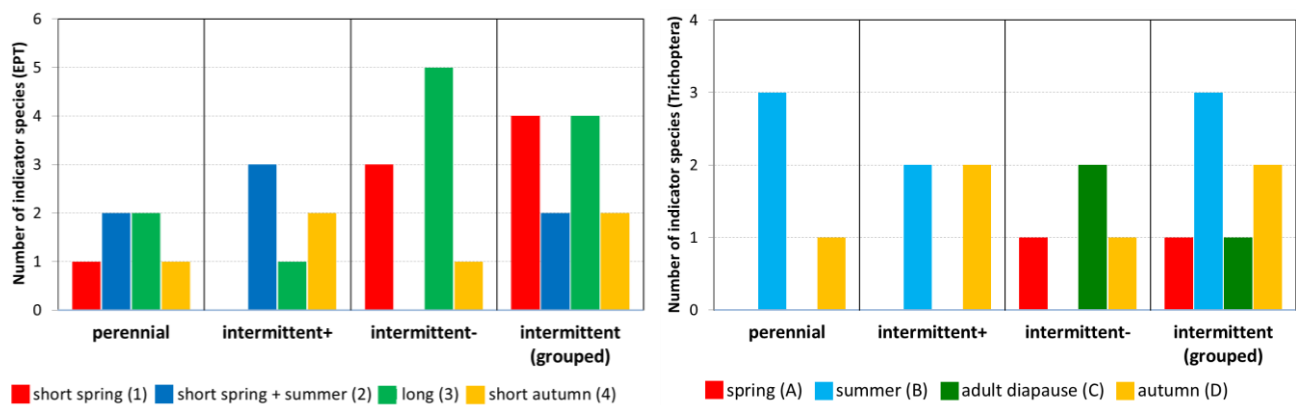


Fig. 56: Flight period types of EPT indicator species (designated by Indicator species analysis) in hydrological stream types; left: flight period types of all EPT indicator species, right: flight period types (sensu Sommerhäuser et al. 1997) of Trichopteran indicator species

5.2.3 Existing indices (DEHLI, Biodrought)

The **DEHLI index** was developed to reflect responses of MI communities to supra-seasonal stream drying of episodically-dry streams (that only dry at extreme events) based on the concept of Boulton (2003, see chapter 2.4.2). No trends in index values between hydrological stream types could be found (Fig. 47). Additionally, for all sites index values indicate that stream drying has proceeded to *early stages of pool shrinking*, which does not correspond to observations.

It is not surprising that the DEHLI index could not show to differentiate between the hydrological stream types tested because it was designed to indicate responses of perennial and not specifically intermittent stream (IS) communities. IS communities respond to seasonal flow cessation in a different way than perennial stream communities. The former communities have specific adaptations evolved in response to seasonal flow cessation, whereas species of the latter are likely to die at different stages of the stream drying process, which is intended to be indicated by the DEHLI index.

A major limitation of the index could also be the high taxonomic level (family and genus) on which drought intolerance scores are attributed. Numerous species found to be even indicators of intermittent streams in this thesis diverge completely from the family classification used in the DEHLI index. For example, *Rhyacophila hirticornis*, *Electrogena ujhelyii*, *Brachyptera risi* and *Lithax obscurus* have high intolerance scores (10, 10, 9 and 8, respectively) indicating virtually no tolerance to reduced flow. This clearly does not apply to these species, but strongly influences the index value. It is generally known that adaptations to flow intermittence can be restricted to specific species (habitat specialists) within higher taxonomic units (Williams 2006, p. 144).

The same drying extent indicated in perennial as in intermittent sites despite the presence of numerous perennial flow indicators confirms that the application of the DEHLI index might not be useful to recognise ISs in Bioregion 14. However, it cannot be excluded that the index is sensitive to flow cessation that might have happened in the years before hydrology was observed. A general trend towards flow cessation or strongly reduced discharge in extremely dry years in Bioregion 14 could exist and should be investigated in future.

The **Biodrought index** (Straka et al. 2019) was developed to recognise antecedent stream drying based on the analysis of MI communities. It integrates total abundance (structural metric), the proportion of species with high body flexibility (biological metric) and the proportion of indicator taxa to calculate a multimetric index from spring samples. Index values can be transformed into probabilities of belonging to three hydrological stream types (perennial, near-perennial, intermittent). These types largely

correspond to those defined in this thesis. Most EPT species found to be hydrological stream indicators in Bioregion 14 are characterised as such for the index calculation.

Classification based on the BI matched the determined hydrological type in seven out of eight sites (Fig. 49). Gepringbach, observed to be perennial in 2018 and 2019, had higher probabilities of belonging to the near-perennial type, but was close to being classified correctly (43% perennial, 56% near-perennial and 1% intermittent; Fig. 49). The main reason for this mismatch was the presence of both perennial (e.g. *Hydropsyche* spp., *Baetis* spp., *Goera pilosa*) and intermittent stream indicators (e.g. *Habrophlebia fusca*, *Lithax obscurus*, *Beraeodes minutus*, *Limnephilus lunatus*). Adult catches of *Zwicknia acuta*, *Chaetopteryx rugulosa* and *Isonychia dubia* even extend the number of intermittent stream indicators. A likely explanation for this is that flow cessation had occurred before the observation period and perennial flow specialists had only partially recolonised the stream. Another cause could be the long deep pool-like sections with hardly visible flow which might favour species with littoral preferences able to survive in remnant pools of intermittent streams. On the other hand, the riffle sections are likely to favour perennial flow specialists. These strongly contrasting conditions within the same stream might enable the coexistence of both perennial and intermittent stream specialists.

Despite the mismatch the index is especially promising as it recognised the intermediate int+ (near-perennial) sites with high certainty. More importantly, its power is expected to be even higher if more taxonomic groups are determined at species level (e.g. Chironomidae). Additionally, the index can be adapted to faunal specifics of bioregions by changing the classifications of hydrological regime indicators (reduce/increase reliability values, add/remove indicators) of predefined ones. For example, for application in Bioregion 14 *Chaetopteryx rugulosa* could be included into the list of intermittent flow indicators and would be assigned a reliability value of 1 (range: 1 and 2). The indicator taxa metric could be calculated without problems involving this adaption.

5.3 Adaption strategies of selected intermittent stream species

5.3.1 *Gammarus fossarum*

Gammarids including *Gammarus fossarum* have often been reported from Central European intermittent streams (e.g. Bohle & Potabgy 1992, Meyer & Meyer 2000, Meyer et al. 2004, Straka et al. 2019). Bohle & Potabgy (1992) and Poznańska et al. (2013) indicate resilience strategies in *Gammarus pulex* based on its ability of upstream migration to permanent springs and downstream migration to perennial stretches. In Bioregion 14 *Gammarus fossarum* is likely to be able to migrate to small permanent spring trickles that sometimes follow intermittent stream courses. Specimens were found in such trickles associated with, for example, typical spring-dwelling Trichoptera species such as *Ernodes articularis*, *Ptilocolepus granulatus*, *Potamophylax nigricornis*, *Wormaldia occipitalis* or *Crunoecia irrorata*, but also *Notidobia ciliaris*.

Iversen et al. (1978) report that *Gammarus* species are able to survive in remnant pools and to withstand drying that does not exceed approximately one month. Vertical migration into the saturated hyporheic interstitial was found to be an explanation of the drying resistance in *G. pulex* and *G. fossarum*, but mortality is high in this habitat (Vadher 2017, Pařil et al. 2019).

In this thesis *G. fossarum* was present in all sites, thus not being defined as indicator species by *Indicator species analysis*. But strong differences in temporal dynamics between hydrological stream types in late spring and early summer might be a potential indication for recognising these types (see chapter 5.1.): Perennial streams were characterised by strong increases from late spring to early summer, int+ sites by some reduction, which were stronger in int- sites.

5.3.2 *Siphonurus armatus*

Siphonurus armatus has a distribution ranging from France eastwards through Central Europe to the Balkans, including Great Britain, Ireland and Fennoscandia.

Siphonurus larvae were found in all intermittent, but not in perennial streams. Pronounced differences in abundance were stated between int+ and int- streams. In all int- streams it was the most abundant EPT taxon in late March by far (1580 – 2440 individuals/m²) and still occurred in samples from early June in low abundances. In int+ sites the highest abundance was only 144 individuals/m². Accordingly, *S. armatus* resulted to be a statistically significant intermittent-flow indicator.

In literature, *S. armatus* is described as habitat specialist of summer-dry standing and running waters with low flow velocity. Egg diapause is the species' pivotal trait for survival in intermittent streams, however, its known life history is known insufficient (Bauernfeind & Soldán 2012). Nevertheless, phenological observations have been described for example in Germany. Fiedler & Bohle (1994) found low numbers of early instars in November and higher numbers as late as in February. After rapid growth in early spring a short main emergence period was observed at the end of May, but low numbers of larvae emerged until early July. They concluded that the duration of egg diapause might differ markedly between individuals (bet hedging strategy). In contrast, Gillies (1990) found first instar larvae only as late as in February in a stream in England. In the course of this thesis first larvae were found only on 6th February 2019. Adults were caught between 8th May and 14th June 2019.

In May 2019 females were observed to oviposit during flight by touching its surface with the rear end of their abdomen. This behaviour corresponds to indications in previous literature (e.g. Elliot & Humpesch

1980, Fiedler & Bohle 1994). It was striking that swarms of females only oviposited at one specific place at the sampling site where flow was turbulent due to tree roots extending throughout the channel width (Fig. 22).

No indications for the presence of other *Siphonurus* species potentially occurring in Bioregion 14, e.g. *S. aestivalis* or *S. croaticus*, were found in the course of this thesis based on larval and adult specimens (determinations by Ernst Bauernfeind). Previously, individuals from Bioregion 14 determined to *S. aestivalis* based on morphology (Leitner et al. 2017) were revealed to be *S. armatus* using genetical analysis (Sumper 2019). Nevertheless, the syntopic occurrence of additional *Siphonurus* species cannot be excluded because the major part of larval specimens could not be determined at species level due to early instars. Moreover, Bauernfeind & Soldán (2012) state that *S. armatus* and *S. aestivalis* occur syntopic in most cases with the minority of specimens being the latter species.

5.3.3 *Habrophlebia fusca* and *H. lauta*

While the occurrence of *H. fusca* in intermittent streams has been reported in several geographical regions: e.g. Germany, Spain, North Africa, Austria and the Czech Republic (Bauernfeind & Soldán 2012), *H. lauta* seems to be limited to perennial streams. Řezníčková et al. (2010) reported *H. fusca* from an intermittent stream in the Czech Republic and attributed numerous advantageous ecological and biological traits for survival in intermittent streams to the species. These traits are broad substrate- and habitat range, high degree of adaption to flow (rheo- to limnophilic), wide distribution, detritivorous and omnivorous feeding type etc. In contrast, they only found *H. lauta* in a perennial stream and classify traits as less advantageous. In Austria, *H. fusca* was reported from intermittent streams in the Vienna Woods (Philipp & Forster 2000, Dienstl 2019). Detailed information on life-history strategies to survive intermittent flow (e.g. dormant stages) do not seem to exist.

In this thesis, both *Habrophlebia* species were found as larvae and adults and a clear trend towards a preference of *H. fusca* for intermittent streams was detected (in int- Pfaffenbach and Glauningbach only *H. fusca* was determined). However, it was also present in lower abundances in both perennial streams. *H. lauta* occurred in all three int+ streams syntopic with *H. fusca*. Its syntopic occurrence with *H. fusca* in int- Reichelgraben could be explained by the vicinity to the int+ site in the same stream (approximately 300 m distance).

Bauernfeind & Humpesch (2001) indicate a flight period from June to August for *H. fusca* and a more extended one (April to October) in *H. lauta*. Imagines and subimagines of *H. fusca* were found from 2nd June to 23th July at the sampling sites, but high abundances of *H. fusca* larvae were still found at the end of this period in remnant pools of int- Glauningbach (Fig. 57). This behaviour corresponds to observations in Germany, where larvae were found in even tiny and muddy puddles in the drying streambed in summer (Fiedler & Bohle 1994, Foltyn et al. 1996). Abundances in late March were very low, which is in accordance with larval findings from April to early August in Fiedler & Bohle (1994). Therefore, egg diapause is likely but was seemingly never investigated. Polyvoltinism does not seem to be known (Bauernfeind & Soldán 2012).



Fig. 57: Larvae of *Habrophlebia cf. fusca* in a remnant pool in int- Glauningbach on 23 July 2019

5.3.4 *Electrogena ujhelyii*

E. ujhelyii occurred in all sites except Pfaffenbach (int-) and was relatively abundant in int+ sites. It preferably inhabits the hypocrenal and epirhithral zone and unlike most Heptageniidae can also be found among submerged roots and logs (Bauernfeind & Soldán 2012). Few researchers report about its occurrence in intermittent streams. Dienstl (2019) and Battisti (2003) found it in the Vienna Woods, Řezníčková et al. (2010) in the Czech Republic. The latter researchers state that life history of this species is not well known. They found few favourable traits for survival in intermittent streams, but the ability to obtain high abundances, relatively long adult life span, respiratory adaption by movable gills and asynchrony in emergence by developing several cohorts are indicated as favourable. The latter trait matches the extended flight period from April to September indicated in Bauernfeind & Humpesch (2001). Own adult catches dated between 8th May and 18th July in 2019 with larvae still being present in samples from early July in five intermittent sites. Therefore, these findings suggest bet-hedging as significant strategy for survival in intermittent streams of Bioregion 14.

5.3.5 *Zwicknia* spp.

The genus *Zwicknia* has recently been erected establishing eight new species previously combined in *Capnia bifrons* (Murányi et al. 2014, Boumans & Murányi 2014, Reding et al. 2016, Vinçon & Reding 2018: *Z. bifrons*, *Z. acuta*, *Z. kovacsi*, *Z. rupprechti*, *Z. westermanni*, *Z. komica*, *Z. ledoarei* and *Z. gattolliati*). *Z. acuta* is widespread in Central and eastern Europe, *Z. bifrons* is additionally present in Scandinavia (Murányi et al. 2014). The other *Zwicknia* species have narrower distributional ranges.

The association with intermittent streams of *Zwicknia* has often been reported (e.g. Foltyn et al. 1996, Westermann 2003, Wolf 2007), but has to be reassessed based on the newly defined species (Boumans & Murányi 2014). For example, Wolf (2016) found *Z. acuta* and *Z. bifrons* in both perennial and intermittent streams and Philipp & Forster (2000) determined *Zwicknia* from both hydrological types in the Vienna Woods, where both *Z. acuta* and *Z. bifrons* were revealed to occur later (Wolfram Graf, personal communication). In this thesis,

Z. acuta was also found as adult at perennial Gepringbach.

In **Austria**, *Z. bifrons* and *Z. acuta* have been recorded so far (Graf et al. 2017), but *Z. westermanni* and *Z. rupprechti* were found relatively close to Austria (Germany, Hesse; Wolf 2016). In the course of this thesis *Zwicknia acuta* was revealed to be a typical inhabitant of intermittent streams in the Eastern Styrian Grabenland (Fig. 60). One male specimen with distinctly shorter forewings corresponding to *Z. bifrons* was found syntopic with *Z. acuta* at one intermittent stream (Schweinsbach, Hütt, 26th March 2019). However, additional specimens should be captured to confirm the determination. Attempts to find larvae and adults through qualitative sampling in the Western Styrian Hügelland were not successful.



Fig. 58: Male of *Zwicknia acuta* (Schweinsbach, St. Nikolai ob Draßling, 15th March 2019)



Fig. 59: Female of *Zwicknia* cf. *acuta* at Glauningbach (9th March 2018)

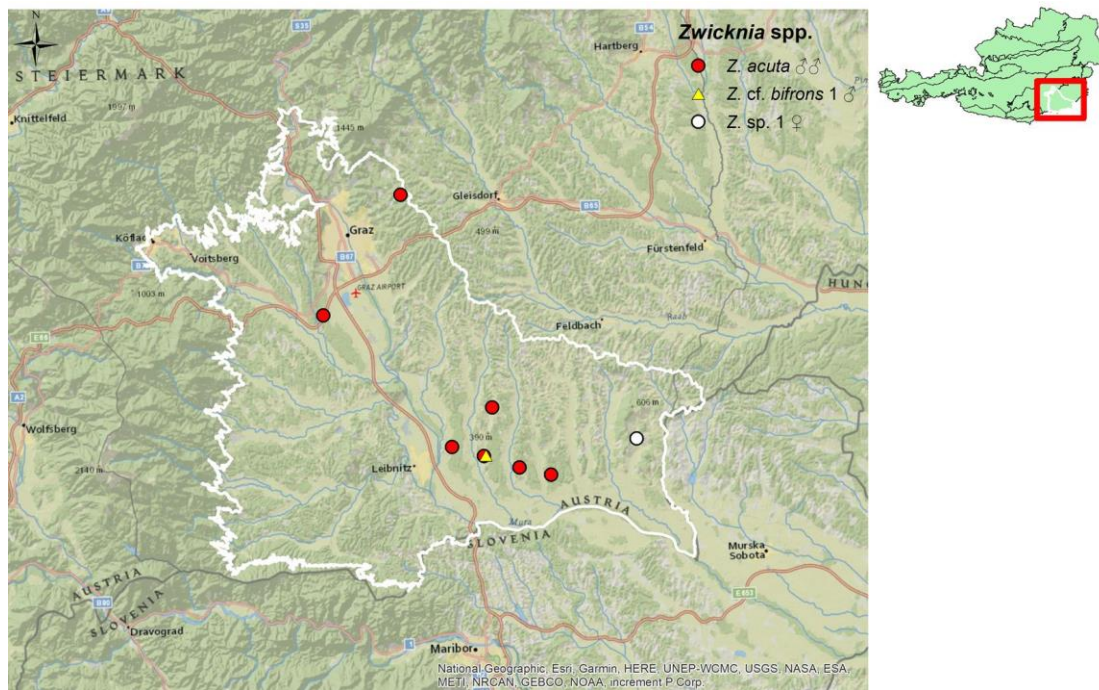


Fig. 60: Distribution of *Zwicknia acuta* in Bioregion 14 (white border)

In the present study, adults of *Zwicknia acuta* were found at five of six intermittent sites, but larvae only at three of them, probably due to preceding emergence. Larvae were only found in the March samples, adults between 8th March and 6th April. *Zwicknia* was designated as a non-statistically significant indicator of intermittent streams (grouped int+ and int- sites) by *Indicator species analysis*, but it would have probably been significant if sampling had been executed in late winter.

Life history of the former *Capnia bifrons* taxon was described by Khoo (1964, 1968a) from populations of the United Kingdom, where only *Z. bifrons* occurs according to recent studies. Emergence of the univoltine species happens very early in the year (from February to May). Females are **ovoviviparous**, which means that eggs hatch immediately after oviposition. Fourth or fifth instar larvae (14 to 16 instars in total) enter diapause for three to four months in June in the hyporheic interstitial. Diapause is accompanied by morphological modifications, of which the head fixed towards the body is most conspicuous (Fig. 61). Diapause terminates between September and November, i.e. when continuous flow usually resumes in intermittent streams. Laboratory experiments showed that diapause is induced by temperatures higher than 9 °C during the first three instars and by increasing photoperiod if temperature fails to increase. Temperature plays a decisive role in termination of diapause as well: Low temperature shortens the diapause duration and is therefore likely to synchronise the further development with flow resumption in autumn.

Larval diapause in *Zwicknia* corresponds to eudiapause sensu Müller (1970; see chapter 2.4.3).

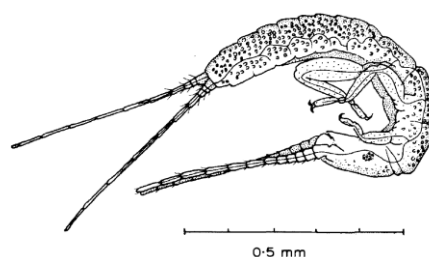


Fig. 61: Fourth instar diapausing larva of *Capnia bifrons* (from Khoo 1968a)

5.3.6 Other Plecoptera species

Larval catches of *Nemoura* were not identified at species level due to delicate distinctive features indicated in literature. However, three species of the genus could be obtained by adult catches at the sampling sites: *Nemoura cinerea*, *N. sciurus* and *N. marginata* (Table 20), with the former in highest numbers. Larvae of the genus were found at all sampling sites except for perennial Gatterbach. Abundances were clearly highest in int- sites, that is why it was designated as statistically significant indicator of int- sites in March. In late April abundances were already substantially reduced in int- sites and close to zero in June and July samples due to emergence.

N. cinerea has frequently been associated with intermittent streams (ISs). It is known to be a generalist species and was even reported to complete its life cycle in an under-cup of a flower pot (Malicky 1989). It has drying-resistant eggs that remain in diapause in the moist stream bed during summer and first larvae hatch in winter (Khoo 1964, in Foltyn et al. 1996). Foltyn et al. (1996) report first mature larvae from mid of April, which is in accordance with first adult catches on 25 April 2019 in this thesis. Single larval and adult specimens were still found in July (flight period from April to September indicated in Graf & Schmidt-Kloiber 2008).

The first adults of *N. sciurus* were captured one month earlier on 23 March, last on 16 May (flight period in April and May indicated in Graf & Schmidt-Kloiber 2008). No catches were obtained along int- sites. The species has a relatively small distributional range in Central Europe (Fochetti 2019) and is not mentioned in viewed IS literature from Austria, Germany and the Czechs Republic. However, the species can be found in intermittent streams of the Vienna Woods (Wolfram Graf, personal communication).

Amphinemura standfussi was found in five of six intermittent sites (not in Pfaffenbach) and in no perennial site. The determination at species level is based on the exclusive presence of *A. standfussi* in adult catches and its association with intermittent streams in literature. It was classified as (non-statistically significant) indicator species for intermittent streams (grouped int+ and int-) in the March samples and for int- streams in June. Generally, abundances were higher in int- than in int+ sites.

The occurrence of *A. standfussi* in ISs was reported previously (e.g. Germany: Foltyn et al. 1996, Vienna Woods: Philipp & Forster 2000). The species' life-cycle seems to show plasticity as flight periods vary substantially between perennial and ISs and several potential adaptations to intermittent flow are indicated (e.g. long incubation period and diapause of eggs in summer, asynchronous egg hatching, cold-stenothermal egg development including thermal quiescence of eggs at temperatures higher than 12 °C; Lillehammer et al. 1989). However, the role of these traits in survival of intermittent flow is equivocal (Bohle 2000).

Larvae occurred in samples from March to early June with relatively high abundances in March and April, which were substantially reduced in June. This was similar to *Nemoura* sp. and *Isoperla tripartita*, which had an early flight period restricted to spring, but adults of *A. standfussi* were captured from May to late August (May and June at int- Reichelgraben and int+ Gatterbach, but in July and August at int-Glauningbach; Table 20)

Brachyptera risi was found in int- Glauningbach and Reichelgraben in low numbers (one and 5 ind., respectively, in late March only) and was defined as (non-statistically significant) indicator of int-streams. In contrast, in Czech intermittent streams it was the most constant EPT species in intermittent streams (Straka et al. 2019). Its occurrence in ISs has been known for a long time (see e.g. Hynes 1941, Bohle & Potabgy 1992) and the underlying egg diapause during summer was studied by Khoo (1968b). Eggs have a long incubation time (2 - 3.5 months of diapause and 1.5 - 2 months of post-diapause

embryonic development), which is likely to favour survival under dry conditions. Larvae develop rapidly in relatively cold water from autumn to early spring. Emergence starts early in the year (flight period from March to July according to Graf & Schmidt-Kloiber 2008). In accordance with these findings, larvae were only found in late March at two int- sites in this thesis. Only one male was captured on 26th April 2019 at one of these sites (Table 20).

Isoperla tripartita occurred at five of six intermittent sites (not in Pfaffenbach) and at perennial Gatterbach (one specimen in March only). In all intermittent sites it was present from March to June in partly high abundances (730 ind./m² in int- Reichelgraben in March, 790 ind./m² in int- Glauningbach in April), but in June abundances were markedly reduced. As it also occurred in perennial Gatterbach it was not defined as indicator of intermittent streams in March, but in April and June. However, high abundances in spring seem to be a good indication for intermittent flow. Differences in abundances between int+ and int- sites were not distinct.

Records of its occurrence in intermittent streams seem to be scarce, but Philipp & Forster (2000) found it in one IS in the Vienna Woods. Řezníčková et al. (2013) and Straka et al. (2019) associated it with intermittent streams in the Czech Republic. In other regions (e.g. Germany, United Kingdom) its congener *I. grammatica* was found in ISs (Hynes 1941, Bohle & Potabgy 1992). For this species Bohle (2000) assumes dormancy of eggs, which seems to be likely in *I. tripartita* as well because most larvae in late March were only very small. Adults were captured in May and June at all intermittent sites (Table 20), which corresponds to the initial part of the flight period indicated by Graf & Schmidt-Kloiber (2008, May to August).

Table 20: Adult catches of Plecoptera species (without *Zwicknia* spp.) at the sampling sites; Roman numerals indicate months, x: present; species: *Nemoura cinerea*, *N. sciurus*, *N. marginata*; *Nemurella pictetii*; *Brachyptera risi*; *Amphinemura standfussi*; *Isoperla tripartita*

Site	<i>N. cinerea</i>				<i>N. sciurus</i>			<i>N. marginata</i>		<i>N. pictetii</i>		<i>B. risi</i>	<i>A. standfussi</i>				<i>I. tripartita</i>	
	IV	V	VI	VII	III	IV	V	IV	V	IV	V	IV	V	VI	VII	VIII	V	VI
perGep		x			x				x									
int+Gla	x	x				x											x	x
int+Gat	x	x	x	x	x	x	x	x		x	x		x				x	x
int-Rei	x	x										x	x	x			x	x
int-Pfa	x	x	x	x									x					
int-Gla	x	x		x											x	x		x

5.3.7 *Lithax obscurus*

L. obscurus is distributed in Central, Eastern and Western Europe (Robert 2015), where it lives in spring-fed streams of low- and mid-altitudes (Waringer & Graf 2011). Its occurrence in intermittent streams (ISs) is documented in several countries (Austria: Philipp & Forster 2000, Germany: e.g. Sommerhäuser 1995, Denmark: Nielsen 1977, Netherlands: Tolkamp 1980). In Austria it was only reported from the Vienna Woods and the area of Bioregion 14 (Malicky 2009). Findings from the present study show that it is distributed in all parts of the Bioregion (Fig. 63) and it is common in upper and middle reaches of nature-like streams. It was also recorded from two streams in the neighbouring Bioregion 13 (*Eastern ridges and Lowlands*, Ecoregion 11 - *Hungarian Plain*).



Fig. 62: Male of *Lithax obscurus* (from sampling site Gepringbach, 6th May 2018)

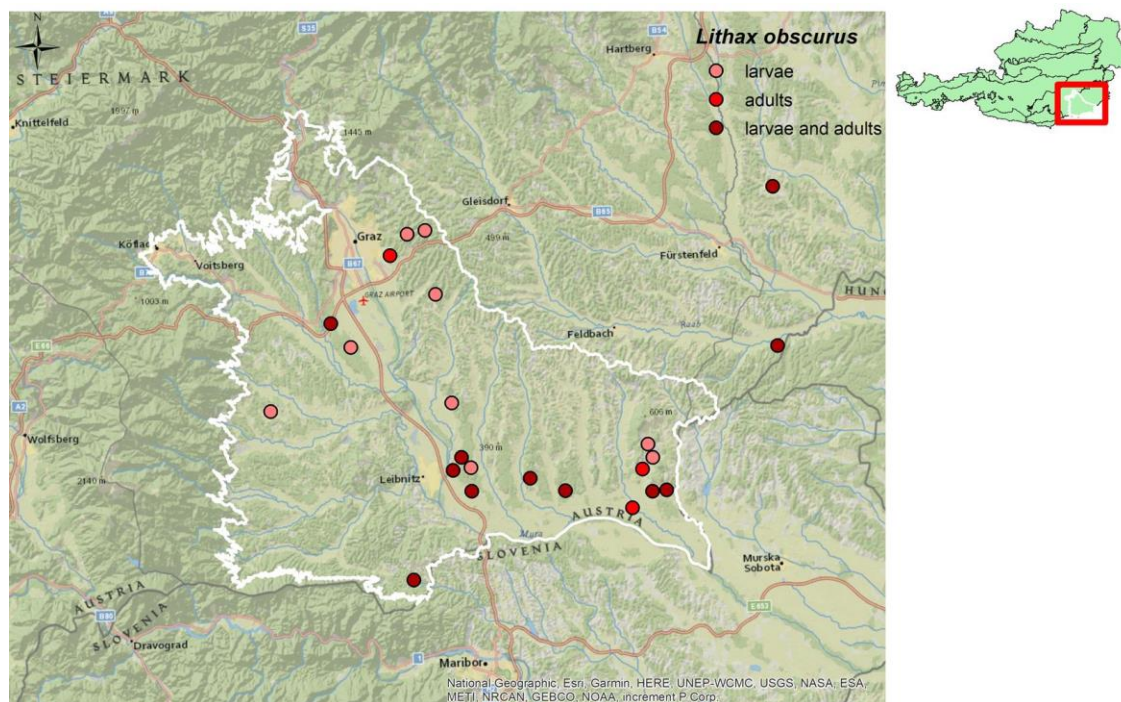


Fig. 63: Distribution of *Lithax obscurus* in Bioregion 14 (white border)

Despite its occurrence in five of six intermittent streams in the MHS samples it was not designated as an indicator species because it was also present in considerable numbers in both perennial sites.

No detailed research on its biology and adaption strategy for survival in ISs seems to exist, but Bohle (2000) and Lorenz (2000) indicate **larval quiescence**. The latter author reports that larvae remain on the dry substrate in summer. In the course of this thesis larvae were observed in remnant pools during

summer, but could not be found in completely dry sections where larvae had been collected during flowing phase (substrate was sieved for this purpose). Furthermore, in small perennial tributaries larvae were found on stones that protruded from the water, but were covered by a barely visible thin layer of flowing water.

A point that might be important for survival of intermittent flow is addressed in Tolkamp (1980): Despite a short flight period in May and June reported by several authors, pupae were found from December to September in Danish streams. Therefore, the author speculates on an extended flight period, which, however, has not been described up to now. A long flight period from spring to autumn would potentially favour survival through bet-hedging as, for example, described for the caddisfly *Plectrocnemia conspersa* (Lillpopp et al. 1998). In Bioregion 14 adults were only found from mid-April to mid-June, but last (together with early) instars were still found in benthic samples in early July and in qualitative samples at the end of July. Semivoltinism is not indicated in previous literature, but is a possible explanation for these observations. However, the life history of *L. obscurus* continues to be obscured.

5.3.8 *Oligostomis reticulata*

O. reticulata occurs in Eurasia and Fennoscandia (Sommerhäuser et al. 1997). It inhabits slow-flowing streams and various types of stagnant waters such as ponds, lakes, drainage channels, moor waters (Tobias & Tobias 1981). It was found in all three landscapes of Bioregion 14 (Grazer Basin, Western Styrian Hügelland, Eastern Styrian Grabenland) and seems to densely populate the Eastern Styrian Grabenland (Fig. 65). Additionally, it was recorded in the neighbouring Bioregion (Hungarian Plain) in the Federal Province of Burgenland.



Fig. 64: Adult specimen of *Oligostomis reticulata* (Thallergraben, Klösch, 15th April 2017)

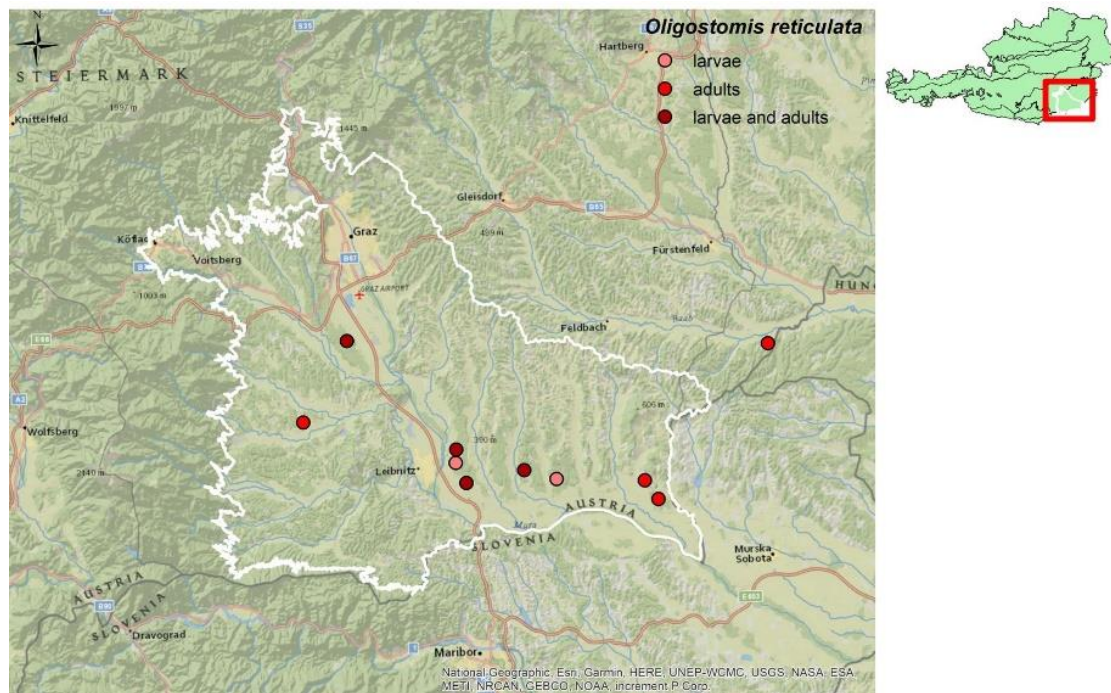


Fig. 65: Distribution of *Oligostomis reticulata* in Bioregion 14 (white border)

O. reticulata was found in five of six intermittent streams (not in int+ Reichelgraben). In all of them early instar larvae were found in the early June samples, whereas it was only found in Pfaffenbach in early March (one pupa). Generally, abundances were higher in int- than in int+ streams. Accordingly, it was defined as an indicator species for intermittent streams (int+ and int- as a group) in early June samples, but also as a statistically significant indicator for int- streams considering all seasons. In the season of late winter/early spring *O. reticulata* can also be regarded as important indicator for intermittent streams, but sampling has to occur in mid-March at the latest.

Biological and ecological characteristics of *O. reticulata* are described in Sommerhäuser & Timm (1994) and Sommerhäuser et al. (1997). The species has various adaptations to drying. Larvae are able to survive on both the dry substrate and in remnant pools. In the latter they remain active, on the dry substrate they close their cases with silken membranes and stay in quiescence until flow returns. An early flight period between March and May enables egg and larval development before habitats dry in summer. Adults were captured from early to late April in the course of this thesis.

5.3.9 *Chaetopteryx fusca*, *C. major* and *C. rugulosa*

Three species of the genus *Chaetopteryx* were determined in benthic samples and from adult catches and will be discussed hereafter.

The distribution of the *C. rugulosa* species group is limited to the Dinaric Western Balkan and the Hungarian Plain (Ecoregions 5 and 11 sensu Illies 1978). The taxonomy of this group is controversial and researchers use species and subspecies rank of some taxa differently. A recent genetic-based investigation established several new species with small-scale distributional patterns suggesting that two species are present in Bioregion 14: *C. rugulosa* Kolenati, 1848 in the surroundings of Graz and *C. prealpensis* Oláh, 2012 in the



Fig. 66: *Chaetopteryx rugulosa* from the sampling site Gatterbach upstream (int+), 15 October 2018

rest of the Bioregion (Oláh et al. 2012). However, Malicky (2014) synonymised *C. prealpensis* with *C. rugulosa*, which is followed here. Within Austria the distribution of *C. rugulosa* is restricted to the southeast (Malicky 2009). Findings of the present study suggest that it is a common species in Bioregion 14 (Fig. 67).

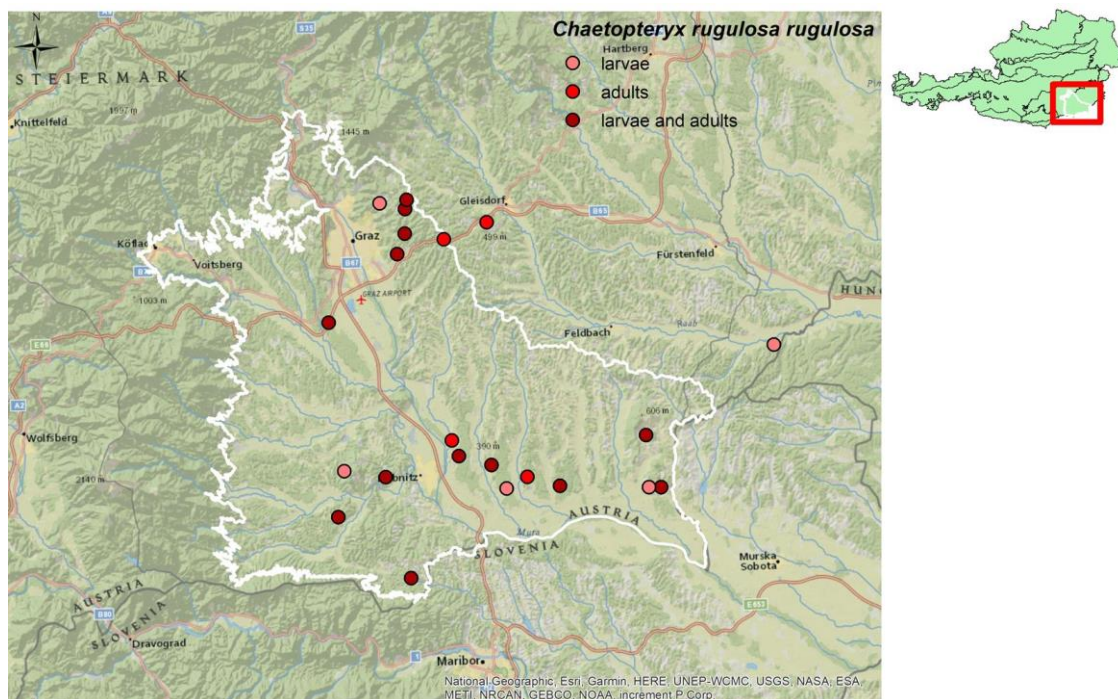


Fig. 67: Distribution of *Chaetopteryx rugulosa* in Bioregion 14 (white border)

In literature *C. rugulosa* like all representatives of the species group is indicated to occur close to springs in small streams of forested areas (Malicky 1986), however, no additional information on its ecology and biology seems to be available. Results from this thesis strongly suggest that the species is able to survive intermittent low and is not restricted to upstream sections of these streams. It was designated as statistically significant indicator for intermittent streams (int+ and grouped int+ and int-) and was more abundant in int+ than in int- streams.

Several observations on *C. rugulosa* during 2018 and 2019 are given here and are intended to constitute a starting point for detailed investigations of its biology.

Larval development proceeded rapidly during spring: Almost exclusively early instar Limnephilidae supposed to be *C. rugulosa* occurred in the March samples, but in early June and July most had reached the final instar (thus enabling determination at species level). During dry phase viable larvae were found beneath big stones, partly in large accumulations (Fig. 68). Where big stones were not available, they were buried in moist fine gravel some centimetres beneath the streambed surface. When transferring them to water, they immediately started to walk around. However, active movement into remnant pools cannot be assumed as in summer motionless larvae were found in similar densities in both remnant pools and in the dry streambed at the same site. Unlike in spring, feeding on plant material or active movement inside the water was not observed in summer. However, larval cases were never found to be attached to in-stream structures as reported from diapausing last instar larvae of *Anabolia*, *Halesus* and the congeneric *C. villosa* (Novák 1960, Denis 1978). Nevertheless, observations indicate a similar life history in *C. rugulosa*. In addition, all species have flight periods in autumn; adults of *C. rugulosa* were captured between 11th October and 7th December during 2018 and 2019 in Bioregion 14. Instead of synchronisation of life cycles postulated to be the purpose of last larval instar diapause in *Anabolia* and *Halesus*, in *C. rugulosa* larval diapause could enable them to survive drying. Larval quiescence as known from *Oligostomis reticulata* (see chapter 5.3.8) seems less likely in *C. rugulosa* according to own observations, but future laboratory experiments are needed for clarification.



Fig. 68: Dormant larvae of *Chaetopteryx rugulosa* from beneath a stone in intermittent Lieberbach (St. Nikolai ob Draßling, 2nd August 2018)

Besides *C. rugulosa*, ***C. major*** and ***C. fusca*** were determined in benthic samples. Separation of these two species in the larval stage are connected with uncertainty, but adult catches confirmed their presence. *C. major* was found in sites of all hydrological stream types, whereas *C. fusca* was limited to perennial and int+ sites. However, not all specimens could be determined at species level.

No research seems to have been done on ***C. fusca***, but the biology of a congener, *C. villosa* is well studied. Additionally, *C. fusca* is considered to be a subspecies of *C. villosa* by Malicky (2009). The existence of a larval diapause in *C. villosa* is discussed in literature (Denis 1978, but see Wagner 1986 and Tolkamp 1980) and could also favour survival of *C. fusca* in intermittent streams. Additionally, *C. fusca* is known to occur in both running and stagnant waters (Graf et al. 2017), which is likely to favour survival in remnant pools of intermittent streams.

The biology of ***C. major*** also seems to be poorly known. In contrast to *C. fusca* occurrence in stagnant waters is not indicated by Graf et al. (2017). Still, it occurs in intermittent streams of the Vienna Woods (Wolfram Graf, personal communication).



Fig. 69: left: Male of *Chaetopteryx major* (Pfaffenbach, St. Veit am Vogau, 30th September 2018); right: Female of *Chaetopteryx fusca* (Kainbach bei Graz, Ankesbach, 25th October 2018)

5.3.10 Trichoptera species with adult diapause

Many Trichoptera of the Limnephilidae family that inhabit intermittent streams have an ovarial diapause (parapause; chapter 2.4.3): Adults emerge in spring or early summer before flow ceases, aestivate with immature/diapausing ovaries in terrestrial habitats and return to the streams in autumn to oviposit. Originally, these species were thought to have two generations due to two peaks in light-trapped adults in spring and autumn, respectively (Ulmer 1925, in Novak & Sehnal 1963). In many of those species additional adaptations to intermittent flow can be present. For example, drying-resistant eggs in gelatinous matrices are deposited into the dry streambed (*Stenophylax* spp., *Limnephilus* spp.; Bouvet 1978; Novák & Sehnal 1963) or are stuck to branches or leaves of plants protruding the stream channel (*G. pellucidus*; Crichton 1987; Fig. 70). First instar larvae hatch inside the gelatinous matrix and quiesce until water returns (Bouvet 1978). In the case of *G. pellucidus* they leave the matrix at long-lasting rainfall and drop into the stream.



Fig. 70: Egg masses probably of *Glyptotaelius pellucidus* containing quiescent first instar larvae deposited on a leaf of the common hazel (*Corylus avellana*) above the streambed at sampling site Reichelgraben upstream (Weinburg, 15th October 2019)

Larvae of *Stenophylax* spp. are also able to be quiescent in more mature larval instars within moist substrate. Additionally, pupae are drying-resistant and adults are able to emerge from the moist substrate (Bouvet 1978). Fey (1984, cited in Bohle 2000) observed downstream migration to remnant water-filled pools in larvae of *Stenophylax permistus*, which does not happen when flow is permanent. Another behavioural adaptation found in this group is adult aestivation in caves. In Europe this is known from several species of the genera *Micropterna*, *Stenophylax* and *Mesophylax* (Décamps & Magné 1966). In Austria, migration into caves is known from *Stenophylax permistus* and *Micropterna nycterobia* (Malicky & Winkler 1974, Malicky 1971), which both occur in Bioregion 14.

Table 21 and Table 22 show adult catches of species of this group from 2017 to 2019. All species that occurred in benthic samples could also be captured in the adult stage. Some species in the tables were only recorded as adults: *Limnephilus flavicornis*, *L. griseus*, *L. ignavus*, *Grammotaulius nigropunctatus*, *Micropterna lateralis* and *M. sequax*.

Table 21: Adult catches of *Limnephilus* species (Trichoptera: Limnephilidae) with adult diapause at the sampling sites; Roman numerals indicate months, x: present

Site	<i>L. auricula</i>				<i>L. flavicornis</i>	<i>L. griseus</i>	<i>L. ignavus</i>		<i>L. lunatus</i>		<i>L. rhombicus</i>			
	V	VI	VIII	XI	IX	IX	VIII	IX	VI	VIII	V	VI	VIII	IX
perGep	x											x		
int+Gla										x				
int+Gat		x							x			x		
int-Rei		x												
int-Pfa		x	x	x	x	x	x		x		x		x	x
int-Gla	x	x						x	x			x		

Table 22: Adult catches of *Trichoptera* species with adult diapause at the sampling sites (all Limnephilidae); Roman numerals indicate months, x: present; species: *G. nigro.* = *Grammotaulius nigropunctatus*, *M.* = *Micropterna*, *G.* = *Glyphotaelius*

Site	<i>G. nigro.</i>			<i>M. lateralis</i>		<i>M. sequax</i>	<i>S. permistus</i>			<i>G. pellucidus</i>					
	VI	IX	X	V	VI	X	VI	IX	X	V	VI	VII	VIII	IX	X
perGep				x	x					x	x	x		x	
int+Gla	x				x										
int+Gat				x	x										
int-Rei	x				x	x			x					x	
int-Pfa		x	x					x	x	x			x	x	
int-Gla	x				x		x		x	x				x	x

For late March *Stenophylax permistus* and *Glyphotaelius pellucidus* were defined as **indicator species** for int- sites and for both intermittent types. *S. permistus* occurred exclusively in the larval stage in the March samples, in April samples the major part was pupae and in early June only one pupa was still present in one site. While larvae of *Limnephilus auricula* and *L. rhombicus* were only found in March and April samples, those of *L. lunatus* were still found in int+ Gatterbach in July, which is surprising. This finding indicates that *L. lunatus* might deviate from the typical phenology in limnephilids with adult diapause by having a more extended emergence period. However, it has to be mentioned that determination of *Limnephilus* larvae at species level is delicate. *L. lunatus* and *L. rhombicus* were not restricted to intermittent sites but also occurred in perennial Gepringbach.

One of the species only captured as adult is ***Micropterna lateralis***. However, fifth instar larvae were found in high numbers in a small intermittent tributary and even in a stagnant water body near the sampling sites in Glauningbach in winter. At that time the tributary was not flowing and larvae were concentrated in remnant pools.

In literature, inconsistent information is given on life cycles. While some authors attribute a long flight period from May to October connected to adult diapause to this species (e.g. Waringer & Graf 2011, Graf et al. 2019a), some authors attribute to it a shorter flight period in spring and summer without diapause (Germany: Hoppeler et al. 2017, Sweden: Svensson 1972, Great Britain: Crichton 1971). Adult catches during this thesis were made between the beginning of May and mid-June, but not in autumn. This is consistent with lacking adult diapause in the species. Hoppeler et al. (2017) found further evidence because they only found adults with mature ovaries in spring and early summer. They suggest dormant eggs as strategy for survival in intermittent streams.

5.3.11 *Synagapetus moselyi* and *S. krawanyi*

Distributional ranges of *S. moselyi* and *S. krawanyi* are similar (Central and East Europe; Malicky 2019, Neu et al. 2018), but *S. moselyi* seems to be restricted to lower altitudes, whereas the main distributional range of *S. krawanyi* are Alpine regions even if records from the Hungarian Plain exist (e.g. Nógrádi 1984).

This pattern seems to be reflected in Bioregion 14 as well: *S. krawanyi* is restricted to regions that are relatively close to prealpine regions (Styrian Randgebirge, Poßruck). In the southern range of the East Styrian Grabenland, where most benthic samples were taken, only *S. moselyi* occurred (Fig. 72). East of the city Graz both species were found, in one case syntopic (Kainbach bei Graz, Ragnitzbach).



Fig. 71: Male of *Synagapetus moselyi* (10th May 2018, Lieberbach, St. Nikolai ob Draßling)

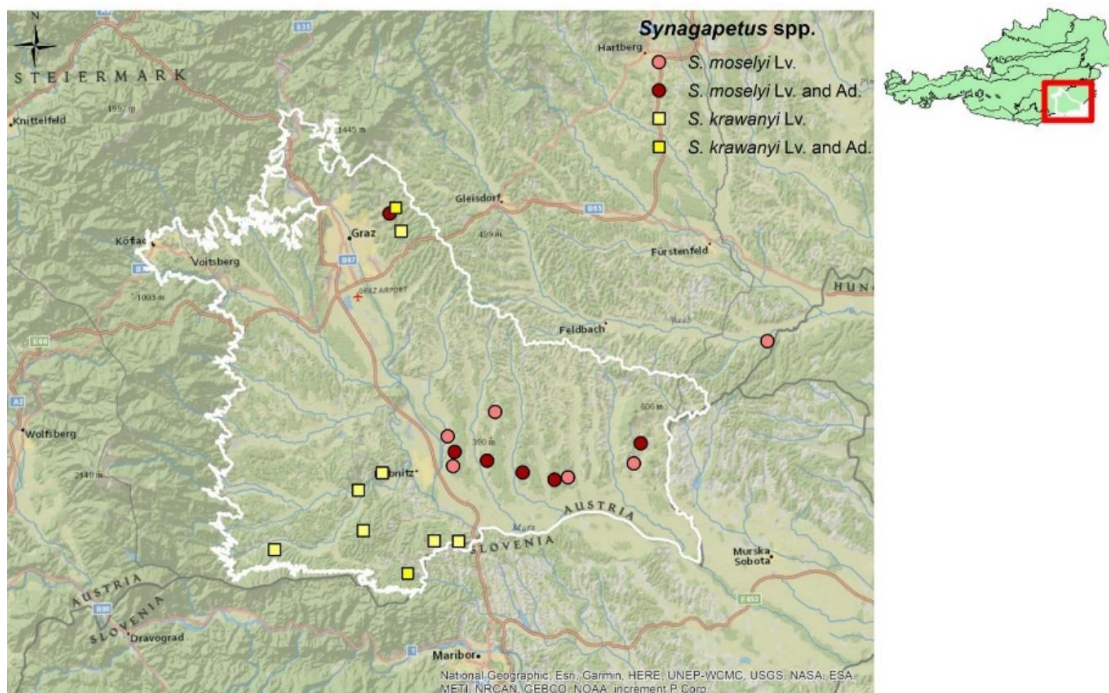


Fig. 72: Distribution of *Synagapetus moselyi* and *S. krawanyi* in Bioregion 14 (white border)

S. moselyi resulted to be a statistically significant **indicator species** for intermittent streams (int+ and grouped intermittent types) and was found in five of six intermittent sites. Abundances were especially high in int+ streams, e.g. about 500 individuals/m² in int+ Gatterbach and Glauningbach in late March (see Fig. 73). However, the species was also found in small perennial headwater streams through qualitative sampling. Such streams constitute their typical habitat in Austria (crenal and hypocrenal zone; Graf et al. 2017). *S. krawanyi* was only sampled qualitatively and was also found in considerable abundance in an intermittent stream in the West Styrian Hügelland (Heimschuh, Wellingbach).



Fig. 73: Pupal cases of *Synagapetus moselyi* and *Potamophylax rotundipennis* from the sampling site Glauningbach upstream (Wittmannsdorf, 24th May 2019)

The presence of *S. moselyi* in intermittent streams was frequently reported and is favoured by **drying-resistant eggs and pupae** (Fischer 2003). In his laboratory experiments, dry conditions of 84 days resulted in only a slight reduction in egg hatching success. Furthermore, asynchronous hatching from eggs (about two months between emergence of first and last larvae of one egg clutch) could constitute a bet-hedging strategy that is adaptive to intermittent flow.

Flight period of *S. moselyi* is indicated from May to August in Waringer & Graf (2011). Catches in the present study were between 10th May and 17th June. Early instar larvae were found in October 2019 in int+ Reichelgraben and int+ Glauningbach, but not in the int- sites, where early instar larvae had been recorded in early February. This is in accordance with the experimental finding that drying extends the embryo development considerably (84 days of dry conditions extend development from 78 to 149 days on average and to a maximum of 200 days; Fischer 2003).

5.3.12 Other Trichoptera species with unknown strategies

Beraeodes minutus (Fig. 74, left) occurred in perennial Gepringbach and int+ Gatterbach, in the latter in relatively high abundances (e.g. 890 ind./m² in early July). Last instar larvae and pupae were present in the March and April and early instars in the June samples indicating that egg diapause does not exist. Its presence in intermittent streams has been reported previously (Lorenz 2000, Sommerhäuser et al. 1997, Philipp & Forster 2000). Lorenz (2000) only found it in remnant pools only but not on the dry substrate indicating that drying-resistance does not exist in the species. An early flight period in May and June (Waringer & Graf 2011) and a tolerance to low oxygen concentrations are likely to favour its survival in intermittent streams that retain remnant pools. Adults were captured between 5th and 16th May, which indicates a short and highly synchronised flight period.

Potamophylax rotundipennis (Fig. 74, right) occurred in sites of all three hydrological stream types. The ability to survive drying of its aquatic habitat in Bioregion 14 is obvious. However, no reports seem to exist on its occurrence in intermittent streams. Straka et al. (2019) even classify it as reliable perennial flow indicator. Indications on its flight period are ambiguous in literature. Graf et al. (2019a) indicate a

short flight period although it extends from spring to autumn (see Table 9). On the other hand, a flight period only in August and September is often indicated (Tobias & Tobias 1981, Sommerhäuser et al. 1997, Waringer & Graf 2011). In the course of this thesis the earliest capture of adults was on 17th June at an int+ stream, the latest on 14th October at perennial Gepringbach. This finding confirms a long flight period, however, the records are insufficient to describe the flight period duration at single sites. It would be worthwhile investigating if flight periods diverge significantly between streams with different hydrological regimes. It seems possible that this euryoecious species has a highly plastic life cycle, which enables survival in intermittent streams.



Fig. 74: left: male of *Beraeodes minutus* (10th May 2018, Lieberbach, Hütt); right: *Potamophylax rotundipennis* (14th September 2018, Petersbach, Graz)

Other caddisfly species which occurred in intermittent stream samples in this thesis, but do not seem to have been reported from such streams are *Lype reducta* and *Rhyacophila hirticornis*. Larvae of *L. reducta* were only sampled from int+ Reichelgraben, but adults were also captured at the other two int+ sites and perennial Gepringbach. Adult catches were obtained between 2nd June and 23th July. *R. hirticornis* (Fig. 75) occurred in benthic samples at two int+ and one int- site and was designated as indicator species for grouped intermittent stream types in March. Adults were captured between 8th May and 14th June. Adults of both species can also be found along intermittent streams of the Vienna Woods (own observation).



Fig. 75: Female of *Rhyacophila hirticornis* (tributary of Halterbach, Vienna, 23th May 2018)

5.3.13 *Ironoquia dubia*

Ironoquia dubia is the only European representative of the Dicosmoecinae subfamily within the Limnephilidae. Worldwide, the genus *Ironoquia* is represented by five additional species from North America (*I. plattensis*, *I. parvula*, *I. punctatissima*, *I. lyrata*, *I. kaskaskia*) which all inhabit temporary standing and running waters (Wissemann 1988).

I. dubia has a **distribution** ranging from Central, East and North Europe to Asia Minor and the Caucasus (Botosaneanu & Malicky 1978; in Roos & Semmler-Elpers 2008), but is often regarded as rare. In Austria, Malicky (1999) reports it from the Federal Provinces of Lower Austria, Burgenland and Styria, but only from few locations (Malicky 2009). In the course of this master thesis the species was found in the Grazer Basin, Western Styrian Hügelland and Eastern Styrian Grabenland and seems to be common in pristine intermittent streams.

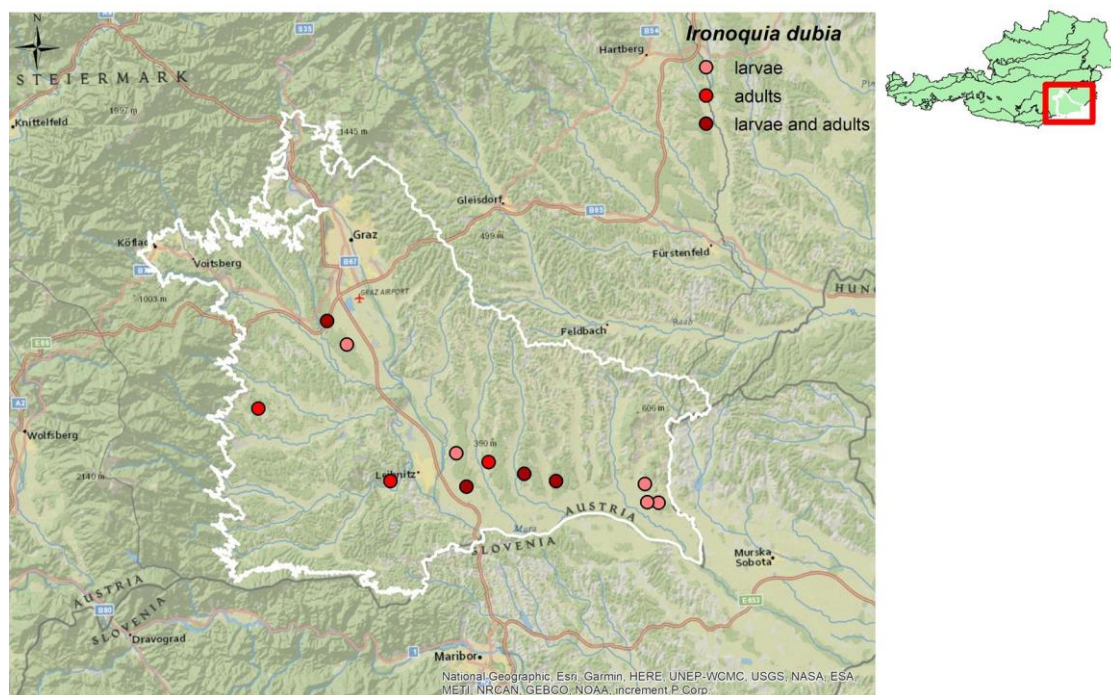


Fig. 76: Distribution of *Ironoquia dubia* in Bioregion 14 (white border)

It occurs in both slow-flowing streams and small stagnant waterbodies rich in submerged vegetation and detritus at low altitudes which can be dry in summer (Schmedtje & Colling 1996). Lepneva (1970) reports larvae from slow-flowing canals, even drainage canals, along the Moscow-Leningrad highway that sometimes have temporary current. Similarly, Cuk & Vuckovic (2010) reported *I. dubia* from heavily modified small streams partly deprived of all woody riparian vegetation in Croatia. So far in Austria, it has been found in oligo- to alpha-mesosaprobic streams from the hypocrenal to the epipotamal, but not in stagnant waters (Graf et al. 2017). Larval findings in this thesis originate from small intermittent streams with clean water, but single larvae were found in perennial Gepringbach through qualitative sampling. This stream, however, had several intermittent-flow indicator species and was classified as near-perennial by the *Biodrought index* (Fig. 49). Whether *I. dubia* occurs in stagnant water bodies was not investigated but seems likely.

Little research has been done on the **life history** of *I. dubia*. Fiedler (1992) reports from Germany that except for some larvae all disappeared from the stream by the end of May and adults were found in mid-September when the streambed was still dry. Fiedler assumed that larvae leave the water to bury in the bank substrate because she had unintentionally transported three specimens attached to a tuft of grass from a stream bank to the laboratory in mid-August. Two of them had closed cases and were reared in aerated water. They emerged to the adult stage in September. It was not clear if these two specimens were larvae or pupae, though. The third specimen, the case of which was not closed, was observed to immediately bury in soil offered to it, but Fiedler gives no information on the “fate” of this specimen. Hiley (1973, in Wallace 2011) attempted to **rear** four larvae confined to an aquatic environment, but was not successful. Failure to rear larvae by standard methods corresponds to reports from North American congeners (Ross 1944, Flint 1958: *I. parvula*, Williams & Williams 1975: *I. punctatissima*) as well as to own attempts. In contrast, rearing larvae of *I. punctatissima* in a moist terrestrial environment enabled them to emerge as adults (Williams & Williams 1975), which also applies to *I. dubia* according to findings of this thesis: Larvae as well as pupae taken from their natural habitat could be reared to adults in this way, but with greater success in the case of pupae. The attempt to rear specimens with closed cases (most probably pupae) in an aquatic environment by offering a water film in a Petri dish failed, but more sophisticated experiments will be required for future experiments to confirm survival of pupae in aquatic environments.

The **life histories** of three of the aforementioned **North American *Ironoquia* species** are relatively well-known. All have dormant last instar larvae that aestivate in moist leaf litter at the margins of their aquatic habitats and emerge as adults in late summer and autumn (*I. parvula*: Flint 1958; *I. punctatissima*: Clifford 1966, *I. plattensis*: Whiles et al. 1999). Another member of the subfamily (Dicosmoecinae) applies the same strategy (*Nothopsyche yamagataensis*; Nozaki 1993, in Bohle 2000). Burying in the soil of bank substrate as assumed by Fiedler (1992) and confirmed by this thesis does not seem to have ever been reported from larvae of North American species.

Bohle (2000) indicates that *I. dubia* might aestivate either as larva or as pupa. Observations during this thesis found evidence for aestivation at the larval stage as in mid-July only unclosed larval cases were found in the unsubmerged bank substrate. However, pupation seems to take place in the same terrestrial habitats and might start by the end of August: On 25th August all collected cases were closed but contained larvae, two weeks later only pupae were found.

Significant **morphological** differences between *Ironoquia* species can be observed in the **pupal stage**. For example, *I. dubia* and *I. punctatissima* have gills (Döhler 1914 and Williams & Williams 1975, respectively) whereas *I. parvula* has none and no lateral line either, features which correspond to the pupa of the truly terrestrial European species *Enoicyla pusilla* (Limnephilinae; Flint 1958). The pupa of *I. plattensis* does not seem to have been described yet but adults are close to *I. parvula* (Alexander & Whiles 2000). These differences among *Ironoquia* species might represent different extents of pupal adaption to the dry habitat with *I. parvula* (and probably *I. plattensis*) having more distinct adaptations to a terrestrial existence (Wallace 2011), thus being more recent species.

The **flight period** observed in Bioregion 14 corresponds to information from most literature sources (September and October: Hickin 1967, Higler 2005, Tobias & Tobias 1981, Ulmer 1909). Literature attributing an extended flight period from spring to autumn including adult diapause (Graf et al. 2019a, Müller & Schönfelder 2010, Dahm & Hering 2015,) does not describe or cite empirical evidence. **Eggs**, presumed to be deposited in the dry streambed in early autumn if water is not available yet (Bohle 2000), could not be found and have to be subject to future research.

Another important point in the life histories of *Ironoquia* species is **larval development**. Fiedler (1992) reports fourth instar larvae from early February, whereas in Pfaffenbach fourth instar larvae dominated

as late as on 6th April. Later seasonal resumption of flow in Pfaffenbach (in fact at the end of October) is a potential explanation for this, but this was not the case. Therefore, own observations suggest that egg diapause could exist. Bohle (2000, p. 55, Table 2) lists egg diapause as possible adaption to flow intermittence, but justification for the presumption is not given; he possibly related it to eggs placed in the dry streambed. First instar larvae in late autumn and slow growth during winter is known from *I. plattensis* and *I. punctatissima* and seems possible in *I. dubia*, too.

Larval growth from different regions and laboratory experiments should be subject to future research to identify detailed adaption mechanisms.

In summary, observations of this thesis suggest that adaption strategies to flow intermittence include **drying resistance** and **diapause** in the **egg stage, larval diapause** (most likely an eudiapause sensu Müller 1970) in terrestrial bank substrate and an **autumnal flight period**. Confirmation and detailed information on environmental drivers of the physiological adaptations have to be obtained by future laboratory experiments.

6 Summary and conclusion

The main aim of this thesis was to test different methods that potentially enable identification of streams with different hydrological regimes in Bioregion 14 (Austria: Styria): perennial, intermittent streams that retain remnant pools (int+) and those which dry completely (int-) (hypotheses [H] 1, 2 and 3). Furthermore, this thesis was intended to verify and extend autecological and distributional knowledge on important macroinvertebrate species that inhabit intermittent streams (Research questions 5 and 6), especially on the caddisfly *Ironoquia dubia* (H4).

It was shown that the three postulated **hydrological regime types can be separated** based on samples in early spring and multivariate statistical methods (H1), namely non-metric multidimension scaling (NMS) and cluster analysis. Subsequently, it was also possible to identify indicator taxa for all three hydrological stream types by means of Indicator species analysis.

Structural community metrics, namely total abundance and taxa richness were shown to change along the hydrological gradient. However, the potential to identify hydrological types based on single samples turned out to be little reliable. Higher potential was found for Plecoptera (abundance and relative abundance), Coleoptera (abundance and number of taxa) and Ephemeroptera (number of taxa) based on early spring samples.

Higher **temporal dynamics** in abundance and taxa richness during spring predicted for intermittent streams (H2) could neither be stated at the community level nor for single higher taxonomic units. Changes were inconsistent within the hydrological types and/or not significantly different between the types. The only relatively distinct trend was a stronger increase in *Gammarus fossarum* in perennial sites from June to July samples, which had a high impact on total abundance too. These trends were weaker or converse in intermittent sites and, most importantly, were not significantly different between int+ and int- sites.

In contrast, several methods using **biological community traits** could be shown to act as relatively effective hydrological stream discriminators (H3). In all of them samples from late March showed most distinctive differences. Among two published MI community-based indices, the **Biodrought index** (Straka et al. 2019) was shown to enable identification of all three hydrological stream types, whereas the **DEHLI index** (Chadd et al. 2017) does not seem to be suitable for this purpose in Bioregion 14. Another published **method according to Leigh et al. (2015)**, which classifies taxa based on their resistance and/or resilience ability at mainly family level, responded to differences in hydrological regime with statistical significance, but differences are thought to be too slight in order to be an effective hydrological regime discriminator.

A method developed in this thesis based on biological **trait classifications** at mainly genus level in **Tachet et al. (2010)** was shown to recognise some of the intermittent sites, but others were similar to perennial sites. Thereby, traits enabling resistance to ceased flow (respiration through plastron and spiracles, resistance forms) proved to be suitable because they had significantly higher scores in intermittent sites (H3a).

Four analyses based on published features of EPT species' **flight periods** (FPs) were developed and tested for the hypothesised existence of earlier and longer FPs in intermittent compared to perennial stream communities (H3b). All analyses had in common that FPs between perennial and int+ sites were relatively similar. Most distinct differences between perennial and int- sites resulted from seasonality of FPs: An index which quantifies the seasonal timing of FPs (the **community flight period index**) showed

significantly earlier FPs (approximately 25 days) in all int- compared to perennial sites. However, the timing of FPs was not significantly different between perennial and int+ streams. Two other analyses categorising EPT species according to **FP types** (short FP in spring, short in spring and summer, long FP, short FP in autumn) showed statistically significant differences in proportions of these types between hydrological stream types, but differences were relatively slight and all FP types were present in all hydrological stream types. Most distinct differences resulted from **FP types of caddisfly species**: Proportions of species with adult diapause during summer were significantly higher in int- streams than in perennial streams. A major limitation of this analysis is that the number of caddisfly species can be low in stream communities.

Despite the mostly weak responses in the four FP analyses no trend towards the predicted longer FPs in intermittent stream types were detected. On the contrary, perennial streams tended to have higher proportions of EPT species with long FPs.

Ironoquia dubia is a poorly studied inhabitant of intermittent streams. Some information on its life-history is available in Fiedler (1992), which was intended to be verified in this thesis (H4). The existence of dormant terrestrial larvae and pupae buried in the bank substrate could be confirmed through field and ex-situ observation. These findings suggest that attribution of adult diapause in literature should be changed in order to improve future works on biological traits on aquatic communities.

Recognition of different hydrological regimes is a crucial task for the assessment of the ecological status and for the management of streams. The occurrence of intermittent stream is expected to increase due to climate change and subsequently the assessment will become more important in future. This master thesis was intended to be a first step towards developing an assessment method for such streams in Austria. Different methods to recognise hydrological regime types were tested based on eight least-impacted streams in Bioregion 14. Some were shown to be effective discriminators; however, verification and application in other biogeographical regions need to be undertaken in future studies. Additionally, autecological knowledge gaps of poorly studied species shown to be inhabitants of intermittent streams in this thesis have to be filled (e.g. *Rhyacophila hirticornis*, *Chaetopteryx rugulosa*, *C. major*, *Lithax obscurus*, *Potamophylax rotundipennis*, *Siphonurus armatus*) in order to make recognition, ecological status assessment and protection more effective. Finally, it should be the task of every researcher of natural sciences to show the value, importance and beauty of ecosystems to the public.

I want to conclude this thesis with a sentence of a famous zoologist:

“We only love what we know and we only protect what we love.” (Konrad Lorenz)

7 References

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8.1 Tables

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9 Appendices

Appendix A

Table A: Mean affinity scores (Tachet et al. 2010) for traits of resistance to intermittent flow

<u>Hydrological type</u>	<u>site</u>	Mean affinity scores (Tachet et al. 2010) for traits of resistance to intermittent flow			
		<u>late March</u>	<u>early June</u>	<u>early July</u>	<u>all dates</u>
perennial	perGep	1.45	1.68	1.53	1.41
	perGat	1.93	1.68	1.38	1.33
intermittent+	int+Rei	2.35	1.40	1.57	1.54
	int+Gla	2.07	1.38	1.24	1.46
	int+Gat	1.71	1.45	1.58	1.37
intermittent-	int-Rei	2.38	1.77	1.57	1.55
	int-Pfa	2.27	1.22	2.05	1.66
	int-Gla	2.33	1.78	1.82	1.74

Appendix B

Table B 1: Proportions of flight period types (%) from samples of late March and early June; type 1: short in spring, type 2: short in spring and summer, type 3: long, type 4: short in autumn

<u>Hydrol. type</u>	<u>site</u>	Proportion of FP type (all dates)					Proportion of FP type (April)				
		<u>type 1</u>	<u>2</u>	<u>3</u>	<u>4</u>	<u>n</u>	<u>type 1</u>	<u>2</u>	<u>3</u>	<u>4</u>	<u>n</u>
perennial	perGep	16%	20%	48%	16%	25	24%	24%	41%	12%	17
	perGat	20%	20%	50%	10%	20	25%	25%	33%	17%	12
intermittent+	int+Rei	18%	18%	45%	18%	22	19%	25%	31%	25%	16
	int+Gla	24%	24%	29%	24%	21	20%	27%	27%	27%	15
	int+Gat	25%	21%	36%	18%	28	13%	31%	38%	19%	16
intermittent-	int-Rei	29%	19%	38%	14%	21	18%	27%	27%	27%	11
	int-Pfa	22%	0%	56%	22%	9	33%	0%	0%	67%	3
	int-Gla	35%	24%	29%	12%	17	30%	30%	30%	10%	10

Table B 2 Proportions of flight period types (%) from samples of early July; type 1: short in spring, type 2: short in spring and summer, type 3: long, type 4: short in autumn

<u>Hydrol. type</u>	<u>site</u>	Proportion of FP type (early July)				<u>n</u>
		<u>type 1</u>	<u>2</u>	<u>3</u>	<u>4</u>	
perennial	perGep	18%	12%	47%	24%	17
	perGat	20%	40%	20%	20%	5
intermittent+	int+Rei	17%	17%	33%	33%	6
	int+Gla	14%	43%	29%	14%	7
	int+Gat	10%	20%	30%	40%	10
intermittent-	int-Rei	33%	17%	17%	33%	6
	int-Pfa	0%	0%	100%	0%	2
	int-Gla	100%	0%	0%	0%	2

Table B 3: Proportions of flight period types (%) of Trichoptera species from late April early June; types according to Sommerhäuser et al. (1997): type A: short in spring, type B: mainly in summer, type C: long with diapause, type D: short in autumn

Hydrol. type	site	Proportion of FP type (late April)					Proportion of FP type (early June)				
		type A	B	C	D	n	type A	B	C	D	n
perennial	perGep	25%	50%	8%	17%	12	11%	56%	0%	33%	9
	perGat	13%	63%	0%	25%	8	17%	67%	0%	17%	6
intermittent+	int+Rei	11%	33%	11%	44%	9	10%	60%	10%	20%	10
	int+Gla	11%	44%	0%	44%	9	25%	38%	0%	38%	8
	int+Gat	10%	40%	20%	30%	10	38%	25%	0%	38%	8
intermittent-	int-Rei	0%	33%	17%	50%	6	17%	33%	17%	33%	6
	int-Pfa	0%	0%	0%	100%	2	50%	0%	0%	50%	2
	int-Gla	17%	33%	33%	17%	6	40%	40%	0%	20%	5

Table B 4: Proportions of flight period types (%) of Trichoptera species from early July; types according to Sommerhäuser et al. (1997): type A: short in spring, type B: mainly in summer, type C: long with diapause, type D: short in autumn

Hydrol. type	site	Proportion of FP type (early July)				
		type A	B	C	D	n
perennial	perGep	20%	40%	0%	40%	10
	perGat	33%	33%	0%	33%	3
intermittent+	int+Rei	33%	0%	0%	67%	3
	int+Gla	25%	50%	0%	25%	4
	int+Gat	14%	14%	14%	57%	7
intermittent-	int-Rei	50%	0%	0%	50%	4
	int-Pfa	0%	0%	0%	0%	0
	int-Gla	100%	0%	0%	0%	2

Appendix C

Table C 1: DEHLI index values according to Chadd et al. (2017) of all samples

Hydrological type	site	DEHLI		
		late March	early June	early July
perennial	perGep	5.30	5.13	4.90
	perGat	5.25	5.25	5.43
intermittent+	int+Rei	5.58	4.94	4.94
	int+Gla	5.09	5.13	5.17
	int+Gat	5.39	5.29	4.85
intermittent-	int-Rei	5.00	4.95	4.89
	int-Pfa	3.86	4.00	3.29
	int-Gla	5.40	5.39	5.50

Table C 2: Drought Intolerance Score (DIS) taxa list for the calculation of the DEHLI index (Chadd et al. 2017) for samples from late March; key to DIS scores: white = 9 or 10, light grey = 7 or 8, medium grey = 5 or 6, dark grey = 3 or 4 and black = 1 or 2; key to hydrological regime types: **blue** = perennial, **orange** = intermittent+, **red** = intermittent-

site	perGep	perGat	int+Rei	int+Gla	int+Gat	int-Rei	int-Pfa	int-Gla
date	26-Mar-2019	26-Mar-2019	26-Mar-2019	26-Mar-2019	23-Mar-2019	26-Mar-2019	26-Mar-2019	29-Mar-2019
	Heptageniidae	Heptageniidae	Heptageniidae	Heptageniidae	Heptageniidae	Rhyacophiliidae		Rhyacophiliidae
		Rhyacophiliidae			Rhyacophiliidae	Brachyperta		Brachyperta
	Aeshnidae	Goeridae	Glossosomatidae	Glossosomatidae	Beraeidae	Glossosomatidae	Phryganeidae	Glossosomatidae
	Beraeidae	Hydraenidae	Goeridae	Goeridae	Glossosomatidae	Goeridae		Goeridae
	Goeridae	Perlodidae	Hydraenidae	Hydraenidae	Goeridae	Hydraenidae		Hydraenidae
	Hydraenidae	Baetis	Perlodidae	Hydrobiidae	Hydraenidae	Hydrobiidae		Hydrobiidae
	Hydrobiidae		Amphin./Nemurella	Perlodidae	Leuctridae	Perlodidae		Perlodidae
	Sericostomatidae		Baetis	Amphin./Nemurella	Perlodidae	Amphin./Nemurella		Amphin./Nemurella
	Baetis				Sericostomatidae			
					Amphin./Nemurella			
	Coenagrionidae	Elmidae	Elmidae	Elmidae	Elmidae	Gammaridae	Elmidae	Elmidae
	Elmidae	Ephemeridae	Gammaridae	Ephemeridae	Gammaridae	Polycentropodidae	Gammaridae	Gammaridae
	Ephemeridae	Gammaridae	Gomphidae	Gammaridae	Polycentropodidae	Simuliidae	Simuliidae	Simuliidae
	Gammaridae	Hydropsychidae	Hydropsychidae	Polycentropodidae	Siphonuridae	Siphonuridae	Siphonuridae	Siphonuridae
	Hydropsychidae	Leptoceridae	Simuliidae	Siphonuridae	Habrophlebia/Leptop.	Habrophlebia/Leptop.	Habrophlebia/Leptop.	Nemoura
	Leptoceridae	Simuliidae	Siphonuridae	Habrophlebia/Leptop.	Nemoura	Nemoura	Nemoura	
	Polycentropodidae	Habrophlebia/Leptop.	Habrophlebia/Leptop.	Nemoura				
	Habrophlebia/Leptop.	Paraleptophlebia	Nemoura	Paraleptophlebia				
	Nemoura							
	Corduli-/Libellulidae	Psychodidae	Physidae	Corduli-/Libellulidae	Scirtidae	Corduli-/Libellulidae	Corduli-/Libellulidae	Corduli-/Libellulidae
	Physidae	Scirtidae	Tabanidae	Tabanidae	Tabanidae	Physidae	Hydrophilidae	Psychodidae
	Cloeon					Psychodidae		Tabanidae
						Tabanidae		
						Tipulidae		
	Ceratopogonidae	Ceratopogonidae	Ceratopogonidae	Ceratopogonidae	Ceratopogonidae	Asellidae	Asellidae	Ceratopogonidae
	Dytiscidae	Lymnaeidae	Pisidiidae	Dytiscidae	Dytiscidae	Ceratopogonidae	Ceratopogonidae	Culicidae
	Planorbidae	Pisidiidae		Lymnaeidae	Lymnaeidae	Dytiscidae	Culicidae	Lymnaeidae
		Planorbidae		Pisidiidae	Pisidiidae	Lymnaeidae	Pisidiidae	Pisidiidae
				Planorbidae	Planorbidae	Pisidiidae	Planorbidae	
Σ DIS taxa	23	20	19	22	23	24	14	20
DEHLI	5,30	5,25	5,58	5,09	5,39	5,00	3,86	5,40

Appendix D

Table D: Calculation results for the *Biodrought index* (Straka et al. 2019); kindly provided by Marek Polášek; BI: Biodrought index, *p*: probability, *npe*: near-perennial, *per*: perennial, *int*: intermittent, *IT*: proportion of perennial-flow indicator taxa (range: -1 to +1), *HBF*: proportion of taxa with high body flexibility, *TA*: total abundance (ln(x+1)-transformed), *POS*: proportion of taxa with a preference for organic substrates, *no. IT*: number of indicator taxa; colours: **blue**: perennial, **orange**: int+/near-perennial, **red**: int-/intermittent

site	date	BI	p per	p npe	p int	IT	HBF	TA	POS	no. IT
perGep	26-3-19	0.569	0.430	0.559	0.010	0.286	0.494	9.039	0.253	28
perGat	26-3-19	2.883	0.997	0.003	0	0.771	0.450	9.569	0.042	35
int+Rei	26-3-19	0.581	0.438	0.552	0.010	0.273	0.450	9.014	0.084	22
int+Gla	26-3-19	-0.569	0.027	0.888	0.086	0.048	0.564	8.949	0.155	21
int+Gat	23-3-19	0.218	0.227	0.752	0.021	0	0.279	9.967	0.129	26
int-Rei	26-3-19	-3.021	0	0.016	0.984	-0.667	0.416	9.391	0.220	24
int-Pfa	26-3-19	-3.348	0	0.005	0.995	-0.636	0.563	8.987	0.259	11
int-Gla	29-3-19	-2.470	0	0.09	0.910	-0.400	0.51	8.603	0.221	20